

CHARTIERS NATURE CONSERVANCY



# *Lower Chartiers Creek Trail and Greenway Feasibility Study*

*April 2006*

**PREPARED BY:**

*Civil & Environmental Consultants, Inc.*



SIMONE JAFFE COLLINS





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Tim Volk - President  
Derek Rykaczewski - Vice President  
Vincent Copolla - Secretary  
Celia Santander - Treasurer

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## ***Executive Summary***

The lower Chartiers Creek valley presents the residents of southwestern Pennsylvania with a wonderful opportunity for an authentic outdoor experience. Although much of the land in the watershed has been impacted by years of industrial activity, wooded hillsides and expansive floodplain open areas abound. It is the goal of this study, and the mission of its sponsors, to construct land and water trails to encourage access to these resources, and to ensure that significant natural areas are afforded protection.

### ***Why Chartiers Creek?***

The Chartiers Creek valley, although located just a few miles from downtown Pittsburgh, offers breathtaking vistas and quite solitude for those who seek its offerings. It also offers cultural and social riches in the same magnitude. Although, recovering from more than a century of impacts that have resulted from coal extraction, the water quality in Chartiers Creek has exhibited significant improvement in the last decade. Much of this improvement has been the result of



*Chartiers Creek near Pittsburgh*

grassroots efforts to inventory and remediate harmful discharges from the vast network of deep coal mines that underlay the watershed. Efforts to reduce other sources of point and non-point water pollution have been initiated by local conservation groups and governmental agencies as well. These efforts are returning large dividends as water-based activities in the Chartiers Creek watershed have gained increasing popularity. These include recreational and sport paddling, fishing, waterfowl observation and development of educational programs. An inter-connected water/land trail will enhance these experiences and provide the basis for the future development of the concepts that are recommended in this study.

### ***What Opportunities are Present?***

Fortunately, there are significant opportunities for the development of a regional trail and greenway system in the Chartiers Valley. At the confluence of the Ohio River and Chartiers Creek will be an important hub that connects the Chartiers Trail with the Three Rivers Heritage Trail. The City of Pittsburgh is currently undertaking a master plan study for a riverfront park along the Ohio River in this area. A connection between these two corridors would provide a large urban population with a direct pedestrian/bicycle route to the city of Pittsburgh, and connect with 37 additional miles of waterfront trail.



Approximately 6 miles upstream on Chartiers Creek from the Ohio River confluence is a possible connection with the Panhandle Trail. The Panhandle, when completed, will extend 29 miles from Walkers Mill in Collier Township to Weirton, West Virginia. The Panhandle Trail is actively being developed and was recently awarded funding from the TEA 21 for construction of Phase III that encompasses 17.3 miles of trail through 9 municipalities. Approximately 6.6 miles now have improved surface between Walker's Mill Road in Collier Township and Laurel Hill Road in McDonald Borough.



*Montour Trail at Coraopolis, PA*

At a point commencing approximately 7 miles downstream on the Ohio River, and progressing south, is the well-used Montour Trail. This rail-trail, when completed, will be a part of a complex of trails that stretch from Coraopolis, PA to Washington, D.C. The trail covers 54 miles of abandoned railroad right-of-way from the Montour Railroad in Allegheny and Washington Counties. Of the 54 miles of trail, approximately 23 miles are located within the lower Chartiers Creek

watershed. A connection can be made at the Montour Trail's northern terminus along the Ohio River, or at the southern terminus of the Chartiers Trail near the Greer Tunnel.

The Chartiers Trail will potentially make numerous intermodal connections with major transportation facilities such as the Port Authority's Southern Busway and the proposed Maglev project. An emerging water trail also adds to the uniqueness of the Chartiers Creek Trail and Greenway system.

The greatest potential for a contiguous trail system throughout the Chartiers Creek valley is the existing Pittsburgh and Ohio Central (P&OC) railroad corridor. This rail line is operated as a short line freight system by the Ohio Central Railroad System. Under this ownership, it is not likely that any portion of the P&OC will be used for the Chartiers Trail. An interim plan has been specified, however, that will offer connections to existing trails and provide neighborhoods with recreational opportunities.

Contained within this study are also recommendations for a greenway system that is formed around the spine of Chartiers Creek and the proposed trail. Due to the steep valley slopes, much of the land adjacent to Chartiers Creek has remained relatively undisturbed. Although not suitable for some active recreational uses, this land is valuable in its provision of ecological and passive recreational function. A portion of this land has been acquired by the Chartiers Nature Conservancy and other conservation organizations, and these parcels will form the basis of future land acquisition and conservation activities.





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## ***What Can be Done?***

The vision of the stakeholders to make this exiting trail and greenway network a reality is fueled by the presence of numerous regional linkages, many of which have drawn the local population into a routine of outdoor activity and recreation. To turn the vision of the Chartiers Creek Trail and Greenway into reality, it is imperative that the stakeholders gain the support of local and state government officials and planning entities. At a grassroots level, the population of the primary trail and greenway service area must be educated of the benefits of hosting a regional trail and greenway facility. Public education, and the willingness of local government and conservation groups to work together towards a common goal, will draw attention to the unprecedented opportunity that is present in the Chartiers valley.

This study has been prepared as a tool for the project sponsors to build consensus and to generate enthusiasm within the adjacent communities. It should also be used as a tool to solicit and secure project funding.





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## ***Lower Chartiers Creek Trail and Greenway Feasibility Study***

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# 1.0 Introduction

## 1.1 Study Purpose, Goals and Objectives

Formed in 1992, the Chartiers Nature Conservancy (CNC) is a community oriented, grass-roots, non-profit organization that is committed to the preservation, protection and promotion of the natural beauty of the Chartiers Valley. CNC also strives to foster environmental awareness through public education. The mission of CNC can be summarized as follows:



*"To preserve, protect and promote  
the natural beauty  
of the Chartiers Valley"*

- **Preserve:** CNC actively seeks to acquire land within the watershed thereby allowing future generations to enjoy the natural resources within the watershed.
- **Protect:** CNC pursues measures to improve water quality and protect other aquatic and terrestrial resources.
- **Promote:** CNC's activities are based on the concept that green space enhances property values of surrounding communities and provides passive recreational opportunity for local residents.

To further this mission, CNC has obtained and administered a Pennsylvania Department of Conservation and Natural Resources (DCNR) grant for the preparation of this Trail and Greenway Feasibility Study. The goal of this study is to define the extent and characteristics of a 23-mile trail and greenway system in the lower Chartiers Creek Valley. The project area is located in Allegheny and Washington Counties.

This study will provide CNC and area stakeholders with a planning and guidance document that may be utilized to gain community consensus and outline concepts for an inter-connected trail system and greenway. This document may also be used as the basis to secure funding for future trail and greenway development plans and implementation projects.

The specific goals of this study are to:

- examine the existing conditions within the proposed trail and greenway corridor;
- summarize the adjacent property ownership based on recent information provided by Allegheny and Washington Counties;





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- identify opportunities and constraints for trail and greenway development;
  - recommend typical conceptual design cross-sections for proposed trail segments based upon site conditions and accepted standards and guidelines;
  - propose a phased plan for trail development; and
  - provide guidelines for the trail's design, maintenance, operation, and security.

The study also addresses:

- considerations for potential project partnering,
- potential easements or parcel acquisitions;
- funding sources.

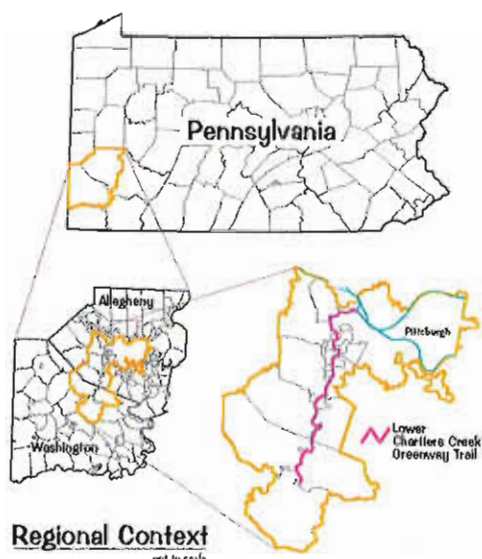
Attached to this report is a graphical representation of the custom Geographical Information Systems (GIS) database that has been prepared for this project and that contains topographic mapping and corresponding tabular data. The GIS has been used to perform powerful planning and reporting functions and will be utilized throughout the trail and greenway development process. As conditions change or opportunities arise, the data contained in this document and the GIS can be modified in an effort to acquire funding for particular segments and/or combinations of greenway or trail segments.

The next logical planning phase for implementation of the concepts contained herein will be the completion of a trail and greenway master plan. A master plan is a document, which, based on the constraints and concepts contained in this report, identifies specific items which will be required for development. The master plan will involve additional consensus building through community involvement, the preparation of detailed cost estimates, and the preparation of schematic drawings.



## 1.2 Regional Context

Allegheny and Washington Counties are host to one of the most extensive trail networks in the country. These trails include the Montour, Three Rivers Heritage, Panhandle, Eliza Furnace, and Arrowhead Trails. The regional trail development activities have been supported by state, county, and local government through project funding and in-kind matches of resources. Allegheny County's vision is to establish a "network of parks, trails, green hillsides and valleys, riverfronts and secured agricultural areas" within the county (Allegheny County Greenways, 1995).



The geographic setting of the project area establishes it as a critical component of future regional trail and greenway development. The Chartiers valley is in an ideal setting to become a connection to the north to the City of Pittsburgh, the Ohio River corridor, and the proposed extension of the Three Rivers Heritage Trail and McKees Rocks Native American burial mound interpretive area; west to the existing Panhandle Trail; and south to the well-used Montour Trail.

As discussed in the following sections of this report, the Chartiers Creek valley provides an excellent opportunity for recreational, cultural, and natural experiences. Much of this potential is being represented by CNC through its sponsorship of community awareness activities and nature outings, its acquisition of key properties in the watershed, and its sponsorship of trail and waterway access projects.

This plan will help bring the attention of the conservation minded community to the natural beauty and rich cultural heritage within the Chartiers Creek valley, and to establish the framework for a regional connection of trails and greenways. If this vision is pursued, the benefits will be realized by the residents of Allegheny and Washington Counties and beyond.





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## **2.0 Resource Inventory and Analysis**

### **2.1 Methodology**

The resource inventory has been compiled from numerous published sources, field reconnaissance, and interviews. It provides support for the trail and greenway recommendations that are found in the following sections of this report. A portion of the following narrative and supporting tables has been excerpted from the Lower Chartiers Creek Rivers Conservation Plan (Skelly and Loy, 2001). The Rivers Conservation Plan (RCP), which was prepared in part under a DCNR grant, has been used to the extent practical in the context of this feasibility study.

### **2.2 Physiographic Setting**

#### **Location**

The project area is located in southwestern Pennsylvania in Allegheny and Washington Counties. The downstream or northern terminus of the project is located in the Borough of McKees Rocks and the City of Pittsburgh at the confluence of Chartiers Creek and the Ohio River in Allegheny County. The upstream boundary of the study area is defined by Canonsburg Lake in Peters Township, Washington County.

#### **Topography**

The project area is varied in topography and is typical of the Appalachian Plateaus Province, which runs in a band from the southwestern to the northeastern part of the state. The study area lies in the Pittsburgh Low Plateau Section of the Province. The topography of the county is typical of plateaus with flat uplands and steep sided V-shaped valleys that have been created by the downcutting action of streams as they attempt to reach the elevation of the deep, broad valleys of the Ohio River (WPC, 1994). The highest stream elevation, (900 feet) in the project area is located at the confluence of Little Chartiers Creek and Chartiers Creek in Washington County. The highest land elevation in the watershed is located at a point near the Village of Champion, North Fayette Township, Allegheny County and has an elevation of 1,358 feet. The lowest elevation in the watershed is located at the point where Chartiers Creek discharges into the Ohio River at the Borough of McKees Rocks and the City of Pittsburgh, Allegheny County and has an elevation of 710 feet (USGS, 1953 – 1993).





## 2.3 Relevant Regional Routes and Linkages

Transportation and recreational facilities have been evaluated for the potential to provide routes and linkages within the Chartiers Trail and Greenway. These potential linkages are important to future trail development for the following reasons:

- Existing transportation routes can be considered for use as potential recreational linkages.
- They provide a means for bringing trail users from secondary service areas.
- Established transportation networks are more effective in attracting funding for development.
- They provide a viable alternative form of transportation.

Bicycle and pedestrian facilities and access to transit in large cities is a tremendous untapped resource for reducing motor vehicle trips. Transit facilities and the surrounding street systems must be carefully designed to accommodate bicycles and pedestrians (PennDOT, 1996). The following potential transportation linkages will provide the opportunity for recreation and transportation, thereby improving the quality of life for surrounding communities in the Chartiers Greenway project area.

### Local Historic Rail Corridors

During the late 1800's and early 1900's the steel boom in Pittsburgh was fueling the industrial revolution across the world. To accommodate the enormous demand for raw materials and to move finished steel products, an extensive network of rail systems was built and operated. Locally, these systems included the Montour and Panhandle Railroads.



*Montour Junction*

The Montour Railroad was chartered in 1877. It was one of many short line railroads built in the late 19th century to haul coal from rural western Pennsylvania to the steel industry of Pittsburgh and beyond. The first segment constructed ran from Montour Junction (near the confluence of Montour Run with the Ohio River east of Coraopolis) south to the Town of Imperial. In 1912, the Montour Railroad undertook major expansion to reach new mines. By 1917, it reached West Mifflin via McDonald,



Cecil and Bethel Park. It was an important link in the rail transportation system serving Pittsburgh, Youngstown, and Cleveland by connecting all five of the trunk railroads serving the area (MTC, 2003).

The Pennsylvania Railroad Panhandle Division mainline ran from Pittsburgh, Pennsylvania to Columbus and Cincinnati, Ohio. The multi-track mainline wound its way through southwestern Pennsylvania, the northern panhandle of West Virginia, and across a large part of southern Ohio. The Pennsylvania Railroad utilized the Panhandle Division with heavy coal traffic, perishable goods and mixed freight trains. Passenger trains ran tight schedules between Cincinnati and Columbus, Ohio and Pittsburgh, Pennsylvania for connections to other major cities. (Johnson, 2001)



*Panhandle Railroad (courtesy of the Ohio Historical Society)*

### **Rails-to-Trails**

The Chartiers Trail will connect with the following existing rail trails. As noted above, the right-of-ways that were established during the construction of industrial railroads in southwestern Pennsylvania provide an excellent opportunity for recreational trails. "Rail Trails" provide an alternative to common transportation options and provide safer recreational opportunities on gentle grades without vehicle traffic. Additionally, many abandoned rail corridors provide beautiful scenery and a relaxing atmosphere through lengthy stretches of rural and wooded areas.

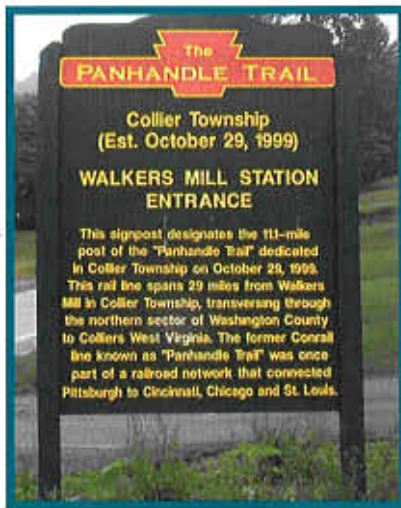


*Montour Trail at Coraopolis, PA*

The Montour Trail, when completed, will be a part of a complex of trails that stretch from Coraopolis, Pennsylvania to Washington, D.C. The trail covers 54 miles of abandoned railroad right-of-way from the Montour Railroad in Allegheny and Washington Counties. Of the 54 miles of trail, approximately 23 miles are located within the lower Chartiers Creek watershed. Several sections of the trail are completed and total approximately 24 miles, with the right-of-way secured

for its overall development. Additionally, feasibility studies have been completed and major funding sources have been secured to complete sections just outside of the watershed. The Montour Trail Council was just awarded additional funding from TEA 21 for construction of 3 additional sections of trail.





The Panhandle Trail is a trail that will extend for 29 miles beginning at Walkers Mill in Collier Township and ending at Weirton, West Virginia. The Panhandle trail was recently awarded funding from the TEA 21 for construction of Phase III that encompasses 17.3 miles of trail through 9 municipalities. Approximately 6.6 miles now have improved surface between Walker's Mill Road in Collier Township and Laurel Hill Road in McDonald Borough (see Map A).

The original rail network once connected Pittsburgh to Cincinnati, Chicago, and St. Louis. The ultimate goal is to have the abandoned corridor converted into a multi-use, non-motorized

recreational trail linking Collier Township to Oakdale, Burgettstown, Midway, and McDonald. The trail will eventually extend from McDonald to Weirton, West Virginia. The corridor acquisition process took a total of five years ending with the donation of the trail's right-of-way by Consolidated Rail Corporation (CSX) on May 5, 1999 (Panhandle Trail, 2002).

The portion of the trail in Washington County will be improved and managed by the Washington County Parks & Recreation Department. A half-mile section of the trail between the McDonald trestle and Primrose was opened in June, 1999 (MTC, 2003).

It is evident that the abundance of abandoned rail tracks is a tremendous asset for the communities in southwest Pennsylvania. "When completed, these trails will provide an exciting opportunity for unique recreation and transportation alternatives for residents of the watershed" (Allegheny Land Trust, 1999).

## **Highway Infrastructure**

### Mon/Fayette Expressway Southern Beltway Transportation Project

The Southern Beltway project is being sponsored by the Pennsylvania Turnpike Commission (PTC) and will form an arc about 30 miles long with a radius approximately 15 miles from downtown Pittsburgh. It is designed to improve access and east-west mobility between the mid-Mon Valley, at the Mon/Fayette Expressway near Finleyville, and Pittsburgh International Airport at the PA Route 60 Expressway.

This highway is among a number of new toll roads the state legislature authorized the PTC to design, construct, and operate when it passed Act 61 of 1985. The long-term benefit of assigning these new highways to the PTC is that once they are built, tax dollars would not be required to maintain them. The PTC

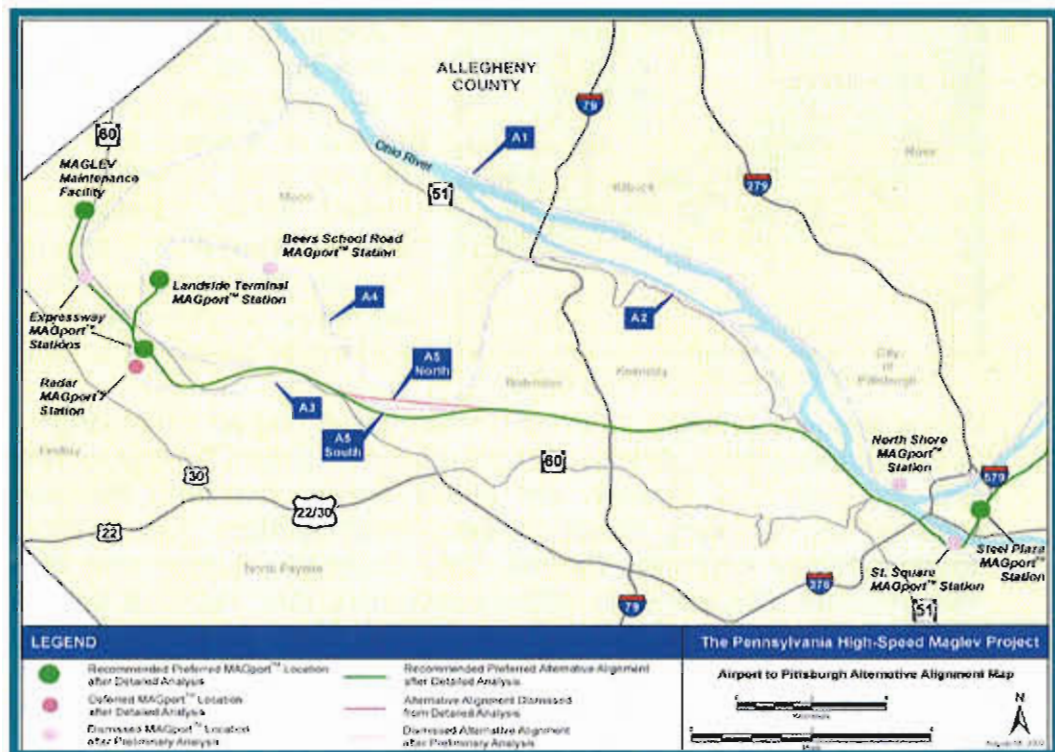




is responsible for 500-plus miles of highway and is able to maintain its system with toll revenue (FAECO, 2003).

## Mass Transit

Maglev is a proposed regional mass transit project that may greatly influence transportation patterns within the study area. The project is being administered by the Federal Railroad Administration (FRA) under TEA-21. Among other national transportation initiatives, the legislation calls for the deployment of a high-speed maglev system as a future means of transportation technology. The FRA requires that the system operate safely at speeds in excess of 240 mph. Additionally, the project must contain 70% American content. An illustration is provided of the "A" Route which connects the Pittsburgh International Airport with downtown Pittsburgh.



*Proposed Maglev Route Alternatives – Allegheny County*

A Maglev project Environmental Impact Statement (EIS) is being conducted by the Port Authority of Allegheny County and the Pennsylvania Department of





Transportation (PennDOT) utilizing the services of a consulting group. Additional information on this project can be obtained at [www.maglevpa.com](http://www.maglevpa.com) or by contacting:

Port Authority of Allegheny County  
Maglev EIS  
345 Sixth Ave.  
Third Floor  
Pittsburgh, PA 15222  
[cbray@portauthority.org](mailto:cbray@portauthority.org)

In addition to 24-hour access to Maglev information via the web site, a toll-free information telephone hotline is also available to the public day or night. The hotline number is 1-866-MAGLEV-411 (1-866-624-5384).

This route will bisect the study area and may provide an excellent opportunity for an alternative transportation corridor that includes bike and pedestrian improvements. Further discussion of the opportunities and threats associated with this project are discussed in Section 2.10 of this report.



*Port Authority's West Busway in Study Area*

The West Busway is the Port Authority of Allegheny County's newest busway facility, opened on September 8, 2000 as an immediate success. The West Busway is a five - mile exclusive bus roadway that provides reliable, fast, efficient public transportation service between downtown Pittsburgh and the Borough of Carnegie. Nearly 40,000 customers experience the busway's rapid transit benefits each week.

With a bridge connection to West Carson Street and an interchange connection to the Parkway West, the busway provides an efficient bypass to Greentree Hill and the Fort Pitt Tunnels for buses serving western Allegheny County communities, including those in the Airport corridor. Direct access to the communities of Carnegie, Crafton, East Carnegie, Ingram, and Sheraden are also provided. Currently, six stations and more than 700 new park and ride lot spaces are making public transportation a more attractive option for many customers. The remaining park and ride lots, which will bring the total to approximately 2,800 spaces, are scheduled to be constructed by the end of 2002 (Allegheny County Port Authority, 2002).

The busway facility will form an inter-modal component of the Chartiers Trail. Field observations reveal that bicycle storage facilities exist at many of the parking areas along the bus corridor, and the proposed alignment passes one

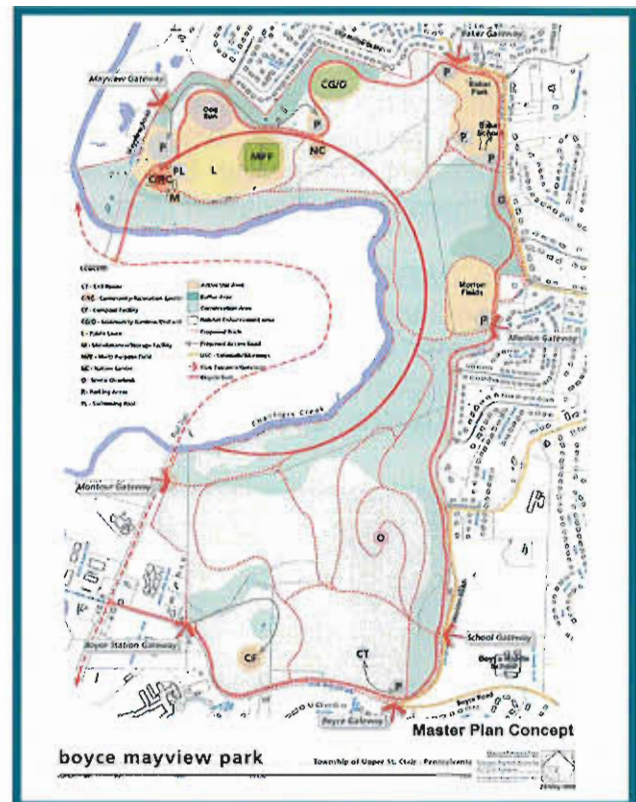


such station at Carnegie (see Map 4). Signage and other public awareness strategies should be employed to increase the usage of the bike/bus dual modal system. Additional information on the use of dual modality is contained in Section 2.10 of this report.

## Local Trail Systems

### Boyce-Mayview Trails

In 1996, Upper St. Clair acquired 238 acres of the Mayview Farm property from the Commonwealth of Pennsylvania. The location of this property directly adjacent to the Township's 236-acre Boyce Park property called for a master plan process to examine potential development and conservation opportunities for the now combined Boyce/Mayview Park. An update of the master plan explains the planning process. These two largely undeveloped properties contain extensive woodlands, level fields, and open spaces as well as stream valleys, hiking trails, and active vegetable gardens. The Boyce-Mayview Park can be seen on the attached Trail Alignment Map 8.



### McLaughlin Run Trail

This trail is a newly proposed trail that was recently awarded TEA 21 funding. It will connect existing trails at Wiltshire Park with trails extending from the Upper St. Clair Township building (USC, 2003).

### Share the Road Bikeways

Share the Road Bikeways can be found on nearly 60 miles of existing roads in the watershed. Most notably, the Carnegie Bikeway encompasses 2.7 miles of that total. Other roadways with considerable bikeway distances are the Airport Toll Corridor (3.3 miles), PA Route 50 (3.4 miles), PA Route 3048 (3.5 miles), PA Route 3041 (3.2 miles), Washington Road (5.7 miles), and McLaughlin Run Road (3.8 miles) (Skelly and Loy, 2001).



The Scott Conservancy Hiking Trails are a series of fairly rugged hiking trails located in the middle section of the study area. The trails are used primarily for wildlife viewing and are generally unimproved. Some of the trail access points are marked with signage.

### **Active Rail Corridors**

Pittsburgh and Ohio Central Railroad (P&OC) The P&OC is owned and operated by the Ohio Central Railroad System (OCRS). The OCRS is a network of nine railroads that operate over 425 miles of rail throughout the eastcentral and northeastern Ohio and Pittsburgh, Pennsylvania area. The OCRS is a full-time freight railroad that also offers passenger excursions powered by vintage equipment throughout the year. The OCRS is divided into three divisions; Northern Lines, Southern Lines and Pittsburgh Lines.

The P&OC commenced operation in year 2000 on 43 miles of line formerly owned and operated by the Pittsburgh Industrial Railroad. The P&OC Main Line extends from Neville Island on the Ohio River near Pittsburgh to Arden, Washington County, Pennsylvania with branches to Esplen and Armstrong Mills (Ohio Central, 2002).



*Typical Section of P&OC Railroad*

The P&OC parallels Chartiers Creek throughout the project area. Generally, one track exists in the right-of-way although evidence of a second track and numerous sidings were observed. The location of this line is illustrated on each of the maps within the study area. Although the P&OC offers a unique opportunity for a trail corridor, the OCRS is currently a viable operating corporation which brings benefit to the local community through the provision of freight and excursion rail service. It should be

assumed therefore, that potential trail alignments could only be viable if they share portions of the P&OC rail with the operating freight and excursion service.

The potential for sharing the P&OC line in a rail with trail scenario is based on the following factors:

- The OCRS's willingness to discuss rail with trail configurations.
- The speed and frequency of the freight and excursion traffic on the P&OC.



- 
- 
- Right-of-way width and the location of rail facilities such as switches, bridges, tunnels, and crossings.
  - Pennsylvania Public Utility Commission (PUC) regulation and criteria.
  - Ability of an entity to enter into a legal agreement with the rail company for easement or purchase of trail right-of-way and to assume liability.

Contact has been made with the P&OC regarding use of their right-of-way for recreational trail use. Details of this conversation are contained within Sections 2.10 and 3.2 of this report.







## **2.4 Cultural and Historic Resources**

The Cultural and Historic Resources present an overview of cultural resources within the Lower Chartiers Creek watershed. Cultural resources are protected under Section 106 of the National Historic Preservation Act of 1966 as amended and implementing Regulation CFR 800 as revised on May 18, 1999; the National Environmental Policy Act of 1969; the Archaeological and Historic Preservation Act of 1974; the regulations of the Advisory Council on Historic Preservation 36 CFR 63 and 36 CFR 800; the Pennsylvania History Code (37 Pa. Cons. Stat. 507 et seq.); and the Commonwealth of Pennsylvania State Acts No. 1970-120 and 1978-273.

The Chartiers Creek watershed encompasses an area of 280 square miles of land within Allegheny and Washington Counties, Pennsylvania. Christine Davis Consultants, Inc. (CDC) have completed the cultural resources section of this report.

### **Methodology**

The collection of data significant to the cultural history and prehistory of the Chartiers Creek watershed involved an examination of historic and archaeological site survey inventories including the Pennsylvania Archaeological Site Survey (PASS), the National Historic Landmarks (NHL), the National Register of Historic Places (NRHP), and the Historic American Buildings Survey/Historic American Engineering Records (HABS/HAER). Local histories, county atlases, cultural resource surveys, planning projects, and other works were examined.

A list of resources was compiled from these inventories and other sources. A team of two persons then conducted a reconnaissance to determine the condition and verify the locations of these resources. A sample survey was undertaken to observe the general visual, cultural, economic, and social characteristics of the study area.

The survey considered previously recorded cultural resources that retain historic, architectural, archaeological, and/or cultural integrity. At this time, no consideration was given to the economic value of the properties or how they would be treated in the planning process. A database was created to provide an efficient way to organize, retrieve, and sort information. The database was developed to keep the inventory open for the addition of new properties and new data fields in response to any possible expansion of the proposed project area.

The survey and data evaluation are an overview and are to be used as part of local decision-making processes regarding public and private preservation activities such as future surveys, revitalization projects, planning, zoning, tax



credits, and other issues. Only properties that were previously listed or determined eligible for the NRHP were identified; the following four criteria, as described in Title 36, Part 60 of the Code of Federal Regulations, were used:

- **Criterion A** - property is associated with historically significant events, trends, activities, or patterns
- **Criterion B** - property is associated with an important individual
- **Criterion C** - property possesses distinctive characteristics in physical design or construction
- **Criterion D** - property has the potential of yielding important prehistoric or historic information

Determining National Register eligibility involves an assessment of a property's historic integrity expressed as the physical characteristics that existed during the period of time that the resource was significant. An overall sense of past time and place is determined by seven factors - location, design, setting, materials, workmanship, feeling, and association.

For the purpose of this study, properties that are not ordinarily recognized by the National Register were not included. The Register typically excludes cemeteries, birthplaces or graves of historical figures as well as properties owned by religious institutions. Other properties not generally listed on the Register are structures that have been moved from their original location, reconstructed historic buildings, commemorative properties, and properties that have achieved significance within the past 50 years. Other exceptions are religious properties deriving their primary significance from architectural or artistic distinction or historical importance.



## Analysis and Formation of Themes

Historic properties and archaeological resources provide a sense of place, a shared past, and provide meaning for the people living and working within the Chartiers Creek Valley. Themes link a property to a place and time within a local, state, or national framework for determining its significance. A total of 17 historic resources and 111 prehistoric archaeological resources were identified within the Chartiers Creek Watershed (Table 2.4.1). To evaluate these resources, three themes were developed. They are:

*Theme I: Railroad Development (5 sites)*

*Theme II: Architecture and Suburban Development  
(4 sites)*

*Theme III: Prehistoric Native Americans (111 sites)*

It should be noted that no known systematic study of historic properties has been completed in the watershed.



*Crafton Historic District Marker*



**Table 2.4.1  
Documented Historic Sites in the  
Chartiers Creek Watershed**

| MUNICIPALITY             | RESOURCE                                                                              | ADDRESS                                             | DESCRIPTION                                                         | THEME                                 | STATUS                     |
|--------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------|---------------------------------------|----------------------------|
| Multiple                 | Panhandle Railroad Historic District                                                  |                                                     | Currently Airport Busway                                            | Railroad Development                  | Listed                     |
| Bridgeville Borough      | Bethany Presbyterian Church                                                           | 720 Washington Avenue                               |                                                                     |                                       | Eligible                   |
|                          | Bridgeville, Pennsylvania Railroad Station                                            | 441 Station Street                                  | Railroad station                                                    | Railroad Development                  | Eligible                   |
|                          | Cloussy Brothers Property                                                             | Washington Avenue at Church Street (Northeast)      |                                                                     |                                       | Eligible                   |
| Carnegie Borough         | Duquesne Light Company Substation Complex                                             | Adjacent to the Panhandle Railroad                  | Neoclassical electric power station                                 | Architecture and Suburban Development | Eligible                   |
|                          | Andrew Carnegie Free Library                                                          | 300 Beechwood Boulevard                             | 1899 Renaissance library                                            |                                       | Listed                     |
|                          | Carnegie Downtown Historic District                                                   | Main Street                                         | Architecture in 19th century suburban neighborhood                  | Architecture and Suburban Development | Eligible                   |
| Collier Township         | Neville House                                                                         | Washington Pike                                     | Owned by significant Pittsburgh family                              |                                       | National Historic Landmark |
|                          | Woodville State Hospital and Farm                                                     | Thom's Run Road                                     | State hospital and farm                                             |                                       | Eligible                   |
| Crafton Borough          | Campbell Building                                                                     | 60 E. Crafton Avenue                                |                                                                     |                                       | Listed                     |
|                          | Crafton Historic District                                                             | Steuben Street to Woodlawn Avenue, Creighton Street | Architecture dating from 1873 through 1940 in suburban neighborhood | Architecture and Suburban Development | Eligible                   |
| Ingram Borough           | <i>No previously recorded historic resources within the Chartiers Creek Watershed</i> |                                                     |                                                                     |                                       |                            |
| McKees Rocks Borough     | Saint Mary's Roman Catholic Church Complex                                            | St. John Street                                     | 1901 Gothic Revival Church and 1930 stone convent                   |                                       | Eligible                   |
| Rosslyn Farms Borough    | Chartiers Creek Bridge No. 15                                                         | Turner Road                                         | 3 components dating from 1854-1865, ca. 1876 and 1920-1945          | Railroad Development                  | Eligible                   |
| Thornburg Borough        | Thornburg Historic District                                                           | Off of S.R. 0060                                    | Historic District laid out in 1899 based on Tuxedo Park in New York | Architecture and Suburban Development | Listed                     |
| Upper St. Clair Township | Boyce Station                                                                         | 1050 Boyce Road                                     | Railroad station                                                    | Railroad Development                  | Listed                     |
| Washington County        |                                                                                       |                                                     |                                                                     |                                       |                            |
| Multiple                 | Montour Railroad Historic District                                                    |                                                     | Includes Greer and Montour Tunnels, now Rails-to-Trails project     | Railroad Development                  | Eligible                   |
| North Strabane Township  | James Thome Farmstead                                                                 | 213 Linwood Road, Eighty Four                       | Architecture                                                        |                                       | Listed                     |







## Railroad Development Theme

The Chartiers Valley provided a significant corridor for transportation necessary to access the valley's rich coal and natural resources, later opening the valley for growth and suburban development. McKees Rocks developed as a prominent railroad transit center for the coal industry partly because of its geographical location at the confluence of Chartiers Creek and the Ohio River.

There are a total of five historic resources within the project area that are associated with the Railroad Development Theme and that have been previously listed or determined eligible for the NRHP: the Panhandle Railroad (including the Chartiers Creek Bridge No. 15), the Montour Railroad, the Bridgeville Pennsylvania Railroad Station, and Boyce Station. The Pittsburgh, Chartiers, and Youghiogheny Railroad also passes through the Chartiers Creek watershed; however, no official National Register eligibility has been determined for this property.

The historic Panhandle Railroad is located across several municipalities within Allegheny County and has been listed on the NRHP. The railroad was as an essential link to the western agricultural and commercial districts for the Pennsylvania Railroad. Over the years, the Panhandle Railroad, originally proposed as a modest route of less than one hundred miles, had been developed into a major division of the Pennsylvania Railroad, and covered over 1,000 miles as it stretched into the American mid-west. Its growth parallels the development of the Pittsburgh region as an industrial and transportation center of international proportion. The Port Authority of Allegheny County transformed the railroad's right-of-way into the Airport Busway.

The Chartiers Creek Bridge No. 15 is an abandoned Pratt truss bridge carrying the Panhandle Railroad over the Pittsburgh, Chartiers, and Youghiogheny Railroad. Constructed in 1854 in Rosslyn Farms Borough, Allegheny County, it consists of three components dating from 1854-1865, ca. 1876, and 1920-1945. The structure has been determined eligible for the NRHP.

The Montour Railroad was the main railway for the coal industry of this region, and it has been determined eligible for the NHRP. The Montour Railroad extends across several municipalities within Washington County. The railroad right-of-way is now a popular rails-to-trails project and includes the Greer and Montour Tunnels.

The Boyce Station is a refurbished farmhouse located at 1050 Boyce Road in Upper St. Clair Township, Allegheny County. Built in 1885, the building housed a grocery store and post office until the Baltimore and Ohio Railroad took over the building as a train station. This property has been listed on the NRHP.





*Bridgeville Railroad Station*

The Bridgeville Pennsylvania Railroad Station is located at 441 Station Street, Bridgeville Borough, Allegheny County, and has been determined eligible for the NRHP.

### **Suburban Development Theme**

Transportation in the form of commuter railroad service and trolley lines opened the Chartiers Valley for development after the Civil War. These new transportation

systems are intrinsically linked to the development of suburbs around the city of Pittsburgh. There are a total of four historic resources within the project area that are associated with the Architecture and Suburban Development Theme and that have been previously listed or determined eligible for the NRHP: the Thornburg Historic District, the Crafton Historic District, the Carnegie Downtown Historic District, and the Duquesne Light Company Substation Complex. A fifth potential district in Rosslyn Farms began in 1903 but was found to be impacted by urban development and unsympathetic intrusions.

The Thornburg Historic District, located in Thornburg Borough, Allegheny County, has been listed on the NRHP under Criterion C for its significant architecture. In 1899, the Thornburg family subdivided their land holdings along the Chartiers Creek to create a suburban community resembling Tuxedo Park in New York. The district has been described as an outstanding example of late nineteenth and early twentieth century residential suburbia. The unusual examples of avant garde architecture reflect designs associated with Frank Lloyd Wright and Gustav Stickley.

The Crafton Historic District, located in Crafton Borough, Allegheny County, has been determined eligible for the NRHP. The District is significant as representing the broad pattern of suburban development and for its late nineteenth and twentieth century architecture. In 1860, James S. Craft, the President of the Pittsburgh & Steubenville Railroad (Panhandle Railroad), purchased large tracts of property in the area and his son laid out Crafton in 1873. A second plan



*Crafton Borough Historic District*

of lots was developed in the period between 1910 and 1930 reflecting changes in the neighborhood after the introduction of the trolley system. The Crafton Historic District also contains samples of earlier Georgian and Federal Styles that precede the suburban nature of the district. It is a good example of multi-component and multi-stage development of a suburban area.





The Carnegie Downtown Historic District, located in Carnegie Borough, Allegheny County, has been determined eligible for the NRHP. It consists of the residential, commercial, public, and social structures dating to the mid-eighteenth to mid-nineteenth centuries. Carnegie was originally known as Mansfield when it began its history as a small industrial town along the banks of Chartiers Creek. Its growth is attributed to its location along the Pittsburgh and Steubenville Railroad (Panhandle Railroad) corridor. Carnegie's history includes its successful transition into a suburb of a major metropolitan area, Pittsburgh, once the industrial age had ended.

The Duquesne Light Company Substation Complex, located in Carnegie Borough, Allegheny County, is an example of a Neoclassical Carnegie Electric Light Building and an excellent portrayal of an early (1905-1927) electric power substation. The building represents the need for public utilities to provide electrification in suburban neighborhoods.

### **Prehistoric Native Americans Theme**

A total of 132 archaeological sites in the Chartiers Creek watershed, including 111 prehistoric sites and 21 historic sites, have been recorded on the Pennsylvania Archeological Site Survey (PASS) forms. These archaeological sites are located along the alluvial terraces and floodplains as well as on upland benches, saddles, and hilltops. The archaeological resources in the Chartiers Creek valley were considered so important that the National Park Service conducted an archaeological survey here in the 1960s in conjunction with the Carnegie Museum of Natural History. It was during this survey that the majority of significant sites in the watershed were documented. Wide, level floodplains and terraces along the Creek that have not been previously disturbed have a very high probability of archaeological sites including villages with human burials.

Among the previously recorded archaeological sites are the famous McKees Rocks Mound and a series of Monongahela village sites. The McKees Rocks Mound remains partially extant above the confluence of Chartiers Creek with the Ohio River. The mound functioned as a burial site for approximately 1,000 years by both the Adena and, later, the Hopewell cultures. Carnegie Museum considered the mound, the largest Native American mound in Western Pennsylvania, so important that excavating the site was the first major archaeological project conducted by the museum in the Pittsburgh area. The project began over a century ago in 1896, and in the fall of 2002 a Pennsylvania Historic Marker was placed near the site to commemorate its significance.

The largest recorded sites along the stream are prehistoric Monongahela village sites. The Monongahela people resided in nucleated villages often surrounded by stockades and fields of corn, beans and squash. In the early 1600s, the Monongahela people left this region during a time when the Iroquois were





searching for new hunting territories. There is a very high percentage of woodland sites in the Chartiers Valley when compared to sites dating from the Archaic period. Middle woodland and late prehistoric components were most often found together suggesting a shared economic and land-use orientation when compared to earlier sites. There are dozens of small campsites and farmsteads throughout the valley's level terraces and upland environments. These fragile cultural resources possess important information about the prehistory and past lifeways of people who lived in our region for thousands of years.

Chartiers Creek was named for Peter Chartiers, the only son of Frenchman Martin Chartiers and his Shawnee wife. Peter Chartiers, who also married a Shawnee woman, played an important role in the emerging French and Indian War when he persuaded the Shawnee to side with the French in 1744. Chartiers built a fortified hamlet near prehistoric McKees Rocks Mound. The commanding location was later viewed by George Washington when he considered placing the first English fort on this site, but he selected Pittsburgh's Point for Fort Prince George. The Native Americans continued to occupy McKees Rocks, and from this site they planned many of the most important conflicts during the French and Indian War.

### **Other Resources Not Associated with the Themes**

There are eight historic resources located within the Chartiers Creek watershed that are not associated with any of the three themes developed for this project. They include: one residence; one farmstead; two churches; one library; two commercial buildings; and one institution.



*The Neville House - Collier*

The Neville House is located in Collier Township, Allegheny County and was built beginning in 1785 by Brigadier General John Neville, an officer in the French and Indian War and the Revolution. Neville's role in the Whiskey Rebellion as an inspector of taxes for the collection of levy on whiskey was significant in the region's history.

The Neville House is a National Historic Landmark now owned by the Pittsburgh History and Landmarks Foundation.

The James Thome Farmstead, located at 213 Linnwood Road in North Strabane Township, Washington County. This property has been listed on the NRHP.







Saint Mary's Roman Catholic Church Complex, a 1901 Gothic Revival Church and 1930 stone convent, located on St. John's Street in McKees Rocks. The complex, located in McKees Rocks Borough, Allegheny County, has been determined eligible for the NRHP. The Bethany Presbyterian Church, located at 740 Washington Avenue, Bridgeville Borough, Allegheny County consists of four extant buildings. The oldest extant structure was built in 1917 with subsequent construction in 1953, 1969, and 1980. The original church was built in 1814 and was located in the area currently occupied by the cemetery. In 1838, the original church was demolished and another structure was built. The 1838 church was subsequently demolished, and the extant church buildings were built atop the site. This property has been determined eligible for the NRHP.

One of Andrew Carnegie's earliest libraries was located at 300 Beachwood Boulevard in Carnegie Borough, Allegheny County. The library exhibits the late nineteenth century Renaissance architecture prominent in other similar buildings of the time.

The Campbell Building is a commercial building located on Crafton Avenue in Crafton Borough, Allegheny County. This property has been listed on the NRHP in the category of architecture.

The Cloussy Brothers Property is a commercial building located at the corner of Washington Avenue and Church Street in Bridgeville, Allegheny County. This property has been determined eligible for the NRHP.

Woodville State Hospital and Farm is located in Collier Township, Allegheny County. It was established in 1852 as a county home for the mentally ill and later for tuberculosis victims. This property was determined eligible for the NRHP. In 1941 the Commonwealth of Pennsylvania took over the hospital and four years later, it merged with Mayview State Hospital, which is located outside the project area.





## **2.5 Recreational Resources**

Based on the available resources that exist within the study area, the most auspicious and economically feasible recreational opportunities of the watershed are outdoor-based activities. Common characteristics of outdoor recreational activities include: (1) performed outside the confines of buildings; (2) do not involve organized competition; (3) can be undertaken without the existence of any built facility or infrastructure; (4) usually require large areas of land, water and/or air (i.e., a watershed-based recreational activity); and (5) may require outdoor areas of predominantly unmodified natural landscape. Some outdoor recreational activities that can easily be performed within the study area include the following:

- Nature Observation and Photography
- Walking, Hiking, and Trail Running
- Camping
- Bicycle Riding, Mountain Biking
- Skiing, Snowboarding, Sledding, and Ice Skating
- Canoeing and Kayaking
- Horseback Riding
- Fishing
- Hunting/Shooting with Firearms and Bow and Arrow
- Trapping

The Izaak Walton League (IWL), recognized for its leadership in building a national outdoor recreation policy, has developed eight principles for outdoor recreation planning. These principles are the following:

- There should be preserved, developed, and made accessible to all American people such quantity and quality of outdoor recreation as is necessary and desirable for individual enjoyment and to assure physical, cultural and spiritual benefits to the nation.
- Natural areas and areas dedicated wholly or largely to recreational use must be adequate in distribution and number to serve the entire public.
- The public interest requires that recreation, fish, and wildlife values be considered fully in the planning and management of all public lands and waters.
- The League recognizes that the overall recreational objective can be achieved only through public and private cooperation in recreation planning and development. The League encourages private landowners to make recreation resources available to the public. Private owners who make their lands available for public recreation should be protected fully from legal liability. State and local agencies should make special enforcement or other arrangements for patrolling and protecting private properties that are open for public recreation.



- Public programs should be designed and conducted to protect and, where possible, enhance recreation and fish and wildlife values.
- Abandoned railroad corridors should be preserved for public recreational use through a national rails-to-trails program.

By adopting some of the IWL principles, municipalities can make best use of their existing outdoor recreational opportunities, ultimately improving the quality of life for residents of the Chartiers Creek watershed.

The following is a list of completed or on-going recreation projects that employ these principles (Skelly and Loy, 2001):

- Lower Chartiers Creek Rivers Conservation Plan
- Boyce/Mayview Park Master Plan
- Montour Trail
- Panhandle Trail
- Chartiers Creek Trail Feasibility Study
- McLaughlin Run Trail
- Crafton Borough Park and Recreation Master Plan
- City of Pittsburgh (Un-named park/mouth to Chartiers Creek)
- Settlers Cabin Park Southwestern Pennsylvania Botanical Gardens Horticultural Society of Western Pennsylvania

## **Parks**

As noted in the Rivers Conservation Plan (RCP) for the Lower Chartiers Creek watershed (Skelly and Loy, 2001), sixty-four local and county parklands exist within the watershed. Most of these parks are found in the municipalities located on the eastern rim of the watershed. These parks can be characterized as community parks associated with schools that have jungle gyms, basketball courts, soccer, softball, baseball, and football fields. These parks are geared towards school or municipal active recreation programs.

There are numerous community and regional parks which are discussed throughout this report. One of the objectives of this report was to identify potential connections for the trail to these recreational centers. The following facilities were considered during this study:

- Chartiers Creek / Ohio River Native American Interpretation Area – City of Pittsburgh (proposed – Map 1)
- Sheredan Park – City of Pittsburgh (Map 1)
- Crafton/Ingram/Thornburg (CIT) Ballfields – Crafton (Map 3)
- Scott Township Municipal Park – Scott Township (Map 5)
- Chartiers Park - Bridgeville (Map 7)
- Boyce Mayview Park – Upper St. Clair (Map 8)
- Canonsburg Lake – Pennsylvania Fish and Boat Commission (Map 11)





The accompanying cost tables and maps illustrate potential linkages for these facilities with the proposed trail and greenway system. They can, for the most part, be used for parking and gathering points for outdoor trail-related activities. Points such as Chartiers Park can also be used as canoe access points for the water trail discussed in Section 4.4.





## 2.6 Wildlife and Natural Resources

The lower Chartiers Creek corridor provides habitat to a variety of terrestrial wildlife species including birds, mammals, reptiles, and amphibians. Most of the preferable habitats for terrestrial wildlife exist within undeveloped forested slopes along the creek or on large park or institutional properties within the watershed. Other areas within the stream corridor, such as parks and backyards, offer suitable habitats for wildlife as well. As the Chartiers Creek watershed becomes increasingly urbanized, wildlife species and their habitats become progressively more displaced by humans. Some wildlife species take advantage of man-made structures, while others adapt to the existing ecosystems found within urban areas.

Wildlife habitats within the study area are continuously changing because of land developments, flood events, and other disturbances. Terrestrial wildlife species that adjust easily to these changes in habitats are called "edge species." Edge species inhabiting urbanized areas can adapt to living near humans, and therefore, predominate over other species of wildlife. For example, an abundant edge species inhabiting the Chartiers Creek valley is the white-tailed deer (*Odocoileus virginianus borealis*). Because they find forage in cleared areas and escape cover in forested areas, white-tailed deer benefit by utilizing habitat edges (i.e., the interface between an open space and forest). Other edge species which can be found in the vicinity of Chartiers Creek include the Virginia opossum (*Didelphis marsupialis*), raccoon (*Procyon lotor*), striped skunk (*Mephitis mephitis*), red-tailed hawk (*Buteo jamaicensis*), and numerous species of songbirds. The occurrence of the eastern coyote (*Canis latrans*) has also been reported in the Chartiers Creek valley. A qualitative analysis of the wildlife impacts that may occur during trail development is contained in Section 2.10 of this report.



White-tailed Deer (*Odocoileus virginianus borealis*)

Although Chartiers Creek is not currently stocked with trout, it does maintain a warm water fishery, supporting many game species that are also indigenous to the Ohio River. Fish community surveys performed by CEC biologists in 1999 and 2000 revealed spotted bass (*Micropterus punctulatus*), smallmouth bass (*Micropterus dolomieu*), rock bass (*Ambloplites rupestris*), sauger (*Stizostedion canadense*), and freshwater drum (*Aplodinotus grunniens*) inhabiting a section of Chartiers Creek flowing through downtown Carnegie. Recent studies indicate the the aquatic community in Chartiers Creek is improving and will contribute to the beneficial uses of the greenway (see Section 4.4 for additional discussion).



## **2.7 Trail Use / Demographic Information**

The following data illustrate potential use patterns for the trail and greenway system. It can be assumed that the data is accurate for use patterns that may be expected for a contiguous trail comprised of varying types of trail surface, grades, and configurations.

### **General Demographic of Potential Users**

The primary service area of a multi-use trail consists of the following municipalities within Allegheny and Washington Counties. Secondary users may be comprised of populations outside of these two counties including Butler, Beaver, and Westmoreland Counties.

#### Allegheny County

|                        |                           |
|------------------------|---------------------------|
| Bethel Park Borough    | North Fayette Township    |
| Bridgeville Borough    | Oakdale Borough           |
| Carnegie Borough       | Pennsbury Village Borough |
| Collier Township       | City of Pittsburgh        |
| Crafton Borough        | Robinson Township         |
| Green Tree Borough     | Rosslyn Farms Borough     |
| Heidelberg Borough     | Scott Township            |
| Ingram Borough         | South Fayette Township    |
| Kennedy Township       | Stowe Township            |
| McDonald Borough       | Thornburg Borough         |
| McKees Rocks Borough   | Upper St. Clair Township  |
| Mount Lebanon Township |                           |

#### Washington County

|                       |                   |
|-----------------------|-------------------|
| Cecil Township        | Peters Township   |
| Midway Borough        | Robinson Township |
| Mt. Pleasant Township | North Strabane    |

Many potential trail users in the communities noted above currently drive to trailheads on the Montour or Youghiogheny trails. Development of the Chartiers Trail would allow direct bicycle access for the majority of these communities.



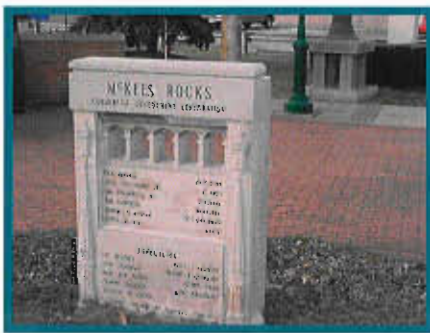
## Community Character

The Chartiers Creek floodplain area has been utilized over the years for major commercial and industrial development. Due to the steeper slopes of the eastern rim sub-basins and the associated developmental limitations, the areas adjacent to Chartiers Creek have been more intensely developed. In an effort to protect real estate and properties along Chartiers Creek, the U.S. Army Corps of Engineers designed, and later constructed, the James G. Fulton Local Flood Protection project from Bridgeville downstream to Crafton (USACOE, 1957). Additional industrial/commercial land use and brownfield (real estate no longer being used or abandoned industrial sites) properties are located primarily along and/or near major transportation corridors such as U.S. Interstates 79 and 279 and State Routes 19, 22, 50, 60, and 980 (Skelly and Loy, 2001).



*Typical Floodplain Industrial Development – Chartiers Creek*

Residential land uses vary throughout the project area. Residential land use is a mixture of duplexes, condominiums, apartment buildings, high-rise apartment buildings, and single-family housing units. Generally, the areas which are distant from the metropolitan centers exhibit rural characteristics. These small communities are interspersed with larger single-family home lot sizes.



*Parklet along Trail Route  
McKees Rocks*

The primary service area for a proposed trail is centered around small communities that were developed during the steel and coal era in southwestern Pennsylvania. Many of these neighborhoods are located along the floodplain of Chartiers Creek and are now included in the suburbs of Pittsburgh.

As the master planning phase of the Chartiers Trail is undertaken, the characteristics and history of each neighborhood should be reflected in the design of trail infrastructure.

This may include providing the location and operating hours of historic places on trail maps, interpretive signs and markers and the incorporation of local architectural themes into trail amenities and structures.



## Current and Projected Population

The following census values were available for portions of the project area in Allegheny County:

**Table 2.7.1**  
**Census Data for Primary Trail Service Area**

| Municipality              | 2000 Population | 1990 Population | % Change    |
|---------------------------|-----------------|-----------------|-------------|
| Bethel Park Borough       | 33,556          | 33,823          | -0.8        |
| Bridgeville Borough       | 5,341           | 5,445           | -1.9        |
| Carnegie Borough          | 8,389           | 9,278           | -9.6        |
| Collier Township          | 5,265           | 4,841           | +8.8        |
| Crafton Borough           | 6,706           | 7,188           | -6.7        |
| Green Tree Borough        | 4,719           | 4,905           | -3.8        |
| Heidelberg Borough        | 1,225           | 1,238           | -1.1        |
| Ingram Borough            | 3,712           | 3,901           | -4.8        |
| Kennedy Township          | 7,504           | 7,265           | +3.3        |
| McDonald Borough          | 415             | 443             | -6.3        |
| McKees Rocks Borough      | 6,622           | 7,691           | -13.9       |
| Mt Lebanon Township       | 33,017          | 33,362          | -1.0        |
| North Fayette Township    | 12,254          | 9,537           | +28.5       |
| Oakdale Borough           | 1,551           | 1,752           | -11.5       |
| Pennsbury Village Borough | 738             | 774             | -4.7        |
| Pittsburgh City           | 334,563         | 369,879         | -9.5        |
| Robinson Township         | 12,289          | 10,830          | +13.5       |
| Rosslyn Farms Borough     | 464             | 483             | -3.9        |
| Scott Township            | 17,288          | 17,118          | +1.0        |
| South Fayette Township    | 12,271          | 10,329          | +18.8       |
| Stowe Township            | 6,706           | 7,681           | -12.7       |
| Thornburg Borough         | 468             | 461             | +1.5        |
| Upper St. Clair Township  | 20,053          | 19,692          | +1.8        |
| <b>Total</b>              | <b>535,116</b>  | <b>567,916</b>  | <b>-5.8</b> |

The analysis of census data population values in the Allegheny County portion of the watershed presents a clear picture of the spatial distribution of population changes. Population growth has been concentrated in the western and southern regions of the watershed; while population decline has been concentrated on the eastern and northern regions. There are many reasons to expect this trend to continue with increasing desires for more rural, low density residential areas. With the standard of living today, two-income households are the norm along with long work days and hectic schedules. Families are looking for a sense of serenity at home, absent of noise and congestion. This is representative of







classic urban sprawl as seen in many urban areas in the U.S., including Pittsburgh.

### **Profile of Potential Users**

Based on the types of potential users and the physical constraints within the study area, the following three trail categories are proposed:

- Hiking
- Multi-use off road
- Share the road bike routes

The proposed surface types may vary from dirt, woodchip, crushed limestone, or asphalt (or macadam) paved, depending on the anticipated uses. It is expected that the following classes of users will utilize the various types of trails that are specified herein.

#### Pedestrians

Walkers, hikers, runners, nature enthusiasts, photographers and educators are types of trail users that travel at relatively low speeds, averaging 3 to 7 miles per hour. Many prefer a soft surface for a more comfortable walk/run/hike and to prevent injury. Strollers and wheelchairs can also utilize rough surfaces when equipped with sturdy tires.

On developed trails, nature enthusiasts and photographers of all ages may enjoy bird watching and observing various plant and animal species without encountering rough terrain. Numerous bird species and mammals occupy the vegetated edges and adjacent lands of corridors in rural and urban areas. The Upper St. Clair – Citizen's for Land Stewardship, and other dedicated birders are involved in the establishment of an Audobon Society Bird Circle in the South Hills. A local trail would prove to be a great resource to such a group (Skelley and Loy, 2001). Educators, ranging from scientist to historians, can utilize the trail as an outdoor classroom to motivate students and further education.

The following schools are located in the project area:

- Bethel Park School District
- Canevin Catholic High School
- Canon-McMillan School District
- Carlynton School District
- Chartiers Valley School District
- City of Pittsburgh School District
- Fort Cherry School District
- Keystone Oaks School District



- Montour School District
- Mount Lebanon School District
- Peters School District
- South Fayette School District
- Sto-Rox School District
- Upper St. Clair School District and
- West Allegheny School District

### Cyclists

Recommendations which are made in Section 4.0 will include a combination of route types, in addition to multi-use. It is important to determine the nature of the routes and how they influence what type of user that is expected to travel each segment.

The American Association of State Highway and Transportation Officials (AASHTO) and PennDOT both classify bicyclists into one of the following three groups:

- **Group A – Advanced Bicyclists** – Generally use their bicycles as they would a motor vehicle. They are riding for convenience and speed and desire direct access to destinations with a minimum of detour or delay. They are typically comfortable riding with motor vehicle traffic; however they need a sufficient operating space on the traveled way or shoulder to eliminate the need for either them or a passing motor vehicle to shift position.
- **Group B – Basic Bicyclists** – Less confident adult riders may also be using their bicycles for transportation purposes, but prefer to avoid roads with fast and busy motor vehicle traffic unless there is ample roadway width to allow easy overtaking by faster motor vehicles. Thus, basic riders are comfortable riding on neighborhood streets and shared use paths and prefer designated facilities such as bike lanes or wide shoulder lanes on busier streets.



*Typical Group B Cyclist*



- 
- **Group C – Child Cyclists** – Riding on their own or with their parents, child cyclists may not travel as fast as their adult counterparts but still require access to key destinations in their community such as schools, convenience stores and recreational facilities. Residential streets with low motor vehicle speeds, linked with shared use paths and busier streets with well-defined pavement markings between bicycles and motor vehicles, can accommodate children without encouraging them to ride in the travel lane of major arterials (AASHTO, 1999).

It is estimated that only 5% of bicyclists overall would qualify as Group A or Advanced Bicyclists, therefore 95% fall into either Group B or C.

### Equestrians

It is becoming increasingly difficult for equestrians to find places to ride in urban communities. Many trails prohibit this activity for fear of conflict with other users and damage to the trail surface. It is unlikely horses will damage a trail surface unless the sub-base is poorly prepared and a softer surface (versus asphalt) is preferred to prevent injury to horse's hooves. It is not anticipated that accommodations for equestrian users will be necessary in the Allegheny County portion of the study area. It may however, based on the experiences of previous trail developers, be desirable for the Washington County sections.

### Cross-Country Skiers

Introducing a winter season alternative allows users to utilize the trail throughout the year. Many multi-use trails with 6 inches of snow cover can be used for skiing during the winter months.

### Commuters

Trails are often viewed solely as pathways for recreational use, but they can be utilized in many different ways. Trails can serve as nonmotorized transportation routes, reducing traffic congestion and improving air quality. The Port Authority of Allegheny County (PAT) buses operate throughout the Allegheny County portion of the project area connecting commuters to downtown, the Pittsburgh International Airport and other local destinations. The majority of PAT's bus routes are concentrated in the eastern and northern developed communities with fewer routes in rural areas.



## In-line Skating

In-line skating has grown substantially in the last decade from a sport that was limited to youthful participants to a viable transportation and exercise alternative for all ages. In Pittsburgh, the Three Rivers In-line Club (TRIC) is a nonprofit,



*TRIC Outing at Northshore Park, Pittsburgh*

volunteer, incorporated organization founded in 1994. It consists of concerned, dedicated skaters that have a wide variety of professions, skating abilities, and skating goals. TRIC's mission is to provide its members opportunities to skate together, represent the interests of skaters in community affairs in Western Pennsylvania, and encourage, promote, and develop the sport of inline skating (TRIC, 2003). In addition to groups such as this, many individual skaters would utilize trail segments that provide an acceptable environment for skating.

## **Potential Demand for Greenway / Trail**

The Statewide Bicycle & Pedestrian Master Plan Executive Summary (PennDOT, 1996) states that nearly as many people rely on walking and bicycling to reach work as the number of people that rely on public transit. This trend may be due, in part, to that fact that suburban population centers are expanding outward from our urban centers. This makes it increasingly difficult to travel between destinations without an automobile.

The lowest level of alternative transportation accessibility has been in the suburbs. Often new residential and commercial development makes little or no provision for facilities to promote these activities. The greatest potential to increase activity in these areas is to promote walking or biking to transit service or by providing trail access that could connect neighborhoods and destinations (e.g., shopping, downtown, etc.). Rural communities are often faced with similar problems. Although there may be ample amounts of paved road in these communities, there are significant safety considerations with the absence of sidewalks and paved shoulders.

A study commissioned by the Pennsylvania Economy League (PEL) and the University of Pittsburgh, in conjunction with the Allegheny Trail Alliance (ATA), provides extensive documentation of trail use and economic benefit in southwestern Pennsylvania. According to this study (Farber, 1999); the six trailheads that were monitored exhibited nearly 360,000 trips in 1998. These trips were conducted by nearly 110,000 individuals. As noted in the previous section, there are nearly 570,000 people in the proposed trail's primary service





area. Based on the immediate proximity of this trail with these population centers, it is not unreasonable to assume that annual users of a multi-use trail could exceed 100,000 individuals.

Many people belong to bicycling, walking and trail clubs throughout Pennsylvania. Based on a study conducted by Seton Hill College and the ATA (ATA, 2002), the Great Allegheny Passage trails were utilized the most on Saturday and Sunday. This indicates that trails are most likely being used more for recreation than for commuting purposes. It should be noted that the urban setting of the Chartiers Trail would lend itself to substantially more commuter traffic. Commuter traffic could be increased with the construction of a connecting trail along the Ohio River to downtown Pittsburgh or Station Square.

This study also noted that older generations (over 60) were using the bike trail more than other age groups. A similar study for trails in the Oil Heritage Region of Venango County, Pennsylvania notes that the mean age of trail users is 49 years. Both studies indicate age groups ranging from 12 to 78 as potential trail users.

The growing popularity of trails in southwestern Pennsylvania is illustrated by the success of trail based fitness and nature events. For example, IKEA sponsors an annual 5K/Half Marathon race which is held primarily on the Montour Trail. The following table illustrates the growth in the number of participants in this event from 1997 to 2002.

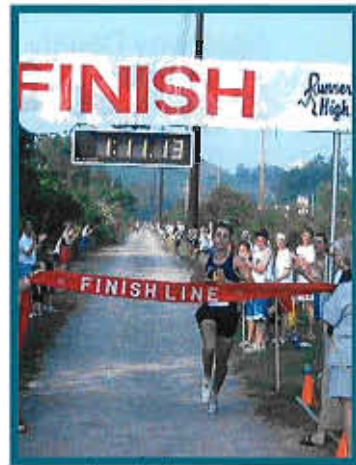
**Table 2.72**  
**Annual Participants in the Ikea Montour Trail Race**

| Year | 5K  | Half Marathon |
|------|-----|---------------|
| 1997 | 242 | 521           |
| 1998 | 181 | 431           |
| 1999 | 210 | 507           |
| 2000 | 278 | 605           |
| 2001 | 285 | 543           |
| 2002 | 291 | 657           |

There are many 5K races held around the Pittsburgh area, but less half-marathons. The Montour Trail offers runners a fairly flat, well-maintained course that is free of traffic. Runners very rarely find these trails in one facility considering local topography and traffic patterns. Based on the popularity exhibited above, area runners would most likely welcome additional trails for variety and convenience.



Rail corridors provide many miles of trail, at gentle grades, with attractive scenery and a relaxing atmosphere. Often, rail corridors are linked together to provide opportunities for extended trips for users. There are two existing trails to which the Chartiers Creek Trail could potentially be linked in the future. The Panhandle Trail is being developed on the Norfolk Southern (previously Conrail) right-of-way along Robinson Run. When complete, the Panhandle Trail will extend from Walkers Mill in Collier Township to Weirton, West Virginia.



*Ikea Half-marathon  
Montour Trail*

The Montour Trail, as previously mentioned, continues to be developed on the Montour right-of-way that bisects the project area through Robinson, Cecil and Peters Townships. When complete, the Montour Trail will become part of a system of trails stretching from Coraopolis, Pennsylvania to Washington, D.C. Two surveys indicate that a typical user covers between 11.98 and 12.7 miles during a trip. It appears that people like to take advantage of local trails, but they are not averse to traveling to experience a scenic trail. The Oil Heritage Region survey indicated 94.3% of users were from Pennsylvania, 3.6% from Ohio, 0.4% from Virginia, 0.4% from New York, and 1.4% from other states including Illinois, Florida, North Carolina, Arizona, and Rhode Island. The Great Allegheny Passage survey indicated 67% of users were from Pennsylvania and 33% were from other states such as Virginia, West Virginia, Ohio, New Jersey, Maryland, Tennessee, Vermont, Indiana, New York, and Minnesota (ATA, 2002).

### **Economic Benefits of Trail Development**

Trail development can provide significant revenue for local businesses by attracting consumers to an area that they may not otherwise visit. The ATA survey results indicate that regional trail development has had a significant impact on the communities through which a trail passes. The study indicates that at least \$5.4 million was spent by the users which visited the six trail heads in a single year (ATA, 2002).

In addition, 33% percent of the respondents to the survey estimated that trail related business accounts for more than ten percent of annual receipts. In addition, the PEL and ATA Business Survey Significant Findings Report includes results from 34 businesses contacted (27 responded or 79%) in Allegheny County and 21 businesses contacted in Washington County (7 responded or 33%). The following responses may be assumed to be comparable to those that would result from the development of the Chartiers Trail.





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### Allegheny County

- 22% of the respondents indicated that the Montour and Youghiogheny North trails strongly influenced their decision to start and/or locate businesses in proximity of the trails.
- Nearly half of the respondents intend to expand their businesses.
- 59% of the respondents estimate that 5% or more of their annual business can be attributed to trail existence.
- Nine businesses reported that their peak season runs from April to November, with estimated annual receipts from trail business averaging 15.6%.
- A tavern owner, whose business can be easily accessed from the Montour Trail, has expanded the menu and business hours to accommodate trail users.
- The owner of a restaurant and bike rental establishment on the Youghiogheny North trail has expressed an interest in opening another establishment on the Montour trail
- Many business owners expressed interest in pursuing advertising along the trail.

### Washington County

- Three of the seven respondents indicated that the Montour trail had a strong influence on decisions to establish businesses located in the proximity of the trails (two of the remaining respondents were established prior to development of the trails).
- 43% of the respondents intend to expand their businesses.
- 87% of the respondents estimate that 5% or more of annual business can be attributed to trail existence.
- Two businesses reported that their peak season runs from April to October, with estimated annual receipts from trail business ranging from 50-75%.



## **2.8 Land Use / Zoning**

A qualitative analysis was performed on land use and zoning patterns within the study area to facilitate an initial determination of trail and greenway feasibility as it relates to land use.

### **Land Use**

Land use in the project area varies as a result of variations in topography, proximity to urban centers, and historic land use practices. The northern portion of the project area is primarily urban residential, with a few pockets of commercial and light industrial uses. The western portion of the project area is primarily agricultural and forested with rural/suburban residential and commercial uses interspersed. Along the main stem of Chartiers Creek in the valley areas, land uses tend to be mixed, exhibiting industrial and commercial uses with urban residential and agricultural/forested areas.

The following table presents a summary of the land use types that were identified during the preparation of the RCP (Skelly and Loy, 2001).

**Table 2.8.1  
Major Land Use Types Within the Lower Chartiers Creek Watershed**

| <b>Land Use Type</b>  | <b>Acreage</b> | <b>Percent</b> |
|-----------------------|----------------|----------------|
| Forested              | 37,689         | 43.0           |
| Agricultural          | 26,385         | 30.0           |
| Residential           | 16,871         | 19.0           |
| Industrial/Commercial | 2,211          | 3.0            |
| Transportation        | 1,767          | 2.0            |
| Brownfields           | 1,293          | 1.5            |
| Strip Mines           | 682            | 1.0            |

Residential land use varies within the project area. Extensive residential land use is found along the eastern communities. These communities consist of a mixture of single-family housing units, duplexes, condominiums, apartment buildings, and high-rise apartment buildings. Lower density residential land use is established in the western portion of the project area. This consists of some rural communities with higher residential density. The communities in this area consist of single family housing units with many less than one acre in size.







## Zoning

Zoning is a critical component in land planning and conservation. Zoning within the project area was evaluated to determine if the projected land use patterns are conducive to the establishment of a contiguous greenway system along Chartiers Creek. The majority of the property in the project area is residential, industrial, forested and open space. Each municipality was contacted and a request submitted for their most recent zoning maps and designations. For those municipalities that provided current zoning documents, a qualitative analysis was performed to determine the zoning types that are found within the proposed greenway boundary. The following zoning classifications are typical of those that were found on the maps. A brief description of the permitted uses follows the zoning classification.

- C-1 Conservation  
Permitted Uses: *Cemeteries, public parks and playgrounds*  
Conditional Uses: *Churches, private recreation facilities, shared parking*
- P-1 Active Public Park:  
Permitted Uses: *Outdoor swimming pool, community gardens, trails, outdoor plant and wildlife sanctuary, fishing and boating, grass and other natural playing surfaces, playgrounds, outdoor courts, outdoor areas devoted to winter sports, picnic facilities*  
Conditional Uses: *Indoor health and fitness, temporary structure*
- P-2 Passive Public Park:  
Permitted Uses: *Public open space, hiking, walking, biking and exercise trails, outdoor plant and wildlife sanctuary or outdoor nature study, fishing and boating, picnic facilities*
- PO – Parks and Open Space:  
Permitted Uses: *Parks and recreation*  
Conditional Uses: *Single unit detached residence, agricultural use, public assembly, cemetery, communication towers, community center, cultural service, excavation/grading/fill associated with mining, golf course, nursery, indoor and outdoor recreation, recycling collection, fast food restaurant.*
- M1- Manufacturing (Planned Industrial)  
Permitted Uses: *Light manufacturing, research laboratory, warehousing offices, distribution center, contractor's yard, auto sales and repairs (other than body shop), printing, restaurant, bar, accessory uses*





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Conditional uses: *Community use, freight terminal, public use, utility structures, auto body shop, animal care and control service, landscape (nurseries)*

- R1-One Family Residence  
Permitted Uses: *One family house, agriculture, parks and recreation, community use, accessory uses*  
Conditional Uses: *Recreation club, public building, dependent dwellings*
- R2-Two Family:  
Permitted Uses: *One family house, two family house, agriculture, community use, accessory uses*  
Conditional Uses: *Garden apartments, townhouses, recreation club, public building, dependent dwelling*
- R-3 Planned Residential:  
Permitted Uses: *Single family detached dwelling, duplex and townhouses; Parks, playgrounds and other recreational uses; Educational or religious uses; Municipal buildings including police, fire, libraries and museums, excluding correctional institutions.*  
Conditional Uses: *Nursing homes and funeral homes, home occupations, private clubs, lodges, recreational buildings or properties, charitable organizations.*
- GI – General Industrial:  
Permitted Uses: *Animal care and veterinarian services, basic industry, car wash, club with or without liquor sales, construction contractor, cultural service, freight terminal, funeral home, grocery store, laboratory, laundry services, manufacturing, medical clinic, nursery, office, recycling collection, safety service.*  
Conditional Uses: *Adult entertainment/bookstore, child care, communication towers, correctional facility, excavation/grading/fill associated with mining, firearms business, hazardous operations, helipad, incinerator, parking garage structure, parks and recreation, religious assembly, fast-food restaurant, retail sales, salvage yard.*

The following specific zoning information was obtained for municipalities within the study area:

#### Bridgeville Borough

The slopes along Chartiers Creek in Bridgeville Borough are zoned primarily C-1 (Conservation). There is a small block of I-1 (Industrial) land on the western side of the borough which seems to be lightly developed and may be a possible option for green space acquisition. A large portion





of the Chartiers Creek floodplain within Bridgeville falls within the I-79 corridor.

#### Carnegie Borough

Chartiers Creek is bounded by land primarily zoned as Manufacturing (M1) and Commercial (C1 and C2) in Carnegie. There is a small area of Residential (R2) on the eastern bank. This land contains high density residential and industrial development. The only open areas bordering Chartiers Creek in Carnegie Borough are the very steep slopes to the west of the creek and adjacent to the P&OC railroad tracks.

#### City of Pittsburgh

According to zoning maps dated September 22, 1997, the land along Chartiers Creek in the City of Pittsburgh is zoned PO (Parks and Open Space District) and GI (General Industrial District)

#### Crafton

The land bordering Chartiers Creek in Crafton Borough is zoned S (Conservancy), R-4 (Urban density, single family), P-2 (Passive public park), and I-1 (Light industrial). The area zoned I-1 Light Industrial appears to be undeveloped. At the time of publication, Crafton Borough is revising its zoning ordinances.

#### McKees Rocks Borough

The Chartiers Creek floodplain in McKees Rock is primarily zoned I-1 (Industrial). The steep slopes of the creek to the west of McKees Rocks are undeveloped and zoned R-1 (Residential).

#### Rosslyn Farms

Chartiers Creek is bordered by land zoned as S (Conservancy District) in Rosslyn Farms Borough. A large plot of land on the flood plane at the southern end of the borough is developed and zoned as industrial.

#### Scott Township

Bordering Chartiers Creek in Scott Township, the land is zoned primarily I (Industrial) and R2 and R5 (Residential). Some of the steep slopes along Chartiers Creek are designated as green space. The areas zoned Industrial and Residential are heavily developed.





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### South Fayette Township

The land bordering Chartiers Creek in South Fayette Township is zoned I-P (Industrial Park), R-1 (Rural Residential), R-3 (Planned Residential), C-2 (Highway Commercial), and I (Industrial). These zoning areas are distributed in large blocks over the length of Chartiers Creek. The land in the area of Mayview State Hospital is zoned R-1 (Rural Residential).

### Thornburg Borough

The zoning in Thornburg Borough along Chartiers Creek is primarily S (Conservation) and P-1 and P-2 (Active and Passive Parks). One area to the north of the borough borders Chartiers Creek and is zoned I-1 (Light Industrial). This area appears to be mostly undeveloped based on review of the municipal maps.

### Upper St. Clair

The steep slopes along Chartiers Creek in Upper St. Clair Township are zoned primarily public and private recreation. There is an isolated parcel of residential land along a bend in the creek to the north of the township.

It is evident that many of the municipalities within the study area have taken steps to protect available open space and implement zoning measures that promote recreation and conservation. Additional examination of zoning and development regulations should be conducted to encourage bicycle and pedestrian trail planning and remove potential impediments to this and other conservation oriented plans. The following action items are recommended to accompany existing zoning and conservation measures.

- Develop a bicycle and pedestrian plan and include it as part of the overall comprehensive and transportation plan of the community.
- Coordinate bicycle and pedestrian planning within:
  - different departments of municipal government
  - agencies serving the same geographical area (e.g., park districts, school districts)
  - municipal government
  - federal, state, county, and township agencies
- Incorporate bicycle and pedestrian needs into other long range planning efforts and capital development programs (e.g., open space plans, recreational plans, infrastructure plans, etc.).
- Adopt or improve bicycle and pedestrian requirements contained in local development regulations (e.g., zoning ordinance and subdivision regulations)







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including requirements and standards for sidewalks, trail connections, street and shoulder width, drainage considerations, bicycling amenities, etc.

- Establish a process which provides for routine review of subdivision plans and site plans for bicycle and pedestrian facilities.
- Perform a bicycle and pedestrian needs assessment as part of the initial engineering on all local road improvement projects (including new construction, upgrading and maintenance).
- Work with state and regional transportation agencies and transit providers to enhance intermodal capability as they implement improvements projects in the community.
- Develop effective local education programs for bicyclists (especially child riders) and pedestrians.
- Promote secondary and tertiary benefits of pedestrian and bicycle use. This may include:
  - using facilities as an economic development tool
  - justifying need and use by special populations (disabled, lower income, adolescents, elderly, etc.)
  - health benefits
  - environmental benefits
  - building community character

Each municipality should be contacted prior to trail development activities to determine the applicability of local zoning ordinances to the proposed development activities.





## **2.9 Property Database**

A project-specific property database has been prepared for this project to provide a means by which property ownership can be readily identified by project stakeholders. The database contains the following components:

- Both digital and hard copy images of Allegheny County's lot and block maps for the project area.
- Downloaded files from the Allegheny County website with up-to date property ownership data.
- Washington County hard copy lot and block tax maps for the project area
- Washington County property ownership data that was obtained from tax records.

The property ownership printouts are contained in Appendix G in a separate binder and the tax maps are attached as Appendix H in a separate binder. It should be noted that the property ownership data is updated by Allegheny County frequently and that the data contained in this report, although accurate at the time of publication, may be outdated in a short amount of time. Methodologies that were employed for researching the different formats in which the property data is available for Allegheny and Washington Counties are as follows.

### **Allegheny County**

To assist the property stakeholders in the identification of parcels and ownership, the Allegheny County property data has been linked to the project GIS. As illustrated on the attached Property Index Map (see figures section), the county tiles, as shown on the figure, are linked to image files of the actual lot and block map. A GIS user may click on the tile which is desired and view or print the associated lot and block number image. To determine ownership of an individual parcel, the user can then look up the owner, referenced by lot number, in the MSWord files that have been created and attached to this plan on CD. The user may also look up the ownership data in the printouts contained in Appendices G and H, which are attached to this report.

### **Washington County**

The Washington County data is not available in electronic format, and therefore is available in hard copy paper format. Copies of the lot and block tax maps have been included in Appendix E along with a spreadsheet which contains the property owners. To identify the owner of an individual property, the hard copy lot and block map must be consulted to obtain a lot and block number, and then the ownership table consulted to identify the owner's name and address.





## Key Property Information

Key property owners are identified by their ownership of a piece of property that may be integral to the development of a trail or greenway. Project stakeholders have already identified several key properties and either acquired or initiated ownership procedures. These properties are listed in Section 4.3, Greenway Options. Additional opportunity for property ownership or control is outlined in Section 5.2, Land Acquisition and Easements.





## **2.10 Issues, Opportunities and Potential Threats**

The following issues, opportunities and threats have been identified through the numerous phases of data collection and discovery described in the previous sections of this report.

### **Issues**

The study corridor may be characterized as an interwoven system of transportation routes including local collector roads, interstate and major highways, a dedicated bus way, and railroad freight lines. These routes weave through areas of steep slopes, dense urban residential development, existing or former heavy industrial sites, regional scale commercial developments, and significant tracts of privately owned open space along the banks of the Chartiers Creek. Creating a continuous trail system that delivers the local and regional populations to the many places of beauty within the Chartiers Creek valley poses a significant challenge, considering the limited land availability and many constraints associated with the existing conditions.

### **Opportunities**

The P&OC railroad is currently an active freight line and has indicated that they will not consider the use of their right-of-way for recreational purposes (see Section 3.2 – Key Person Interviews). However, this right-of-way could become a very valuable rail-trail / recreation resource/non-motorized transportation route should it ever become available for such future uses. The CNC should establish and maintain a relationship with the P&OC to monitor the status of the line and be notified if and when the line should ever become available for purchase. This existing rail line would be the obvious choice for the location and development of a multi-use trail in the Lower Chartiers Creek valley for many reasons. These reasons are explained in further detail in Chapter 4.3 – Trail Alignment and Connection Alternatives.

The Borough of Crafton has received a commitment of \$180,000 of TEA-21 funds which are administered by PennDOT. These funds may be used to develop a section of trail located between Pts. 32 and 36 on Maps 3 and 4. Receipt of these funds is contingent on the Borough raising a local match. A DCNR grant application has been submitted to achieve the match.

Connections to other regional trail systems such as the Panhandle and Montour Trails will also increase the value of the Chartiers Creek Trail by expanding its recreation and transportation potential. A short connection along Robinson Run will provide the connection to the Panhandle, which now terminates at Walker's Mill. For more information about this potential connection, see Map A and the description of alternatives found in Section 4.3 of this report. The recommended routes will intersect the Montour Trail alignment in Washington County. At the







time of this report, this segment of the Montour Trail alignment remains undeveloped.

The Chartiers Valley District Flood Control Authority (CVDFCA) owns the land that surrounds Chartiers Creek in a large portion of the study area. The CVDFCA is comprised of representatives from each municipality through which the lower Chartiers Creek flood control project passes. Their primary function is to maintain the flood control channel, which was constructed by the U.S. Army Corps of Engineers, and now carries Chartiers Creek for the majority of its length downstream of Bridgeville. This presents a unique opportunity for trail and greenway sponsors to coordinate with a primary property owner for activities that are proposed within the immediate stream corridor.

The acquisition of the Turner Street Bridge is a critical step in the development of the Chartiers Trail. This structure (see Section 4.3) will connect the Borough of Carnegie with trail sections in the City of Pittsburgh and Rosslyn Farms. Currently, the bridge is owned by the City of Pittsburgh, and will require at least \$40,000 of improvements prior to potential acquisition by CNC.

### **Potential Threats**

Continued development within the corridor threatens the goal to create a continuous trail system by incrementally eliminating land where a trail may be placed. CNC has the ability to acquire, either through easements or fee simple acquisition, properties for the segmental development of a multi-use trail for the proposed routes explained in Section 4.3. However, acquisition of land in this area is a significant task, due to lack of available properties, cost of acquisition, and coordination issues.

Thus, a major problem that will confront the developers of this trail and greenway system is the complexity of property acquisition issues. Due to the extensive industrial history within the study area, the majority of the properties along Chartiers Creek have been subdivided or bisected with utility or drainage easements. The limited amount of property work that has been performed to date has revealed many complex property issues which require significant expenditure to resolve. Property acquisition in this type of environment becomes tedious and costly. Unfortunately, without an established right-of-way alternative to utilize, the developers of the Chartiers Creek trail should expect to spend significant resources to acquire access to the necessary properties for trail development.

What affects a proposed *Maglev* passenger rail system through the Chartiers Valley would have on a trail system has yet to be determined. The current *Maglev* 'Recommended Preferred Alignment Alternative' passes into the Chartiers Creek valley from its confluence with the Ohio River in Pittsburgh and runs in an east-west direction towards Interstate 79 and Kennedy Township.





Further investigation is necessary to assess the potential impacts this unique transportation system will have on the proposed trail. The Maglev alignment could possibly become an opportunity for a trail by using an intermodal or shared right-of-way. Trail and greenway stakeholders should coordinate with the Maglev system developers in its next stage of development to assess these potential effects and possibilities.

### **Potential Impacts to Wildlife**

As part of this plan, an impact analysis was performed to examine possible impacts of the greenway and trail development on the study area's indigenous wildlife. The impact analysis focused on typical issues of importance concerning wildlife management in urbanizing areas.

#### **Habitat Fragmentation**

Habitat fragmentation is a process of both natural and man-induced processes which occurs when large contiguous habitat types are disrupted. Natural processes may include fire, disease or climactic changes, while man-made disturbance are comprised of deliberate changes in landuse such as timber harvesting or land development which occurs on a significant scale. These occurrences fragment large habitat areas and influence the composition and abundance of wildlife species. While ecosystem fragmentation creates more edge habitat along road and other clearings, it decreases the forest's interior habitats. Such fragmentation benefits edge species (e.g., the white-tailed deer), but are detrimental to wildlife species that live within interior forested habitats. Chuck Cottrell, in an article prepared for the Wildlands Center for Preventing Roads (WCPR) reinforces this concern. He states the "fragmentation of temperate forests is a major factor in the decline in numbers of some neo-tropical migrant songbirds because of the decline in overall forest habitat and the corresponding increase in edge habitat (Wilcove, 1985). Edge habitats act as "ecological traps" for some songbirds (Gates and Gysel, 1978). The dense and varied vegetation of edges offers excellent feeding and nesting opportunities for opportunistic animals. The increased density of birds and eggs can then attract greater number of predators and brood parasites, which affects nesting success (Cottrell, 2002)."

It should be noted that the issue of habitat fragmentation, as discussed by Cottrell, is primarily directed toward larger, intact ecosystems that do not exhibit the level of disturbance that is present within the Chartiers Creek watershed. Thus development of the greenway or trail will cause little, if any, significant additional habitat fragmentation. Important riparian habitat should however, be preserved to the greatest extent possible, as should important aquatic habitat such as wetlands or streams. Conservation measures to ensure that wildlife habitat is maintained are included in Section 4.3 of this report.



## Soils and Plant Communities

With the exception of the steep slopes adjacent to Chartiers Creek, most of the native soils within the greenway have been disturbed by land development practices such as grading and paving, or by historic industrial activities related to the coal and steel industries. These activities have changed soil conditions and influenced the plant species composition, ultimately influencing wildlife habitat. Changes in the habitat have occurred when: (1) forests were cleared during land development; (2) native vegetation was removed, mixed, or replaced with exotic or undesirable species (which may impact the density and diversity of the vegetation); and (3) drainage patterns and soil conditions were changed.



*Invasive Specie - Japanese Knotweed  
(Polygonum cuspidatum)*



*Invasive Plant Specie Management with  
Herbicides*

For properties that are controlled by conservation agencies or municipalities, a land management plan as outlined in Section 4.3 should be implemented. This may include the management of the timber resources for sustainability, invasive plant control, and establishment of native plant species. During construction activities associated with the trail or trailheads, the clearing of vegetation should be kept to a minimum as required to maintain the intended use and safety of the trail users. Only non-invasive plant species

should be planted within disturbed construction rights-of-way and it is desirable to use a conservation mix comprised of warm and cool season grasses and native wildflowers. A typical conservation seed mix is provided in the following table and can be obtained from Ernst Conservation Seeds at 9006 Mercer Pike, Meadville, PA 16335-9299 or by calling (800-873-3321).





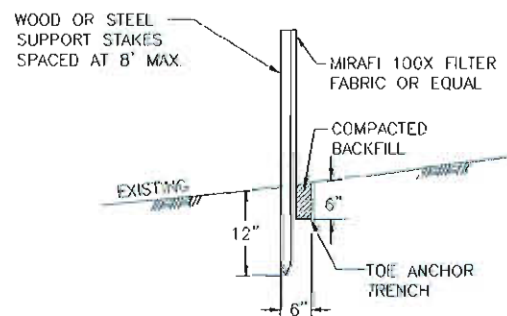
**Table 2.10.1**  
**Low growing Wildflower and Grass Seed Mix**

| Percent by Weight | Scientific Name                   | Common Name        |
|-------------------|-----------------------------------|--------------------|
| 34%               | <i>Festuca ovina</i>              | Sheep Fescue       |
| 31%               | <i>Lolium multiflorum</i>         | Annual Ryegrass    |
| 6%                | <i>Anthoxanthum odoratum</i>      | Sweet Vernal Grass |
| 0.2%              | <i>Cichorium intybus</i>          | Blue Chicory       |
| 9%                | <i>Hesperis matronalis</i>        | Dame's Rocket      |
| 6%                | <i>Chamaecrista fasciculata</i>   | Partridge Pea      |
| 4%                | <i>Achillea millefolium</i>       | White Yarrow       |
| 4%                | <i>Chrysanthemum leucanthemum</i> | Ox Eye Daisy       |
| 0.1%              | <i>Aster prenanthoides</i>        | ZigZag Aster       |
| 0.4%              | <i>Verbena hastata</i>            | Blue Vervain       |
| 0.2%              | <i>Asclepias tuberosa</i>         | Butterfly Milkweed |
| 0.2%              | <i>Liatris spicata</i>            | Spiked Gayfeather  |

### Water Quality

Numerous point and non-point sources of pollution exist within the lower Chartiers Creek watershed and they are detrimental to aquatic wildlife (e.g., benthic macroinvertebrates, fish, and amphibians). Also, the large amount of impervious cover (e.g., roads and parking lots) found within the study area increases stormwater runoff that may increase sedimentation and change nutrient levels of Chartiers Creek. Development of the greenway and trail within the study area will have little, if any, influence on existing point and non-point sources of pollution, or the amount of existing impervious cover existing within the corridor.

Best Management Practices (BMPs) should be implemented during construction as specified in the Pennsylvania Department of Environmental Protection (PADEP) Erosion and Sediment Pollution Control Program Manual, dated March 2000.







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Measures to be implemented during construction and maintenance of the trail system include the following:

- Timely revegetation of disturbed areas;
- Installation of Silt fence or hay bales;
- Placement of rip-rap for drainage control;
- Geotextiles use in trail design;
- Erosion control blanket installation;

Erosion control should be a component of all design packages for the proposed trail. If the project will disturb more than one acre of land, a site-specific Erosion and Sedimentation Control Plan should be prepared and submitted to the proper county conservation district office.





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## **3.0 Public Participation**

### **3.1 Meetings**

Three public meetings were held during the Feasibility Study Process. Public Meeting #1 was held on August 1, 2002 and focused on trail programming. Meeting #2 was held on December 12, 2002 and presented the initial alignment options and findings from the initial site reconnaissance. The third Public Meeting was held in May 2003 to present the final report.

Public participation is a key ingredient in the success of any community project. The meetings were designed to inform the public of the project status, and to receive public input about desired facilities and concerns relative to the trail development within each community.

Each of the Public Meetings was well attended, and the attendance records and notes from each of these meetings are included in Appendix A of this report. Steering Committee meetings took place in June, July, September, November of 2002, and April 10, 2003.

#### ***Public Meeting 1 – Press Release***

July 26, 2002 - The Chartiers Nature Conservancy will hold the first Public Meeting regarding the Chartiers Creek Trail and Greenway Feasibility Study on Thursday evening, August 1st, at 7:00 pm at the new Carnegie Borough Building on Mansfield Avenue. The purpose of this public meeting is to explain the feasibility study process and to invite public comment from interested parties.

Conducted in conjunction with Civil & Environmental Consultants and Simone Jaffe Collins Landscape Architects, the meeting will last about one hour, will be interactive in nature, and will focus on substantial public involvement to address local concerns and conditions related to Greenway and Trail development. The purpose of the study is to assess the feasibility of developing a trail and greenway, identify obstacles and opportunities, and to provide a planning tool for trail and conservation groups, municipalities, and all interested parties.

The study area begins at the confluence of Chartiers Creek and the Ohio River in McKees Rocks and follows the main channel upstream for 26-½ miles to Canonsburg Lake. The study will also assess the feasibility of a trail and greenway connection along Robinson Run from Chartiers Creek to the existing Panhandle Trail at Walker's Mill in Collier Township.





The Chartiers Creek Greenway and Trail could become the centerpiece of the entire Lower Chartiers Creek watershed, connecting residents with water-related recreational activities. A trail and greenway could potentially provide 14 communities with a continuous natural green and blue recreational thread as well as acting as an important wildlife corridor.

The following items were identified during the first public meeting as items which warranted attention in the preparation of the plan. The items were identified using the "card technique" employed by the project team. Using this technique, each attendee at the meeting was given the opportunity to express ideas, concerns and suggestions for the study. Each response was posted on a card and placed on a large board

### **Goals**

- Greenway Trail Feasibility
- Contiguous Trail
- Develop Priorities
- Estimate Costs
- Rail with Trail
- Involve Community
- Greenway
- Trail Alternatives

### **Facts**

- Multiple Municipalities
- Steep Terrain
- Existing Railroad
- On-Road & Off-Road Conditions
- Limited Right-of-way
- Penndot Standards
- ADA Standards
- AASHTO Standards
- Train Excursion Line
- Volunteer Groups
- Train Speed & Frequency
- Wildlife Habitat
- ALCOSAN
- Flood Authority
- Seniors
- Maglev
- DCED, Fish & Game
- DEP, DCNR





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## Concepts

- Connect Neighborhoods
- Trail = Transportation System
- Trail = Recreational Activity
- Trail Alignment Options
- Safe Connections
- Trail Surfaces and Widths
- Funding Strategy
- Security/Monitoring
- Directional Signage
- Combine Engineering Efforts
- Utility Corridors
- Control Measures
- Parking
- Property Values
- Municipal Resources
- Interpretive Signage
- Mile Markers
- Historic Features
- Trail Ownership
- Liability
- Economic Study (ATA)
- Floodplain & Wetlands
- Report as Sales Tool for Funding
- Existing Conditions Inventory
- Re-Zoning
- Greenspace – Preservation, Conservation
- Political Support
- Intermunicipal Cooperation
- Involve Businesses
- Vegetated Buffer

### ***Public Meeting 2 – Press Release***

December 7, 2002 - The Chartiers Nature Conservancy will hold the 2nd Public Meeting regarding the Chartiers Creek Trail and Greenway Feasibility Study on Thursday evening, December 12th, at 7:00 pm at the new Carnegie Borough Building on Mansfield Avenue. The purpose of this public meeting is to present the Draft Greenway and Trail Feasibility Study.

The Draft Plan will present for public comment a proposed principal and alternative trail alignment and a proposed greenway focus area. Chartiers Nature Conservancy and the study consultants, Civil & Environmental Consultants and Simone Jaffee Collins Landscape Architects, will present this plan to the public for the express purpose of inviting constructively critical







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comments. These comments will then be woven into the fabric of the Final Plan which will be released in early 2003.

Public participation is crucial to the acceptance and success of the Greenway and Trail Feasibility Study. The 26-1/2 mile long corridor is simply too big and complex for all the local opinions, opportunities, obstacles and variables to be discovered by the consultants. The Steering Committee and the consultants want to produce a Final Plan that accurately reflects both the feasibility of a greenway and trail and the wishes of the local population. In order to do that, we need feedback from the residents of the Lower Chartiers Creek Watershed. The 2nd Public Meeting will offer an opportunity for the people who live here to say what they think ought to happen.



### **3.2 Key Person Interviews**

The purpose of conducting the key person interviews was to initiate the process of building stakeholder involvement and to gain perspective on the activities of local government and trail organizations. The following individuals were contacted during the study process:

#### **Mr. Dave Wright – Allegheny County**

Mr. Wright is employed with the Allegheny County Department of Engineering and is a local trail advocate who has provided invaluable expertise and guidance for many local trail projects. He has been an integral part of the development of the successful Montour Trail. Mr. Wright was contacted by the project team to determine the feasibility of connecting the Montour and Chartiers trails. He expressed much enthusiasm to make this connection happen, but noted that the Montour Trail Council has significant engineering challenges for trail development in the area of the connection. Due to the bridges and tunnels in this area, development of this section has been delayed until additional funding can be obtained.

Mr. Wright also noted that the P&OC may not be willing to consider a rail-with-trail due to problems that they have had in Ohio. He did note, however, that the railroad is running numerous excursion passenger train trips from Carnegie to the Trolley Museum and Washington County Fairgrounds during the fair. He provided a contact at the P&OC railroad.

#### **Chuck Prine – Chartiers Nature Conservancy**

Mr. Prine has been a substantial supporter of CNC and its efforts to develop a trail and greenway system. Mr. Prine contacted the project team to provide comments and guidance on the direction of this study and to assist in the identification of study priorities. He suggested that the project team put emphasis on the protection of greenspace and natural areas within the study area. This should include the preservation of natural areas that contain mature vegetation and indigenous wildlife populations. Mr. Prine also noted that the greenway should be a component of a larger system that provides regional benefit and use and allows users to experience the natural beauty of Chartiers Creek. The project team would like to thank Mr. Prine for the assistance and enthusiasm that he exhibited during this project.

#### **William Strawn – P&OC**

Mr. Strawn is the president of the P&OC railroad and was contacted to determine their willingness to use a portion of their right-of-way for recreational trail purposes. Mr. Strawn stated that the policy of the P&OC is to prohibit recreational and pedestrian activities on their property. He noted that the





introduction of pedestrians or recreational trail users would require a 30-40% increase in their corporate insurance rates. He added that the P&OC will not consider purchase of, or easement through, any part of the right-of-way.

### **Chartiers Valley District Flood Control Authority (CVDFCA)**

The project team has had extensive contact with the CVDFCA through work with CNC on previous trail sections. Their representatives have indicated that the CVDFCA generally will accommodate development of trail sections on property which they control. They do, however, require that a land survey accompany any trail plans that are submitted to them. They also require a fee for use of their property.

### **Marlon Bartos – Collier Township Engineer**

Mr. Bartos represents Nira Consulting Engineers, Inc. in Coraopolis, Pennsylvania and serves as the borough engineer for Collier Township (Map A). Mr. Bartos indicated that Collier Township is currently pursuing alternative to connect the Panhandle Trail with the Borough of Carnegie, and eventually the Chartiers Trail. Collier is also planning to upgrade the trailhead facilities at Walkers Mill to include comfort facilities and additional parking.





## **4.0 Trail Alignment and Greenway Alternatives**

### **4.1 Methodology**

The Allegheny County Division of Computer Services provided GIS data for the project through the CNC by executing a data exchange agreement. This data serves as the base mapping for exhibits of the trail alignment and greenway alternatives that are discussed in this section. Mapping features include topography at a five foot (5') contour interval, roadway edges, buildings, hydrology, railroad lines, vegetation, and municipal boundaries. The detailed nature of the GIS base mapping allowed the consultant team to view the study corridor from a unique planning perspective, and provided a powerful tool for trail and greenway feasibility analysis.

After the compilation of the base mapping and identification of initial trail alignment alternatives, the project team performed a series of field investigations. The initial reconnaissance was focused on the potential to use the P&OC Railroad right-of-way. As it became apparent that this may not be feasible in the short-term, additional reconnaissance was performed to identify routes that did not use the P&OC Railroad right-of-way.

Development of the greenway plan was based on the results of the data collection, review of published literature and GIS data analysis.

### **4.2 Applicable Trail Design Criteria**

Based on the information that was compiled during the literature research and field view, next step in the feasibility process was to determine trail alignment alternatives and development costs for numerous trail alignments. This was done by applying nationally accepted trail design guidelines to the existing site constraints. The documents listed below were referenced to create a series of typical trail development sections for the project:

- *Guide For Development of Bicycle Facilities*, American Association of State Highway and Transportation Officials (AASHTO), 1999;
- *Trails for the Twenty-First Century: Planning, Design, and Management Manual for Multi-Use Trails*, Rails to Trails Conservancy (RTC), 1993;





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- *Statewide Bicycle & Pedestrian Master Plan, Bicycling & Walking in Pennsylvania – A Contract for the 21<sup>st</sup> Century: Bicycle Guidelines*, Commonwealth of Pennsylvania Department of Transportation (PennDOT), 1996.

The typical sections were assigned a unit cost per foot based on the needed improvements for each trail segment. The length of each proposed trail segment was tabulated using geometric functions of the GIS base mapping software. These unit costs were then multiplied by the tabulated lengths to generate each trail segment's estimated development cost. The typical sections and associated costs are detailed in the following sections.

## Trail Types

### Multi-Use Trail (Off-Road)

AASHTO defines a multi-use trail or shared use path as: *a bikeway physically separated from motorized vehicular traffic by an open space or barrier and either within the highway right-of-way or within an independent right-of-way. Shared use paths may also be used by pedestrians, skaters, wheelchair users, joggers, and other non-motorized users.* Transportation engineers also commonly refer to this trail type as a 'Class 1' bikeway.

As the definition suggests, this type of trail promotes usage by a wide variety of trail users. These types of multi-use trails also encourage universal accessibility for those with disabilities. This is due to their gentle slopes, adequate widths, and smooth surfaces. Parking areas for multi-use trail segments are generally designed to provide facility access in accordance with the federal Americans with Disabilities Act (ADA) guidelines.

The RTC and AASHTO recommended multi-use trail width is ten feet wide with the minimum recommended width for a two-way trail being eight feet, and five feet for a one-way trail. If it is determined that the trails may be high volume, widths of twelve feet or fourteen feet are recommended to avoid potential conflicts between users. An additional two-foot shoulder is recommended on either side of the trail surface to provide clearance from trees, poles, walls, fences or any other lateral obstruction. Site conditions may warrant additional safety measures such as fencing and increased shoulder widths.

In keeping with the mentioned standards, it is recommended that a minimum width of ten feet be applied to the typical design configurations. Site-specific conditions may call for the increasing or decreasing of this width, but for general estimating purposes, the recommended width will be used.



### Multi-Use Typical Section 1

Existing Conditions: Level cross slope in non-wooded areas. Some low-level or understory clearing necessary. Generally found in developed areas.

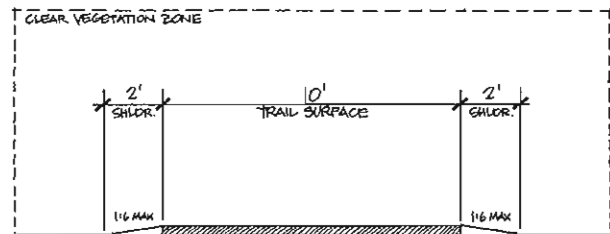


Proposed Improvements (cost (\$) per l.f. installed):

- Clearing and Grubbing (non-wooded): \$1
- Grade and Surface Trail (crushed limestone): \$6
- TOTAL: \$7/l.f.**

### Multi-Use Typical Section 2

Existing Conditions: Level cross slope in wooded areas. Generally found along stream banks or through natural or developed areas. Shoulders cleared.

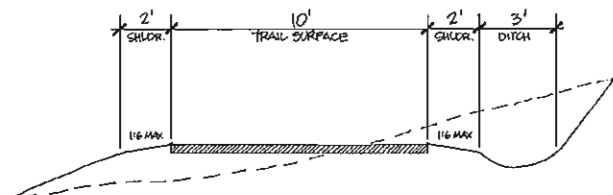


Proposed Improvements (\$/l.f.):

- Clearing and Grubbing (wooded): \$5
- Grade and Surface Trail (crushed limestone): \$6
- Stabilization Planting: \$1
- TOTAL: \$12 /l.f.**

### Multi-Use Typical Section 3

Existing Conditions: Existing cross slope requires benching and periodic drainage conduits in non-wooded areas. Some low-level or understory clearing necessary.



This is the least prevalent of the existing conditions since most alignments requiring benching occur at the base of wooded slopes within the study corridor.

#### Proposed Improvements (\$/l.f.):

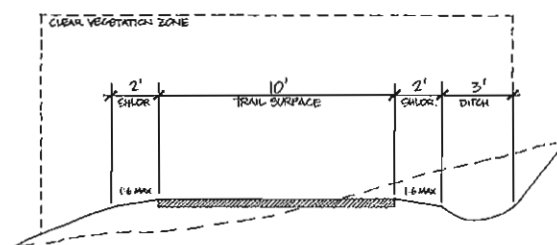
- Clearing and Grubbing (non-wooded): \$1
- Grade and Surface Trail (crushed limestone): \$6
- Additional Earthwork: \$5
- Drainage Swales and Pipes: \$3.50
- Stabilization Planting: \$1.50
- **TOTAL: \$17 / l.f.**

### **Multi-Use Typical Section 4**

Existing Conditions: Existing cross slope requires benching and ditching in wooded areas.

#### Proposed Improvements (\$/l.f.):

- Clearing and Grubbing (wooded): \$6
- Grade and Surface Trail (crushed limestone): \$6
- Additional Earthwork: \$5
- Drainage Swales and Pipes: \$3.50
- Stabilization Planting: \$1.50
- **TOTAL: \$22 / l.f.**



#### On-Road Bikeways

There are two general classifications of on-road bikeways. The first, *Bicycle Lanes*, or *Class 2 Bikeways*, are designated bicycle facilities within an existing roadway. These lanes are one-way traveling in the same direction as motor vehicles.

Bicycle lanes are a minimum 4 feet to a standard 5 feet wide, and are distinctly marked with a combination of signage and pavement markings. Bicycle lanes are commonly found in urban areas where the traffic volume is higher and the potential conflicts between cyclists and motorists are more likely. They are also appropriate on high-speed arterial roads.



*Typical On-Road Bike Lane*



The site reconnaissance and mapping revealed that the majority of existing roadways within the study area are fully developed, and therefore, the feasibility of retrofitting a bike lane on existing roads becomes problematic. To establish this type of trail, a concerted effort would need to be initiated between PennDOT, county and local government and the trail sponsors. Significant expenditures would be required to determine road geometry, utility, traffic, drainage patterns, environmental concerns, property issues, and how these constraints influence the development of a shared bike lane trail system. These items are generally not addressed in a feasibility study and the costs incurred for this type of trail development may not be justified with the intention of ultimately using the P&OC Railroad right-of-way. This report should, however, be forwarded to PennDOT, Allegheny and Washington Counties, and the local municipalities for consideration in their road planning activities.



The second type of on-road bikeway is a *Shared Roadway, or Class 3 Bikeway*, and is defined as: *a roadway which is open to both bicycle and motor vehicle travel. This may be on an existing roadway, street with wide curb lanes, or road paved with shoulders* (AASHTO, 1999). Often, the only improvements necessary for a Class 3 Bikeway are shoulder improvements and 'Share the Road' signage. With the exception of limited access highways, bicycles are permitted to share the roadway with motor vehicles and must adhere to vehicular traffic laws established by the Commonwealth of Pennsylvania.

Shared Roadways comprise a significant portion of the Preferred Intermediate Route, and are classified into two separate categories. The categories are based on the appropriate user type; Group A - Advanced or Group B - Basic bicyclists as would be the most appropriate for the amount of traffic volume or speed associated with the individual segments.

Typical costs were derived based on the following assumptions:

- the sample roadway is 2640' in length (1/2 mile);
- 20% of the roadway, on average, would receive shoulder improvements - 3' wide asphalt on both sides of the road (530' @ \$10/l.f. = \$5300);
- 2 bicycle safe grates (2 @ \$1000 ea = \$2000)
- 2 'Share the Road' signs (2 @ \$250 ea = \$500)

SUBTOTAL: \$7800 / 2,640 l.f.

TOTAL: \$2.95 or \$3.00 / l.f.





## Hiking Trails

For the purposes of this report, the hiking trail classification is defined as a recreational trail that does not meet the design requirements for width, slopes & surfacing of a multi-use trail. Some advantages of hiking trails are they can allow for access and recreational use of the land for a relatively low cost. A disadvantage of hiking trails is that they generally limit the number and type of trail users as a result of their minimal width, steeper slopes, and softer surfaces, and they generally do not meet ADA requirements.

### **Trail Surfaces**

Asphalt or macadam surfaces provide the most versatile surface for the widest variety of trail users including bicyclist, walkers, joggers, wheelchair users, and in-line skaters. Initial installation costs are relatively high compared to other trail surface types. Long term maintenance costs, however, are lower than others if the trail surface is properly installed and maintained.



*Typical Hiking Trail (segment 32A)*



*Typical Crushed Limestone Trail Surface –  
Chartiers Trail (segment 36-37)*

Crushed limestone surfaces can accommodate all trail user types with the exception of in-line skaters. Initial installation costs for this trail surface are relatively low, however long term maintenance costs may be higher than other surfaces due to the surface's susceptibility to heavy use and erosion, especially if not properly installed. A crushed limestone surface can also serve as base material for an asphalt surface if trail use increases or funds become available for a surfacing upgrade.

Compact earth surfaces are the least expensive to install, however they limit the types and number of trail users. Compact earthen surfaces are primarily used for hiking only or horse trails adjacent to multi-use trails that receive significantly less trail user volume.

For the purposes of cost estimating, it is assumed that the Chartiers Trail will be developed with a crushed limestone surface, which may be upgraded to asphalt as warranted by trail use.



## Trailheads

Trailheads are facilities that are primarily provided for users to access a trail system. Many also provide ancillary accommodations that allow for vehicle parking, trip planning and preparation, as well as areas for rest. In many cases, trailheads (for the Chartiers Trail) will be new facilities with parking areas. The size of each facility, land requirements, parking spaces and utility requirements will depend on projected use for each specific segment. The trailheads should be located at the ends of major trail segments, in population centers, at trail connections, and at points of interest. Trailheads size and the level of service will be determined during the master planning and design phases of development. Typical amenities which can be included in trailhead design are:



*Momon Trail – Indianapolis, Indiana  
Photo Courtesy David Gorden*

- Structure with restroom accommodations.
- Accessible Parking
- Drinking fountains
- Canoe Access
- Telephone(s) for emergency and coordination of events
- Seating
- Lighting
- Trail and Informational signage (i.e. wall mounted or kiosk)
- Bicycle racks
- Shade (i.e. structures/ plant material)
- Waste receptacles
- Landscaping
- Overflow parking allowances
- ADA accessibility

The locations of potential trailheads and parking areas are presented on the attached alignment maps.





### **4.3 Trail Alignment and Connection Alternatives**

This chapter includes narrative descriptions of the trail alignment alternatives located within the study corridor. The focus of these descriptions is on what has been designated the Preferred Intermediate Route (PIR).

The attached alignment maps 1-12 show the alternative route locations. The table entitled Estimated Development Cost Matrix, attached to this report, outlines the conceptual-level costs of improvements necessary for each segment of trail. The cost matrix includes proposed improvements for each segment, as well as the PIR, which is broken down by each development phase. A summary cost table is presented in Section 5.1 of this document.

Costs are not included for the Preferred Ultimate Route (PUR) because the consultant team was unable to obtain permission from the owner of the railroad right-of-way to perform a site reconnaissance. The costs for each of the PIR trail segments will allow the CNC to assemble estimates for combinations of trail segments to target particular grant applications or select alternative routes based on future changes in land availability, transportation improvements, or other opportunities that may arise.

#### **Preferred Ultimate Route**

The PUR follows the existing P&OC Railroad right-of-way through the study corridor and is indicated as the green heavy-dashed line on the attached alignment maps. There are many reasons why this existing railroad right-of-way should be considered the ultimate location for a multi-use trail through the Lower Chartiers Creek Valley, including:

- It would provide for non-motorized transportation route, or multi-use trail, allowing use by all user groups (A,B, & C);
- The right-of-way is continuous, with one landowner to negotiate acquisition, as opposed to multiple landowners under a segmented approach;
- Level grade will allow for a universally accessible trail;
- Development costs will be significantly less due to the existing level grades, in-place drainage systems, and bridge structures for crossings;
- The right-of-way provides direct and indirect access to and through the existing business districts, residential neighborhoods, and open natural areas within the valley; and,



- The adaptive reuse of the line will incur minimal additional development or disturbance within the corridor.

Because the existing railroad is active and servicing commercial and recreational customers, the owner currently is not amiable to negotiating the development of any portion this right-of-way as a trail. However, if this right-of-way should ever become available, either through a fee simple acquisition or a rail banking option, the trail sponsors should actively pursue its acquisition. In the interim, trail sponsors should maintain periodic contact with P&OC Railroad.

### **Preferred Intermediate Route**

The PIR should be considered a short term, interim solution to the achievement of the long term goal of providing a multi-use trail within the study corridor. The primary purpose of this intermediate route is to meet the current demands for a local trail network and to connect existing trail segments. If fully developed as described herein, the PIR will amount to a network of trail segments comprised of different trail types and trail users. This network will have individual multi-use segments suitable for recreational use by an entire family, while only the advanced cyclist will be able to use other segments as a transportation route.

The criteria for selecting the PIR from the trail segments included in this study are as follows:

- Most direct route to achieve the goal of connecting Canonsburg Lake and McKees Rocks.
- Land availability - The number of properties affected resulting in the need to acquire land or right-of-ways through purchase or easements.
- Trail Type and (User Group) priority:
  - Multi-Use (A,B,C,)
  - Shared Roadway – Basic Bicyclist (A,B)
  - Shared Roadway – Advanced Bicyclist (A only)
- Connectivity to adjacent parks, schools, neighborhoods, and business districts.
- Future use after the development of the PUR. Segments that may become obsolete with the development of the PUR may be less desirable than those that may contribute to and enhance the PUR.
- Potential development costs – Significant costs for engineering and construction are weighed against other criteria to determine a segment's economic feasibility.

### **Trail Segment Descriptions**

Trail segment alternatives that were studied have been organized using a "point" and "segment" labeling system. Points are generally located at logical beginning or ending points of a trail segment occurring at surface changes, municipal





boundaries, access points, junctions, and roadway crossings or intersections. Segments are labeled based on the numbers found at their endpoints. For example, a trail segment between points 37 and 39 is labeled segment 37-39. These map labels correspond to the costs outlined for each point and segment in the Estimate of Development Cost Matrix following the map exhibits.

The following narrative summarizes the existing conditions and proposed improvements for trail segment alternatives selected as part of PIR and correspond to the map sheet number found at the end of this chapter for reference. Descriptions of the other alternatives, as well as the PUR are included, but the focus is primarily on the PIR. More detailed information pertaining to individual segments including lengths, existing conditions, ultimate improvements, and associated costs can be found in the Estimated Development Cost Matrix.

#### Map Sheet 1 - Ohio River to CNC Windgap Property (Points 1 thru 22)



*West Carson Street Bridge  
(Segment 01-02)*

Starting at the confluence of the Chartiers Creek and the Ohio River and a proposed parking area, the PIR will cross the West Carson St. bridge, on western sidewalk. Paralleling this bridge is where the PUR will eventually join the proposed Three Rivers Heritage Trail and connect the Chartiers Valley to downtown Pittsburgh. A crossing of West Carson Street will be required for this linkage.

The PIR will then become a multi-use trail that will parallel Stanhope St. through Duquesne Light property requiring an easement for approximately 1000 feet in length to the point it meets Linden Street. A continuation of the multi-use trail is suggested under the Linden Street Bridge and along the southern bank of the creek through property controlled by the Duquesne Light Company, with a hiking trail spur to Sheraden Park. Considerable design constraints must be addressed for the potential of a multi-use trail passing under this bridge.

At the time of this study, this alignment is being considered as a right-of way for the proposed Maglev commuter rail line (see Section 2.3). Other concerns for a multi-use trail in this area are steep slopes, the need for easement acquisitions through privately owned industrial properties, and the need for a new trail bridge, between locations 09-10.



*Duquesne Light right-of-way adjacent to  
Stanhope Street (Segment 02-03)*



The PIR will cross the Linden St. Bridge on-road then follow the northern creek bank behind a restaurant and through a municipally owned parking lot until the available land area runs out, forcing it back



*Existing Windgap Pedestrian Bridge  
(Segment 11-12)*

onto Furnace St. The shared roadway route will then continue along Furnace St./Thompson Ave. through McKees Rocks until it reaches an existing pedestrian bridge (see photograph). Cyclists will be required to dismount and walk their bikes across this narrow wood deck bridge which will require its renovation. Once trail users reach the southern bank, the shared roadway PIR will follow Township Rd. utilizing an existing at-grade railroad crossing, then proceed along Youghioghenny St. until it meets the existing

active railroad right-of-way and the northern end of the CNC Windgap property (near location 16).

#### Map Sheet 2 - CNC Windgap Property to Scully Yard (Points 16 thru 23)

The PIR will continue along the base of the slope through the CNC Windgap Property adjacent to the active railroad right-of-way until it meets a spur line servicing an industrial building near the southern end of Scully Yard. A new at-grade crossing of this spur line will be necessary to connect to the northern end of the Industrial Highway.

The alternative routes that include a shared roadway along Creek Road, the rehabilitation of an abandoned railroad yard access bridge (locations 17-18), and a multi-use trail through Scully Yard were not selected as part of the PIR primarily because they would impact railroad property and require additional at-grade R.R. crossings (locations 20-21) and/or a new bridge or R.R. bridge rehabilitation over the creek at the southern end of Scully Yard.

#### Map Sheet 3 - Scully Yard to Crafton (Points 19 thru 35)

Adaptive reuse of the Pittsburgh Redevelopment Authority's Industrial Highway will provide for a significant portion of a multi-use trail at relatively low cost for trail development. Converting the two lanes nearest to the Chartiers Creek to trail use while leaving the other two lanes open for two-directional vehicle traffic will make full use of this under-utilized 'highway' (locations 23-25). A traffic



*Industrial Highway  
(Segment 23-25)*



study should be performed to verify this condition and support the conversion of the highway to trail use. A schematic of this concept is provided in Section 5.1.

The PIR will continue from the southern end of the Industrial Highway at location 35, via an easement the CNC has attempted to acquire along the creek bank. The multi-use trail will terminate in the parking lot of an adjacent business where it will become a shared roadway.



*Turner Street Bridge  
(Segment 38-39)*

Map Sheet 4 – City of Pittsburgh to Carnegie (Points 36 thru 51)

From the Crafton Borough/City of Pittsburgh line, the multi-use PIR continues south until it converts back to a shared roadway approximately 500 feet west of the Turner Street Bridge (location 38-39). Before crossing the bridge, users will have the option to take a hiking trail spur into the CNC's Idlewood property.

Acquisition of the Turner Street Bridge for trail use is a key element in the development of the PIR. The question remains whether the rehabilitation of this bridge for trail use will be less costly than the construction of a new bridge to create the necessary connection from Crafton and the City of Pittsburgh into the Borough of Carnegie. An engineering study has been performed to determine the costs that are associated with renovation of this structure. The analysis indicates that as much as \$40,000 will be necessary to convert the bridge to pedestrian use and enable CNC to obtain liability insurance for the structure.

The PIR through the Borough of Carnegie, with the exception of a short segment along Mansfield Boulevard, consists primarily of a shared roadway for the basic bicyclist. The PIR follows Arch St., James St., Broadway Ave., Mary St., the Mall Plaza, Kinney St., and 1<sup>st</sup> Ave. until it meets Carothers St. and the entrance to Scott Township. Although much of this route is for the basic bicyclist, it will most likely be used only as a transportation route by the advanced cyclist. This is because once the PIR leaves Carnegie, the majority of the PIR is recommended for the



*Mall Plaza – Carnegie  
(Segment 49-50)*





advanced cyclist only all the way through Bridgeville and into South Fayette Township.

An advanced cycling route spur following West Main Street is proposed to connect with Old Noblestown Road. This spur is in lieu of using the currently active railroad spur right-of-way leading to the terminus of the Panhandle Trail at Walkers Mill. Refer to the narrative of Map Sheet A for further description of the alternative routes connecting the Chartiers and Panhandle trails.

#### Map Sheet 5 Carnegie – Collier Township (Points 52 thru 62)

An alternative multi-use segment following the southern and eastern bank of the Chartiers trail would provide an alternative connection from location 54 to the Panhandle Trail. However, this route was not selected as part of the PIR for two reasons; it would require the acquisition of the former spur through an industrial complex, and it would require the rehabilitation of two existing railroad bridges to provide connections to both the Panhandle and Heidelberg Borough. The PIR continues through Scott Township via Carothers St. and joins Route 50 at location 56 before it enters the Borough of Heidelberg.



*Rt. 50 Heidelberg  
(Segment 57-59)*

Soon after the PIR enters Heidelberg, it utilizes the less congested and less steep roadway of Washington Street until it rejoins Route 50 at the southern end of the Borough. From this point, the valley becomes bottlenecked by the steep slopes along the eastern bank of Chartiers Creek and the man-made landform of the Interstate 79 right-of-way, leaving the only available route, other than the PUR, to follow Route 50.

This particular area of the study corridor would be best served by the acquisition of the PUR for a continuous trail connection due the physical features of the valley and congested roadways associated with the alternate shared roadway segments.

#### Map Sheet A – Panhandle Connection Alternatives (Points PH1 To PH2a)

This map exhibit shows the alternate routes considered viable to provide a connection between the Chartiers and the Panhandle Trail. The ultimate and best connection between the two would be to acquire and develop the existing continuation of the railroad right-of-way used by the Panhandle along Robinson





Run into Carnegie. The active use of this line serving a recycling center prohibits this potential at the time of this study.



*Undeveloped section of Panhandle Trail  
(Segment PH1-PH1a)*

Therefore, the analysis of the alternates suggested include the use of Old Noblestown Road as a shared roadway for the advanced cyclist, or a multi-use trail along the southern side of the railroad right-of-way combined with a shared roadway along Cluxton Road. This multi-use segment will require a shared right-of-way with the active railroad for approximately 1500 feet due to the adjacent hillside landform. Once the multi-use segment meets Cluxton Rd., the alternate route becomes a shared roadway until it connects to Old Noblestown

Road west of I-79. From this point, the only two physically possible routes that may provide a connection to the Chartiers through the I-79 right-of-way are via the existing and currently unavailable rail line or via Old Noblestown Road.

#### Map Sheet 6 Collier Township – Bridgeville (Points 63 thru 74)

The PIR continues through Collier Township along Route 50 as a shared roadway for the advanced cyclist. Alternate routes suggested on this exhibit may be considered as spurs to the main route connecting to significant natural and recreational areas on the northern and western flanks of I-79. A multi-use route along the Chartiers main channel would require land acquisition and the construction of up to three new bridges.

This alternative was not considered as part of the PIR due to the associated costs of such a development, the lack of directness for a continuous trail, and because the southern terminus is the reconnection to Route 50.



*Rt. 50 Bridgeville  
(Segment 66f-76)*

This is another location within the study corridor where the PUR is clearly the best choice for trail development due to the existing natural and man-made features.

#### Map Sheet 7 Bridgeville – Upper St. Clair Township (Points 73 thru 79)

The PIR continues to follow its only viable alternative through Bridgeville along Washington Rd./Route 50 as a shared roadway for the advanced bicyclist, as





determined by the existing conditions. It is not until the PIR meets Bursca Drive in South Fayette Township that an opportunity to develop a multi-use trail is feasible. The acquisition of an at-grade crossing and the construction of a new trail bridge as a natural extension of Bursca Rd. would take the PIR into Upper St. Clair Township's Chartiers Park.

After leaving Chartiers Park, the PIR would then pass through land owned by the Allegheny Land Trust and south to Mayview Road. Development of this multi-use trail could serve as one half of a significant loop system once the PUR is developed on the western bank of the creek. The alternate to this system, and perhaps the short-term solution for the advanced bicyclist only would be to continue along Washington Road.

Map Sheet 8 Upper St. Clair Township – Mayview State Hospital (Points 81 thru 88)

The PIR and PUR converge on the south side of Mayview Road. Signs could be installed at this location to direct trail users to Upper St. Clair's Boyce-Mayview Park atop the hills on the eastern bank of the Chartiers. The master plan for the Boyce Mayview Park identifies the PUR as a future rail trail.

The PIR will follow along the southern creek bank, make use of an existing service road, and stay to the east of the active rail line around the Mayview State Hospital adjacent to a wetland mitigation area constructed along the creek. Alternate shared roadway alternative is to utilize the existing ring road of the State Hospital to take advantage of the more gradual slopes and the lower traffic volumes as compared to Mayview Road. The PIR will rejoin Mayview Road south of the hospital where a parking area is proposed.

Map Sheet 9 - South Fayette Township to Allegheny/Washington County Line (Points 87 thru 90)

As the PIR continues south along Mayview Road into Washington County, other alternate segments converge onto Boyce Road. A continuation of the multi-use trail impacts private property and would require land acquisition to continue along the western creek bank crossing Boyce Road and continuing into Washington County. The advanced bicyclist shared roadway alternative following Boyce Road rejoins the PIR from the western hills and Washington Road at the Mayview Road intersection.

Map Sheet 10 - County Line to Valley Brook Country Club (Points 90 thru 103)

The natural character of the Chartiers Valley is significantly different in Washington County than it is in Bridgeville and areas to the north. Much of the land has been developed as residential or for private recreational use. The landscape is more hilly offering less opportunity for trail development on level



lands. The study area also becomes wedged between the I-79 and US 19 transportation corridors as it moves south towards Canonsburg Lake.

The PIR continues along Mayview Road as a shared roadway for the advanced bicyclist until it reaches Georgetown Road. The PIR then travels west approximately 1900 feet to an intersection where it turns left and continues south on Georgetown Road.

The alternative multi-use trail extending into Washington County from Boyce Road (91-98) would involve the construction of two new bridges; require a land acquisition through private property; and impact private residences to provide a connection to Georgetown Road. The continuation of this alternative involves a shared roadway on Georgetown Road and then Buckeye Road to parallel the creek between the PUR and the undeveloped alignment of the Montour Trail. To avoid the knob landform and forcing the alignment to go through the railroad tunnel found along the PUR, the proposed alternative is to acquire an at-grade crossing of the Penn Ohio Central Line (future PUR) and construct a multi-use trail around the perimeter of the private Valley Brook Country Club golf course within the oxbow curve of the creek. A new bridge (101-102) will be necessary to continue the trail southward.



*Montour Railroad Bridge over P&OC  
(Point 100)*

The PIR continues on Georgetown Road until it intersects Hahn Road and the Montour Trail alignment. This is the only location where the Montour and the proposed PIR intersect at the same grade. The Montour alignment passes overhead of the PUR in an area that contains three bridges and two tunnels. A ramp system connection may be possible in the area north of the Greer Tunnel (PUR) to provide a link between the Montour and the PUR (near pt. 100).

#### Map Sheets 11 & 12 - Montour Trail connection to Canonsburg Lake (Points 103 to 112)

Traveling eastward on Hahn Road, the PIR crosses the PUR and heads south between the Chartiers and active railroad right-of-way as a multi-use trail. This will require land acquisition and potentially a short segment of having a rail with trail development. The reason this particular segment is valuable is because it connects to an abandoned rail spur that leads directly to Canonsburg Lake. This abandoned segment would also be part of the PUR once it is developed.





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The shorter term alternative to this multi-use segment would be to use a combination of Hahn Road and West McMurray road as a shared roadway for the basic bicyclist and complete the connection to Canonsburg Lake and the endpoint of the study corridor.





## 4.4 Greenway Development Plan



*"On the bank of Chartiers Creek...there lies a valley of unusual beauty. The creek meanders in a graceful curve along its western bounds, and precipitous hills, rising on the east, and stretching for a short distance to the north and south, at length change their course at each extremity of the range, and slope gradually to the margin of the water." – Thomas Cushing, 1889*

In the book *Greenways for America*, Charles Little describes a greenway as "a linear open space established along either a natural corridor, such as a riverfront, stream valley, or ridgeline, or overland along a railroad right-of-way converted to recreational use, a canal, scenic road, or other route." This description will form the basis of the identification of a working greenway within the Chartiers Creek watershed.

The following sections of this report will present a brief description of a greenway, specify the criteria used to establish the greenway boundaries, and provide an outline for a greenway management plan. The greenway boundary is illustrated on the attached alignment Maps 1 through 12. It should be noted that the greenway boundary indicated on these figures was defined based on criteria stated later in this section, and was meant only to be a guideline for developing the greenway management plan. This boundary was not meant to be exclusive, and should be modified by project stakeholders as necessary to meet conservation goals.

### Greenway Characteristics

The following definition of a greenway has been obtained from the National Park Service, Rivers and Trails Conservation Assistance Program publication *How Greenways Work* (NPS, 1992). Some basic concepts which are important for understanding the role which riparian corridors and greenways play in the landscape are as follows; "Nature is a series of interactions among humans, plants and animals, and our natural resources such as air, soil, and water. A healthy environment depends upon keeping those interactions intact. The challenge for conservationists is to ensure that we orient future development in a way that sustains natural processes. Protecting environmental corridors through



establishing and managing greenways represents one method (to be used in conjunction with other approaches) to safeguard vital ecological processes”.

Planners and conservation managers should therefore target the most important functions and design the greenway boundaries and management alternatives around those functions.

### **Greenway Development Criteria**

The greenway development criteria have been designated based on the topographic setting, geographic location and land use patterns of the Chartiers Creek valley. To provide a more focused approach to greenway management and how it applies to trail and waterway connectivity, the greenway management options have been limited to the Chartiers Creek main stem corridor. The greenway management plan may be expanded to include major tributaries based on additional funding and need.

The Rivers Conservation Plans that have been prepared for both the upper and lower Chartiers Creek watersheds may be used as guidance documents to address greenway issues which occur outside of this study area.



*Chartiers Creek Greenway – Kennedy Township*

The following criteria were applied to the study area to determine an appropriate greenway boundary.

- Current land use patterns and presence of contiguous vegetated or low impact use parcels.
- The presence of parks, hiking trails or other cultural landmarks or facilities.
- Proximity to the Chartiers Creek stream channel.
- Inclusion of property ownership including large parcels owned by institutions or public entities.
- Land which has been designated as critical or unique for the purposes of protecting threatened or endangered terrestrial or aquatic species.
- Land which may serve important ecological or water quality functions such as buffers or water treatment.



- 
- Parcels that have been zoned for conservation or other low impact land zoning classifications.
  - Previously designated greenways and conservation areas.

In this decade, both Allegheny County and the City of Pittsburgh have moved toward improving the natural/aesthetic aspects of local communities by completing and establishing programs to aid in developing greenways. These programs were started with the hope that the county and city would see the economic, environmental, recreation, transportation, and educational benefits of greenways (Skelly and Loy, 2001).

Coal was discovered in the Chartiers valley in the early 1800's and commercial extraction operations began in the 1880's. The extraction of this resource fueled the great Pittsburgh steel industry in the late 1800's and early 1900's. This activity lead to the industrial development of large portions of the Chartiers Creek floodplain. The development was primarily realized around the population centers of Carnegie, Heidelberg, and Bridgeville.

Although most of these parcels are not currently being used for heavy industrial purposes, many are still intact and are either abandoned or are being used for light industrial or commercial uses. Therefore, the broad floodplain areas surrounding Chartiers Creek, although primarily industrial, commercial, or urban, have been included within the greenway boundaries. They have, and will continue to be, an integral part of the Chartiers Creek valley landscape.

### **Greenway Development Options**

The greenway which is illustrated on the attached maps contains approximately 6800 acres of land. The majority of this land is comprised of steep, wooded hillsides and the floodplain of Chartiers Creek.

The following guidelines are offered to assist the project stakeholders in the building of a greenway network. The network must consist of private and public entities working together to establish protection for the resources that are located within the greenway boundary. Funding considerations are addressed in Section 5.0 of this document. The following components of the development of a greenway network should be considered as this plan is implemented. These concepts are currently being utilized by the PA DCNR in the implementation of its statewide greenway plan.





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- Purchase or acquire rights to land within the greenway boundaries.
  - Work with local, county, state and federal entities to establish a physical network of greenways throughout the region. These may include greenways of statewide significance, as well as local and regional greenway networks.
  - Promote the development of contiguous greenway plans by county and local governments as an integral part of their comprehensive planning and implementation efforts. Provide education and encouragement for them to link greenway concerns with programs that address sound land use, community revitalization, recreation needs and open space protection.
  - Develop a relationship with community-based health improvement partnerships to support and promote greenways with a physical activity focus.
  - Launch a campaign to promote alternative transportation. This will include development of a trails system that provides transportation alternatives to the automobile, and is part of a comprehensive multi-modal transportation system.
  - Create a greenways organizational framework which includes coordination of the activities of public agencies and greenway partners. This may include visioning meetings in which stakeholders may share ideas and resources. A charter may be drafted to solicit stakeholder buy-in and to clearly define their roles.
  - Develop and implement a greenways funding strategy that includes assembling public and private funds to assist in planning, building and maintaining greenways, and in meeting grant requirements for local matches.
  - Provide technical training and guidance by building the capacity of all greenways partners by creating a broad program of technical assistance and support. This can be accomplished by matching experts in water quality, biology, planning and recreation with the appropriate stakeholders. Information such as the Greenways Toolbox available from the PA DCNR may be disseminated to the public and other stakeholders.
  - Implement a greenways education and training program by incorporating greenways training into existing education programs for greenway partners. This may include establishing a strategy to integrate greenways into environment and ecology coursework for teachers and students in local school districts. A list of school districts within the study area is contained in Section 2.7 of this report.





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- Implement a plan that promotes the Chartiers Greenway and the potential for positive impacts on the local economy. This may be accomplished by showcasing economic opportunities and benefits of greenways.
  - Initiate a Greenways Volunteer Network with a corps of volunteers to maintain and promote local greenways through an 'Adopt-a-Greenway' or 'Adopt-a-Stream' program.

### **Strategies for Property Acquisition**

Property acquisition is the most effective method of preserving and conserving property within a greenway. This section presents numerous alternative methods for the acquisition of property for conservation purposes. The methods range from relatively simple methods such as receiving rights on a conservation easement to arranging a development rights transfer. Numerous parcels within the greenway have been acquired by non-profit groups such as the Chartiers Nature Conservancy, Allegheny Land Trust, and Scott Conservancy.

#### Fee Simple Acquisition

Although probably the most costly method of acquiring control of a property for conservation purposes, fee simple deed acquisition is the most effective means to practice land conservation. This method of ownership transfer is the most common means of real estate transfer and entails the purchase and recording of a deed in the conservation group's name. The conservation group may then conduct any conservation activities in accordance with local, state and federal regulation. There are, however, certain concerns that should be addressed if an conservation group or agency is going to complete fee simple transactions. Often, legal representation may be required to reduce liability both prior to, and after, any real estate transfer.

Prior to a fee simple transaction, some level of assessment should be done to identify property boundaries, title information, and environmental conditions at the site. Property boundaries should be surveyed prior to the purchase of a fee simple parcel. This will allow the prospective owner to accurately assess the amount of land that is being purchased and to ensure that it will meet the anticipated uses. The research of title information will ensure that there are not liens or restrictions associated with a parcel.

Significant liability could be inherited if a parcel contains environmental issues such as underground storage tanks, asbestos, or contaminated soil or groundwater. "Due diligence" investigation should be performed prior to a transaction to ensure these conditions do not exist at a site. The prospective landowner should also contact its insurance carrier to determine if additional coverage is required for structures or for coverage of visitors and workers at the site.



## Conservation Easements



*Typical Conservation Easement Sign*

A conservation easement is a recorded agreement by which landowners may voluntarily restrict the use of their land. These agreements may be executed as a condition of a local, state, or federal permit or as a condition of municipal zoning ordinances. Many times the easement is purchased by a third party and then relinquished to a conservation agency for ownership and maintenance. Mandatory conditions for easements include the requirement for an established, recognizable public benefit,

such as protecting rare species, public water supplies, streams or wetlands, or scenic vistas.

A conservation easement generally limits ecological impacts to the parcel but retains rights to use the land for other purposes. In some cases, the landowner retains the title to the property and the right to sell it. In the case of transfer of the property ownership, the easement conditions are recorded with the deed and transferred to the new owner.

If the easement is granted voluntarily, there are tax benefits which may be realized by the land owner. Local, state and federal tax codes should be researched to determine the possibility of receiving tax credit for donations of easement property. Generally, to be eligible for most of the above tax benefits, the agreement must be entered into with a qualified conservation organization, such as a land trust or a local unit of government. In addition, the terms of the easement must be perpetual and they must meet other IRS requirements. (1000FOM, 2003).

Sources of funding for the purchase of conservation easements may be obtained from state and federal conservation programs, private foundations, and from land developers. In Pennsylvania, many easements are obtained through the PA Code 25, Chapter 105, wetland and stream permitting process. Under this regulation, land owners who impact aquatic resources during a land development project must mitigate the impacts through preservation or creation of additional water bodies or wetlands. Often these developers purchase easements in which they may conduct their restoration or creation activities. Mechanisms exist for conservation groups or local agencies to express their desire to become involved in this process. By law, there is a 30-day public comment period upon permit receipt, and a 30 day comment period upon permit issuance. The status of these permit applications may be tracked in the PA Bulletin at [www.pabulletin.com](http://www.pabulletin.com).





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### Purchase of Development Rights

Purchase of Development Rights (PDRs) programs have been used successfully in many areas around the nation. They were pioneered in Suffolk County, New York in 1974 and have since been used across the nation to preserve an estimated 400,000 acres of farmland alone (1000FOM, 2003). With this process, a landowner voluntarily sells his or her rights to develop a parcel of land to a public agency or a charitable organization interested in natural resource conservation. The landowner retains all other ownership rights attached to the land, and a conservation easement is placed on the land and recorded on the title. The buyer (often a local unit of government) essentially purchases the right to develop the land and retires that right permanently, thereby assuring that development will not occur on that particular property. The landowner is generally compensated for the value of the right to develop the land through the calculation of the revenue lost by preserving the land minus the cost of the easement.

Although the implementation of a PDR program is very effective in protecting critical ecological or cultural resources, there is significant cost associated with the process. The cost of PDRs is directly related to the growth pressure that is being realized in a community. The resulting situation, in which the land that requires the most immediate protection is the most expensive to acquire, becomes a critical factor in determining the applicability of the PDRs.

One of the most successful PDR programs in the country is run by the Agriculture Preserve Board of Lancaster County, Pennsylvania. It has preserved over 23,500 acres of farmland since 1981 (1000FOM, 2003).

### Transfer Development Rights

Transfer of Development Rights (TDRs) are defined as the process by which development rights are transferred from one lot, parcel, or area of land in a sending district to another lot, parcel, or area of land in one or more receiving districts. Local governments are allowed great flexibility in designing a TDR program; they can establish conditions that they deem necessary and appropriate to achieve the purposes of the TDR program (Pace, 2003).

A TDR must be implemented by a municipality or other governmental agency that has legal jurisdiction over land use. To accomplish this, the local legislature identifies a "sending district" where land conservation is sought and a "receiving district" where development of property is desired and can be serviced properly. In many TDR programs, the zoning provisions applicable to the sending district are amended to reduce the density at which land can be developed. While losing their right to develop their properties at the formerly permitted densities, property owners in the sending district are awarded development rights.

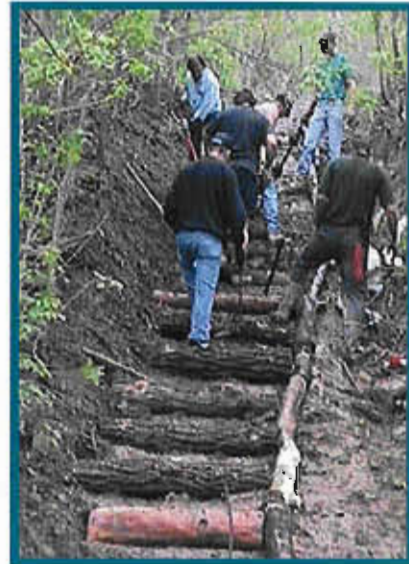




As with PDRs, TDR programs must establish a method of valuing the development rights that are to be transferred from the sending to the receiving district. Some communities establish development rights "banks" which purchase development rights from land owners in sending districts and sell them to landowners in receiving districts (Pace, 2003).

### **Strategies for Property Management**

The following guidelines have been developed to assist conservation and public agencies in the management of land parcels which are under their control. The goals of a property management plan should be tailored for the specific needs of the organization as they relate to the intended use of a property. For example, a group that has acquired a parcel to protect an endangered plant species would want to manage that parcel to limit access and reduce the potential for impacts to the critical habitat. In contrast, a parcel that was acquired along a stream for canoe access, or a trail for pedestrian and bicycle access, may need to be improved with infrastructures such as roads, and utilities to support amenities such as a comfort facility. The following goals, therefore, may be adopted for parcels on a site by site basis.



*Volunteers building hiking trail  
at CNC's Idlewood property*

### **Access Control**

Access control is necessary to prevent unwanted or unauthorized use of conservation or recreation land. Although an overall policy may be drafted by conservation groups to address multiple properties, it will be necessary to address some issues on a site by site basis. Typical unauthorized use of conservation properties include:

- Use of all terrain vehicles (ATVs)
- Dumping and littering
- Unlawful resource extraction (timbering)
- Vandalism

Measures to control these types of activities may include the installation and maintenance of physical controls and barriers as well as the drafting of formal policy and method of enforcement. The following measure may be utilized to control unauthorized trespass.





- Properly post boundary signs and diligently prosecute violators;
- Indicate on a map, and on the specific property, which areas may be open for certain uses;
- Have a set of carefully thought-out rules that may be enforced if necessary;
- Encourage users to report damage to property and trespass signs or unauthorized uses;
- Install gates and fences as necessary to control access;
- Initiate a land stewardship plan as specified in the following section.



*Interpretive Sign*

Some uses, although not illegal or unauthorized, may require control because they are not conducive to the management goals of a property. These uses may include excessive trail use or shortcutting, damage to sensitive plant or animal habitats, uncontrolled pets and incompatible uses. These types of management issues are generally addressed through physical means such as informational signage or trail markers or barriers. Educational measures may also be used to control uses at the property. Educational measures include conducting nature walks and hikes during which the public may be given instruction on proper uses of the land.

Natural resource interpretation may be used to provide guidance on acceptable conduct and reduce impacts to sensitive habitats. Self guided tours are commonly used to control foot traffic and direct users through a predetermined route.

### Land Stewardship

A stewardship program is essential for any conservation group or agency that owns land or holds an easement for which it is responsible. The program should be crafted to meet the management needs of the agency while protecting the public and ensuring that the management goals of a parcel are met. A stewardship plan should provide the means for consistent and effective monitoring and stewardship of protected properties over time. The plan itself can vary depending upon many factors, such as the type of property (land or structure), how it is protected (fee owned or easement) and the grantee itself (town or not-for-profit organization) (LCHIP, 2003).

A typical land stewardship monitoring report is attached to this report in Appendix B and should be an integral component of a land stewardship plan. The land stewardship monitoring plan may be implemented by a committee or stewardship coordinator that is appointed by the organization.





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## Create or Enhance Wildlife Habitat

Regardless of the intended use of a parcel, most land managers will agree that it is important to enhance wildlife habitat on land that they own. It is not certain, however, which species should be targeted and how to implement and monitor a habitat management plan. The following guidelines have been prepared to provide the starting point for the management of properties for wildlife within the Chartiers Creek watershed.

### *Step 1 – Determine Objectives*

The objectives of a wildlife management plan must be defined at the onset of the planning process. Some general goals for a wildlife management plan are as follows:

- Increase wildlife viewing opportunities
- Enhance aesthetic value
- Create nature trails for hiking
- Enhance property values
- Attract the public to a property
- Provide educational opportunities

### *Step 2 – Assess Physical Conditions*

Based on the objective which is chosen for a property, a land manager should then assess the physical characteristics of a property or greenway segment to determine the conditions which may influence the types of wildlife that are targeted. Physical characteristics which should be assessed include:

- Hydrologic conditions
- Soil types
- Streams or wetlands
- Vegetative habitat types
- Existing wildlife patterns and occurrences

A GIS could be used at this point to compile the results of the assessment and analyze trends in the physical characteristics.

### *Step 3 – Identify Target Species*

As objectives and physical conditions are determined, target species may be identified. Typical species that may be targeted in southwestern Pennsylvania include the following:



- Bats
- Frogs, Turtles, and Snakes
- Songbirds
- Bluebirds
- Grassland Birds
- Woodland Birds
- Wetland Birds
- Waterfowl
- Wild Turkeys
- Ruffed Grouse
- Woodcock
- Pheasants
- Bobwhite Quail
- Rabbits
- Squirrels
- Black Bears
- White-tailed Deer



*Wood Duck (Aix sponsa)*

### *Step 3 – Identify Target Habitat Types*

- Conifers
- Hardwoods
- Forest Openings
- Wetlands
- Streams and Rivers
- Grasslands
- Cropland



*Emergent Wetland  
Habitat*

### *Step 4 – Create Habitat Management Plan*

Prepare a habitat management plan that will specify the necessary measures that are required to meet the goals developed in the planning phases. The plan may be prepared with the assistance of a consulting forester or biologist. The plan may be forwarded to natural resource agencies such as the Pennsylvania Game Commission or the Pennsylvania Fish and Boat Commission for review and comment. The plan, at a minimum, should contain the following items:

- A statement which clearly defines the plans goals and objectives;
- Mapping or aerial photography which clearly defines the location and extent of proposed features and measures;
- A list of plant species to be installed and their availability;
- Procedures for installation of feature and plants including specifications;
- A list of necessary permits (i.e., Chapter 105 for waterway work);
- A maintenance or monitoring plan.



### Step 5 - Implement Plan

Depending on the scale, implementation of a habitat management plan may require significant resources. It may be necessary to contract portions of the work; or, if the project is smaller, a corps of volunteers may be adequate to perform the necessary tasks. It is recommended that the forester or biologist that is involved in the development of the plan maintain that involvement through the implementation phases.

Local contractors who are capable of implementing larger habitat creation or restoration projects include the following:

*Meadville Land Company, Inc.*  
Contact: Robin Ernst  
9006 Mercer Pike Meadville, PA 16335-9299  
800-873-3321  
814-336-2404  
Fax: 814-336-5191

*Ecological Restoration, Inc.*  
Contact: Dave Hails  
RR#2, Box 135BB  
Silvis Road  
Export, PA 15632  
(412) 334-3790  
Fax: (724) 334-3781



*Bat Box Installation by  
CEC Scientist*





### Step 6 - Monitor Plan

A habitat monitoring plan should be implemented to determine the success of the project and to determine the need for additional measures. Monitoring of habitat improvements may be integrated with the land stewardship program discussed previously. Monitoring should be continued for a minimum of three years.

### **Water Trails**

Also called blue routes or blueways, water trails are primarily recreational routes through scenic areas where rivers form the corridor and are used for trips by canoe, kayak, or raft. In some cases, protected areas of wilderness habitat along shorelines enhance the water trail effect. The project team has identified the following issues to be addressed during master planning which will ensure the continued popularity of the Chartiers Creek water trail.

### Watercraft Recreation

CNC has organized an annual two day canoe sojourn on both the upper and lower portions of Chartiers Creek. This event, which is held annually in May, is sponsored by the DCNR. The purpose of the sojourn is to highlight the recreational value of Pennsylvania's rivers and streams, and the natural beauty of the valley. The first day of the sojourn the paddlers canoe the upper reach from Canonsburg to Bridgeville. CNC includes an explanation of the route, a shuttle between launch and take-out sites, and an overnight campout at Bridgeville Park, with a bonfire (Maps 7-12).



*Chartiers Creek River Sojourn Paddlers (2002)*

The second day the paddlers float from Bridgeville to the mouth of Chartiers Creek near the Ohio River (Maps 1-7). The highlights of this trip include sightings of great blue heron (*Ardea herodias*), various duck species and spring wildflowers. The sojourn is typically enjoyed by over 20 individuals. Water levels are usually elevated and the paddlers have time to take in the scenery and make new friends. Also, the recreational guide group Outside Adventures, located in Tarentum, PA, is currently offering kayak lessons on both the upper and lower sections of Chartiers Creek.





## Fishing

Recreational fishing is an important component of Pennsylvania's economy. According to the Pennsylvania Fish and Boat Commission (PAFBC, 2003), recreational fishing is a \$1.34 billion/year business that supports 16,677 jobs and provides \$357 million in employee earnings, \$49 million in state sales and income taxes, and \$37 million in federal income taxes each year. Pennsylvania waterways are the destination for nearly 21 million fishing trips each year. Some 11.1 million of those trips are for trout (coldwater game fish), the most of any state in the nation. In addition, anglers take nearly 7.2 million trips in Pennsylvania for bass (warmwater game fish), ranking the state third in the nation.

On an annual basis, more fishing licenses are sold in Allegheny County (68,826 licenses were sold in 2001, corresponding to 516,110 trips for bass/year) than any other county in Pennsylvania (PAFBC, 2003). Given the number of local anglers and Chartiers Creek's improving water quality and indigenous warmwater game fish species, it seems likely that this stream could provide for a significant recreational warmwater fishery. Such a fishery could be developed concurrently with continuing improvements in stream water quality. The initial project for the development of such a warmwater fishery could involve providing access areas for recreational anglers and water quality and habitat improvements.

Over the years, the main stem of Chartiers Creek, from its confluence with Little Chartiers Creek in Canonsburg, downstream to its confluence with the Ohio River in McKees Rocks, is regularly issued a fish consumption advisory by the Pennsylvania Department of Environmental Protection. Under the most recent advisory (PADEP, 2003), PADEP recommends that largemouth bass (*Micropterus salmoides*) only be consumed once a month due to possible PCB and chlordane contamination. Due to the environmental persistence and longevity of these organic pollutants, fish consumption advisories issued on account of PCBs or chlordane are usually indicative of some type of historical water quality impairment for their respective water body. CEC biologists, on the other hand, have documented several pollution sensitive fish species, including longnose dace (*Rhinichthys cataractae*), northern hogsucker (*Hypentelium nigricans*), sand shiner (*Notropis stramineus*), greenside darter (*Etheostoma blennioides*), and banded darter (*Etheostoma zonale*) inhabiting the main stem of Chartiers Creek during their fish community surveys. The occurrence of these sensitive fish species within Chartiers Creek may signify recent improvements in overall stream water quality.

Much of recreational spending is tied to fish and wildlife that both require high water quality and healthy habitats. Activities include hunting, fishing, and viewing wildlife (bird watching, photography, etc.). A healthier Chartiers Creek and tributaries would promote more fishing activity, and potentially generate substantial revenue in nearby communities.



## Access

Numerous unimproved access points are currently being used by recreational boaters, fishermen and paddling enthusiasts. The locations of these launches, along with the location of additional launches, are presented on the attached maps. The following criteria should be applied to properties under consideration for boat launches.

- There must be a demonstrated community need and benefit for the project.
- The project must be consistent with the mission of the CNC and other conservation groups and agencies.
- Personnel must be available to assist with general upkeep of the facility.
- Topography must be appropriate for development of a boat or canoe launching facility.
- If no improvements are planned, the land must be suitable for foot passage to the shoreline.

The property inventory of potential easement and acquisition contained in Section 5.2 contains properties that may be utilized for water access. Development of these properties may include the following items:

- Waste receptacles
- Parking
- Signage
- Comfort facilities
- Ramps / steps
- Habitat enhancement for fishing
- Access for Fishing



*Potential Launch Area - Heidelberg*

It may be necessary to make arrangements with the local municipalities to assist with maintenance activities. These areas should also be patrolled by police or trail security volunteers. Users should be encouraged to report maintenance, safety, or security issues. Permits for stream or floodplain encroachments may be necessary and require services of engineers or scientists.





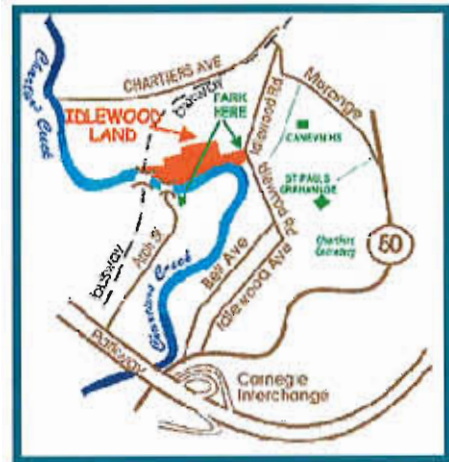
## 4.5 Key Properties Under Ownership

The following properties are currently held by local conservation groups and land trusts. These properties form the basis for local conservation effort and are integral in the introduction of the public to the natural resources of the Chartiers Creek valley. These properties were obtained through the generosity and hard work of local citizens, and are a reflection of the interest and spirit of those involved in the local conservation efforts.

### Chartiers Nature Conservancy Idlewood Property

CNC obtained these 9 acres of wooded slope land in June of 1998. The land is directly adjacent to the northern side of Chartiers Creek in the 28th ward of the City of Pittsburgh. It is between Idlewood Road and the new Airport Busway bridge.

Pittsburgh City Councilman Alan Hertzberg has been a strong supporter of the CNC and its efforts to develop a Greenway that stretches along scenic Chartiers Creek from the Ohio River to Washington County.

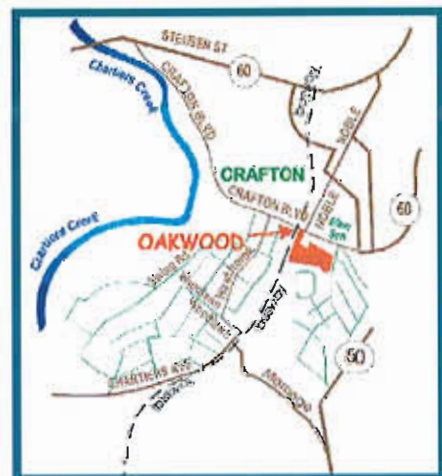


The first link in the Greenway is a walking, hiking, and biking trail on the donated land. This portion of the Chartiers Creek valley is stunningly beautiful and scenic, especially given its proximity to a major city. The CNC is tirelessly pursuing every avenue available to preserve and promote the conservation of this natural treasure.

### Chartiers Nature Conservancy Oakwood Property

The Oakwood Valley property consists of 4.9 acres and is located across the street from Crafton Elementary School, bordered by Crafton Boulevard, the new Port Authority busway, and the Oakwood Bridge. Conservancy volunteers plant and maintain a flowerbed at its very visible entrance.

Bell's Run is a tributary of Chartiers Creek, flows directly through the property creating a small flat floodplain surrounded by steep, heavily wooded hillsides. According to former Conservancy Board Member and local





historian Betsy Martin, a trolley company dammed the stream and flooded the Oakwood Valley in the early part of this century to create a lake meant to encourage riders to use the trolley line on summer days and weekends. Bell's Run is now an impaired stream because of water quality problems.

A principal goal for CNC in the Oakwood Valley will be to identify and eliminate the sources of pollution to Bell's Run and thereby reduce the amount of pollution carried into Chartiers Creek. The Laurel Foundation and Katherine Mabis McKenna Foundation provided funding for this land acquisition.

#### Chartiers Nature Conservancy Ingram Property



The Ingram property consists of 8.22 acres of gently sloping woodlands bordered by Ingram Avenue, North Duane Avenue, and the Ryder Truck facility. This property, purchased from the Urban Redevelopment Authority of Pittsburgh in September of 1999, has been under considerable development pressure in the past decade.

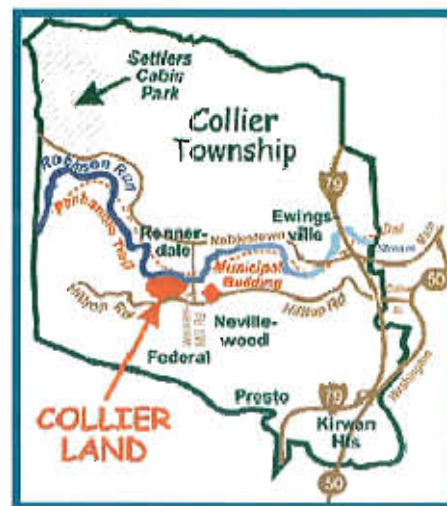
The property will remain an undeveloped green space dedicated to the enjoyment of the public through passive recreation like hiking and biking.

The acquisition was financed with land acquisition grants from the Laurel Foundation and the Katherine Mabis McKenna Foundation.

#### Chartiers Nature Conservancy Collier Land

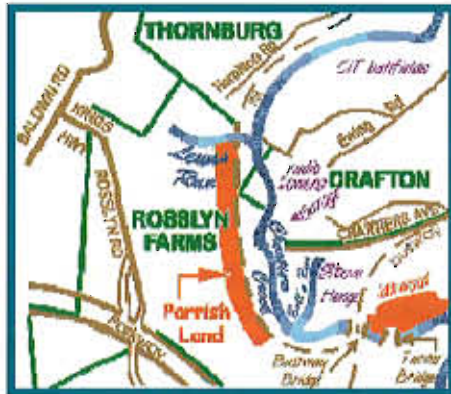
Located on Hilltop Road in Collier Township near the Nevillewood community, the 42.5 acre Collier property joined the CNC's land holdings on June 17, 2002. Criss-crossed with paths and wide grass trails with gentle slopes, this unique parcel of land is bordered by Robinson Run, a tributary of Chartiers Creek. The Panhandle Trail is just on the other side.

Acquisition of this property was made possible in large part by a \$275,000 Keystone Land Acquisition Grant from the Pennsylvania Department of Conservation and Natural Resources (PA DCNR). It marks the culmination of a three year effort to secure funding for the \$550,000 purchase.



The Hopwood Foundation, Environmental Trust, Eden Hall and the Scaife Family Foundation also contributed financially and the conservancy had the support of State Senator Tim Murphy, State Representative John Pippy and the Collier Township Commissioners.

#### Chartiers Nature Conservancy Parrish Property



The CNC extends thanks to Sally Ford for her generous donation of four acres in Rosslyn Farms. Running parallel with Chartiers Creek, the two long, narrow wooded parcels will now always remain as a riparian buffer.

The 2,600 foot long by 165 foot wide property, with a value of over \$110,000, is situated between the hillside and railroad tracks and is located directly across the creek from the confluence of Bells Run and Chartiers Creek. The land extends for a half

mile along the hillside to Lewis Run, approximately across from Crafton Borough's Maintenance Facility.

#### Allegheny Land Trust - Wingfield Pines Property

This property is comprised of 80 acres in Upper St. Clair and South Fayette Townships and contains a section of the proposed recreational trail (See Map 8). Formerly the Wingfield Pines Golf and Swim Club, the property is now habitat for a wide range of wildlife including, fox, coyote, great horned owl, beaver, great blue heron. Future projects include treating abandoned mine drainage, rehabilitating buildings and perhaps providing educational programming. The success of this project is a result of a partnership with the USC Citizens for Land Stewardship.

#### Scott Conservancy - Kumpf Land

Thanks to the generosity of Marilyn and Bill Kumpf, one of the Conservancy's long time goals, land acquisition, is a reality. The Kumpfs donated two parcels of land on the cul-de-sac at the end of Jaycee Drive to the Scott Conservancy. The transfer was completed on August 2, 2001. CNC thanks the Kumpfs for this donation, which amounts to 4-5 acres and is adjacent to the Kane Woods. Bill Kumpf says he always enjoyed walking in the area and had a special rock that he liked to sit on to enjoy the Woods. He has now made it possible for others to enjoy that area as well.





## **5.0 Implementation Strategies**

### **5.1 Phased Implementation Plan**

The implementation of any trail system is dependent upon many factors including; ownership of land; availability of funding; cooperation of partners and the ability to commit to long-term maintenance of the trail. The fact that the proposed Chartiers Trail system traverses multiple municipalities could be viewed as an obstacle or strength to the logical implementation of the trail system. Many funding entities view inter-municipal cooperation as an important and positive factor that is required for the successful outcome of these types of projects.

In deciding which areas of the trail system should be initially targeted for development, there are key strategic decisions that trail stakeholders should consider. Therefore, the following general recommendations are provided for a basis to initiate the development process. A phased implementation plan, which provides greater detail of development options, is provided for the PIR. A summary of estimated development costs, based on the conceptual development alternatives, is also provided.

#### **General Recommendations**

##### Perform an Economic Study of the P&OC Railroad right-of-way.

The PIR as described in Section 4.3 is an interim solution to reaching the long-term goal of a continuous multi-use trail through the lower Chartiers Creek valley. The long-term goal is to acquire and develop the PUR or the P&OC railroad right-of-way as this multi-use trail.

The projected timeframe for development of the alternatives may vary greatly depending on funding, planning, and developmental issues. Realistically, full development of the PIR, or short-term solution could take as long as 10 years. To better define the timeframe for development of the alternative alignments, it is recommended that trial stakeholders perform an economic study of the PUR's active railroad right-of-way. This study should determine:

- a) the right-of-way's value in terms of economic contribution to the community;
- b) the line's profitability with respect to the annual operation and maintenance costs;
- c) potential for expanded freight rail or commuter service on this line;
- d) an appraised value and potential acquisition price for the right-of-way;



- 
- e) comparison of estimated right-of-way value to other regional railroad right-of-way acquisitions.

Part of this same study should show the economic, recreation, and transportation benefits to having this line developed as a multi-use trail. Based upon the amount of people living in the primary service area, and general lack of passive recreational facilities and/or safe areas available to ride a bike within this valley, this rail line has enormous potential as both a recreation entity and a non-motorized transportation route.

#### Initiate Contact with the Penn Ohio Central Railroad

Establishing a good working relationship with the P&OC Railroad owners will help trail stakeholders monitor the status of the active line and potentially help ensure a future opportunity to purchase or acquire the right-of-way for trail use.

For example, the railroad's owners have expressed concerns over trespassing and liability insurance costs. Trail stakeholders can work with the railroad to identify and/or assist with measures that may prevent this trespassing, or to assist with the right-of-way's maintenance operations, such as performing an annual trash cleanup. In any case, a good working relationship will be beneficial to both parties, the condition of right-of-way, and ultimately the residents of the Chartiers Valley. Stakeholders may also request an agreement of first refusal from the P&OC Railroad.

#### Coordinate with PennDOT & Local Municipalities

A significant portion of the PIR utilizes shared roadways for bicyclists. Prior to officially designating any of these roadways as bicycle routes, both the local PennDOT district and the municipalities through which these routes pass should be consulted. All roadways within the State of Pennsylvania, with the exception of limited access highways, permit use by bicyclists, however designating a roadway as a bicycle route requires the following:

- In its present condition the roadway is bicycle compatible. The primary considerations in determining bicycle compatibility are suitable traffic volumes, curb lane widths, and traffic speeds.
- The road has no significant barriers and traffic does not regularly exceed posted speed limits.
- No significant bicycle hazards are present or if the hazards cannot be removed, warning signage is posted as per the Manual of Uniform Traffic Control Devices (MUTCD).
- The bicycle route is designated as one segment within an interconnected system of bicycle facilities (PennDOT, 1996).







PennDOT has the potential to assist with the planning and design of these routes and to assist with improvements through their 'Betterments' program which may be used for things such as shoulder improvements, bicycle safe drainage grates and signage.

The local municipalities should also approve and support these routes prior to designation to help ensure and perhaps assist with the monitoring and maintenance of facilities.

### **Phased Recommendations**

Each of the following phases includes action items listed in general order of priority to develop specified sections of the PIR. These development phases were selected according to their proximities to population centers, anticipated use, and potential value once combined with future development of the PUR. These phasing plans should be combined with the aforementioned recommendations regarding shared roadways and coordination with PennDOT.

#### Phase 1 – Northern Section - McKees Rocks to Rosslyn Farms/Carnegie (Points 01 to 48)

This particular section of the PIR will serve the most people within the corridor and provide the potential for two significant loop systems if fully implemented and combined with the PUR. The first and northernmost loop would be from the south end of Scully Yard (Pt. 19) to the Rutgers Road at-grade crossing with the PUR (west of Pt. 33). The second loop, or bottom half of a potential figure eight system would be from Rutgers Rd. south to the Turner Street Bridge (Pt. 39).

Action items for this phase include:

1. Properties Assessment - Investigate land acquisition possibilities for Multi-use segments. These segments include:
  - The Duquesne Light property adjacent to Stanhope Street in McKees Rocks;
  - The Stanhope Street Extension; and,
  - The CIT ball fields area in Crafton Borough.

With the exception of the Duquesne Light parcel, acquisition of these lands for multi-use trail development is essential to the continuity of the PIR system.

2. Existing Bridges - Secure the safety and use of the existing bridges associated with the PIR. Both the existing suspension footbridge in McKees Rocks and the Turner Street Bridge in Rosslyn Farms should be inspected. These inspections should include recommendations for





- 
- TRAIL SEPARATION  
2' WIDTH, BELGIAN BLOCK
- PEDESTRIAN TRAIL  
6' WIDTH  
ASPHALT PAVED  
HANDICAPPED  
ACCESSIBLE
- EXISTING WOODS  
Woods ADJ TO CHARTERS CREEK
- CHARTERS CREEK
- BIKEWAY  
EXISTING CONCRETE PAVING  
8' WIDTH
- ROADWAY SEPARATION  
REMOVE 6' SECTION OF EXISTING  
CONCRETE PAVING; REPLACE WITH  
LAWN BERM, TREE PLANTING
- EXISTING
- CONCRETE
- SKATING LANE  
EXISTING CONCRETE PAVING  
12' WIDTH  
INCL. WALKING, BICYCLING, ETC.
- PARKING
- EXISTING MEDIAN  
PROPOSED SHOULDER  
BUFFER
- PROPOSED SHOULDER BUFFER
- EXISTING ASPHALT
- EXISTING CONCRETE PAVING  
PROPOSED 2 WAY CIRCULATION
- 34'
- 6'
- 2'
- 8'
- 12'
- 2'
- SECTION
- 11-25

### Industrial Highway Pedestrian Conversion Concept

- Phase 2 – Southern Section - Chartiers Park to Mayview State Hospital thru ALT lands (Points 77 to 82).



Action items for this phase include:

- 
1. Coordination with the Allegheny Land Trust (ALT) – The CNC should discuss the possibility of developing a multi-use trail through this beautiful natural area. This trail in combination has the potential to be not only recreational, but also educational to describe the many natural features of multiple ecosystems found within this land. The CNC in conjunction with the ALT could develop a master plan for the development of a trail or trail network system that could stand alone as a destination, as well as be part of the Lower Chartiers Valley's Greenway Trail System.
  2. South Fayette Township / Chartiers Park Bridge – To provide a direct connection between South Fayette and the Upper St. Clair Chartiers Park, a new bridge crossing is proposed as a natural extension of Bursca Drive. In the event this bridge is constructed prior to the PUR being developed, an at-grade crossing will need to be acquired to allow for people to safely cross the active railroad at this location. A feasibility study for this bridge could be completed in conjunction with a master plan for the ALT tract and trail system.

Subsequent phases of the PIR implementation will be subjected to inevitable changing future conditions including potential PUR development. A conservative estimate to implement these two phases is 5 to 10 years. Trail stakeholders could advance both Phase 1 and Phase 2 concurrently if the resources are available to do so.



## Estimated Development Costs

To assist trail shareholders in the initial process of securing trail funding for development, the project team prepared an estimate of costs. The figures were calculated based on the unit costs assigned to each individual segment that is illustrated on the alignment maps. Supporting documentation for segment costs are attached in the Estimated Development Cost Matrix found in the Tables Section of the report. Unit cost justification, accompanied by typical cross sections is contained in the previous Section 4.3.

**Table 5.1.1**  
**Estimated Development Cost Summary**  
**Preferred Intermediate Route**

| Item                      | Cost         |
|---------------------------|--------------|
| Master Planning           | \$75,000     |
| Land Acquisition/Easement | \$500,000    |
| Engineering/Permitting    | \$250,000    |
| Construction              | \$1,820,830  |
| Total                     | \$2,645,8930 |

It should be noted that there is significant uncertainty associated with cost estimates which are developed at this point in the planning process. These uncertainties include property acquisition issues as well as engineering and permitting constraints. Engineering constraints which may be encountered include structural or geotechnical issues associated with the construction of bridges and design issues associated with trail grades and surfacing. Permitting associated with stream crossings and highway use requires coordination with state agencies and may also escalate development costs. The estimated costs, therefore, should be used only to determine level of funding estimates and for the purposes of developing long-term budget parameters.





## 5.2 Land Acquisition and Easements

Utilizing the property database review and lot and block information that is attached to this report in Appendices G and H. A review of the project greenway boundary and trail alignment GIS mapping was performed to determine which properties should become part of the greenway and trail system. Property ownership data and parcel size were then obtained from the Allegheny County Tax Assessment website.

**Table 5.2.1**  
**Key Properties for Trail and Greenway Development**

| Owner                             | Lot and Block No.    | Size (Ac.) | Use     | Municipality  |
|-----------------------------------|----------------------|------------|---------|---------------|
| Cieslak                           | 0072-N-00180-0000-00 | 25         | C       |               |
| Duncan                            | 0068-F-00250-0000-00 | >10        | C       |               |
| Captline                          | 0108-D-00225-0000-00 | 25         | C       | Kennedy       |
| Trebilock                         | 0105-L-00102-0000-00 | >10        | C       | Rosslyn Farms |
| Orr                               | 0105-L-00200-0000-00 | 9          | C       | Rosslyn Farms |
| Idlewood Road<br>Property Company | 0067-B-00001-0000-00 | 12         | C       | Pittsburgh    |
| Borough of<br>Thornburg           | 0105-G-00300-0000-00 | 13         | C       | Rosslyn Farms |
| Buncher Family<br>Foundation      | 0067-A-00010-0000-00 | 2          | C       | Pittsburgh    |
| Pasquale                          | 0067-E-00250-0000-00 | 1          | C/T/N   | Pittsburgh    |
| Pasquale                          | 0104-D-00020-0000-00 | 5          | C/ T/ N | Pittsburgh    |

C= Conservation  
T=Trail Development  
N=Nature / Habitat

The properties that are listed above may be investigated for a variety of uses as they relate to the implementation of the greenway and trail plan. Property owners may be approached to accommodate simple habitat improvements such as bird or bat boxes, or control of invasive plant species. Others may be approached for easement or acquisition of necessary connections for trail segments or waterway access. These properties will require that trail and developers enter into an agreement and compensate the landowner for their land. Possible intended uses have been noted in the above table.



### **5.3 Potential Funding Sources**

The following sources of funding have been identified by the project team as potential sponsors of the concepts presented in this plan. A matrix which contains detailed contact information for each of the groups mentioned in the following narrative is attached in this report.

#### **TEA-21 Enhancements**

The Transportation Equity Act for the 21<sup>st</sup> Century (TEA-21) is a federal program that funds transportation related projects and is a direct successor to the 1991 Intermodal Surface Transportation Efficiency Act (ISTEA). The bill allocated approximately \$217 billion nationwide over six years and includes funding for trails, and rehabilitation of historic transportation assets.



In Pennsylvania, the Department of Transportation (PennDOT) administers several TEA-21 Enhancements. In the most recent funding cycle, PennDOT allocated its remaining funds in the spring of 2002. New federal legislation is due to consider reauthorization of a similar transportation bill in 2003-4. At the earliest, applications will be due in fall of 2003 with grants awarded in the spring of 2004. The Chartiers Nature Conservancy should position itself to apply for this funding when it again becomes available.

Typically, the non-federal match is required to be 20% of the total project. A strategy preferred by PennDOT is for the local partner to prepare construction documents and obtain necessary environmental clearances, property control documents and utility relocations plans as "preconstruction" tasks as the local match so that the project is ready for construction bidding when TEA-21 funding is released. The costs to prepare these documents can be the non-federal match to the TEA-21 funds, without the need to be exactly 20% if all needed documentation can be completed for less.

#### **Surface Transportation Program (STP):**

Eligible projects include the construction of bicycle transportation facilities; construction of pedestrian walkways; bicycle safety brochures, maps, public service announcements. Any bicycle project must be primarily a transportation project; STP projects should encourage desirable traffic patterns; and should sensitize people to environmental and social concerns.





## **DCNR – Community Conservation Partnerships Program (C2P2)**

This program is a combination of several funding sources and grant programs: the Commonwealth's Keystone Recreation, Park and Conservation Fund (Key 93), the Environmental Stewardship and Watershed Protection Act (Growing Greener), and Act 68 Snowmobile and ATV Trails Fund. The Program also includes federal funding from the Land and Water Conservation Fund (LWCF) and the Recreational Trails component of the Transportation Equity Act for the Twenty-first Century (TEA-21).

The C2P2 contains the following grant components: Community Recreation, Land Trusts, Rails-to-Trails, Rivers Conservation, Snowmobile/ATV, Heritage Parks, LWCF and Recreational Trails. Except for the Heritage Parks grants, all other components have been combined into one annual application cycle (generally late summer/early fall), and use a single application format and process with one grant manual and one set of application forms. Applications selected for federal LWCF funding require some supplemental information to enable submission of the application to the National Park Service (NPS). Generally, all components require a match, usually 50% of cash or in-kind contributions.

### **DCNR - Pennsylvania Recreational Trails Program**

The Pennsylvania Recreational Trails Program (PRTP) provides funds to develop and maintain recreational trails and trail related facilities for motorized and non-motorized recreational trail use. Federal funding for the program is through the Federal Highway Administration (FHWA) and the Transportation Equity Act for the 21st Century (TEA 21).

In Pennsylvania, the Recreational Trails Program is administered by the Department of Conservation & Natural Resources (DCNR), Bureau of Recreation & Conservation (BRC) in consultation with the Pennsylvania Recreational Trails Advisory Board (PARTAB), which is composed of both motorized and non motorized recreational trail users.

For this grant round the Department has approximately \$1 million available for grants. This funding must be distributed among motorized, nonmotorized, and diverse trail use, as follows:

- 40% minimum for diverse trail use
- 30% minimum for motorized recreation
- 30% minimum for non-motorized recreation





Match requirements for Pennsylvania Recreational Trails Program Grants are 80% grant money, up to a maximum of \$100,000, and 20% project applicant money.

"Soft match" (credit for donations of funds, materials, services, or new right-of-way) is permitted from any project sponsor, whether a private organization or public agency. Eligible applicants include federal and state agencies, local governments and private organizations.

Eligible project categories are:

- maintenance and restoration of existing recreational trails;
- development and rehabilitation of trailside and trailhead facilities and trail linkages;
- purchase and lease of recreational trail construction and maintenance equipment;
- construction of new recreational trails (with restrictions on new trails on Federal land);
- acquisition of easements or property for recreational trails or recreational trail corridors.

The Commonwealth may also use up to 5 percent of its funds for the operation of educational programs to promote safety and environmental protection related to the use of recreational trails.

The Department will also give consideration to projects that provide for the redesign, reconstruction, non-routine maintenance, or relocation of recreational trails to benefit the natural environment.

Project sponsors are encouraged to enter into contracts and cooperative agreements with qualified youth conservation or service corps to perform trail construction and maintenance.

Further information on obtaining DCNR Grants can be obtained at:

<http://www.dcnr.state.pa.us/grants.htm>

### **KEYstone Community Grant Program**

A bond issue approved in a statewide referendum initially funded this program. Perennial funding is through a dedicated percentage of the statewide real estate transfer tax. Funding from the program is dedicated toward recreation, environmental and cultural heritage resources throughout the state. Several agencies distribute funds through competitive grants, including; the PA Fish and Boat Commission, PA Historic and Museum Commission, and the PA DCNR. Trail projects are eligible in the DCNR funding application rounds annually







(usually mid-October). State funds can be used for discrete projects or as a match to federal funds. DCNR requires a 50-50 match (cash or in kind) to its grant awards for trails.

### **Allegheny Regional Asset District (ARAD)**

The mission of the District is to support and finance regional assets in the areas of libraries, parks and recreation, cultural, sports and civic facilities and programs. In addition to providing grants from half of the proceeds of the Allegheny County Sales and Use Tax, the District works with citizen boards and government officials monitoring the assets for effective operation and development.

Applicants for ARAD funding must be governmental entities or be non-profit, tax-exempt corporations, duly organized and in good standing under Pennsylvania laws, and provide programming or services of a regional nature in Allegheny County.

The applicant must be the owner and operator of the program or facility for which funds are sought. No "pass through" or fiscal agent arrangements are permitted. Non-profit organizations must have 501(c)3 status at the time of application. Applicants must have a board adopted budget and regular financial reporting to its board.

### **DCNR - Geologic Survey**

The DCNR Geologic Survey has expressed an interest in assisting with the geological interpretation along trails projects. Technical assistance would be provided to develop the text for signage features at significant geologic sites.

### **PA Historic and Museum Commission (PHMC)**

The competition under this program is considerable for relatively small grant awards. Application and administration for these grants are time consuming and this should be considered when deciding whether to apply. Perhaps the best target for PHMC KEYstone funds is for the museum function of preparing an interpretive plan for a trail highlighting the many historical interpretive sites along the trail. Applicants should closely review eligibility requirements.

### **PA Growing Greener**

The Growing Greener Program signed into law by Gov. Tom Ridge in 1999 is for investment to preserve farmland and protect open space; eliminate the maintenance backlog in State Parks; clean up abandoned mines; restore watersheds; and provide new and upgraded water and sewer systems. In 2002, the state legislature added additional monies to the program due to its great





popularity. Four different agencies are involved in helping communities "grow greener" under the Environmental Stewardship & Watershed Protection Act: Departments of Environmental Protection, Agriculture, Conservation and Natural Resources and PENNVEST. Of these four agencies, projects that may be applicable to the trail development will most likely be funded by the Department of Conservation and Natural Resources.

The Act authorizes grants for acid mine drainage abatement, mine cleanup efforts, abandoned oil and gas well plugging and local watershed-based conservation projects. These projects can include: watershed assessments and development of watershed restoration or protection plans; implementation of watershed restoration or protection projects, stormwater management wetlands, riparian buffer fencing and planting, streambank restoration and agricultural BMPs.

These grants are available to a variety of eligible applicants, including: counties, authorities and other municipalities; county conservation districts; watershed organizations; and other organizations involved in the restoration and protection of Pennsylvania's environment. These grants will support local projects to clean up "non-point" sources of pollution throughout Pennsylvania.

Since the proposed trail follows along the Chartiers Creek, there will be many opportunities for re-establishment of riparian buffers along the creek. It may be possible to blend Growing Greener grants with other grants for trail construction.

### **DCED Community Revitalization Funds**

The Department of Community and Economic Development (DCED) Community Revitalization Fund is a state program that supports local initiatives that improve the stability of communities and enhance local economies. The grant program covers a wide range of eligible uses including: acquisition of land, buildings, and right-of-ways; recreation projects; programs and developments that build capacity of the local community and relevant local organizations to better serve the needs of the community, and other reasonable and necessary expenses related to community-based activities. The Chartiers Creek Trail would be eligible for application for these funds. Active support of the district's state senator and / or state representative is critical in a successful grant application.

### **Legislative Funding**

State and federal elected officials can often include items into legislation for worthy projects in their districts. A conversation between county and municipal officials and legislators is the way to begin this process. This should be a well coordinated effort between the county and the municipalities. This type of funding should be targeted toward capital improvement projects.





## **County Government Programs**

Allegheny and/or Washington County may be able to provide matching funds (50/50) to state, federal or local funds through some of their programs. These funds may be recognized as materials or in-kind services.

## **Local Municipalities**

Any of the affected municipalities might contribute toward various trail implementation phases. Often, grant sources allow services in-kind in place of cash to count as a match.

It is strongly suggested that each municipality immediately begin and keep a detailed inventory of municipal staff and/or official time spent on the trail project. Occasionally, grantors may allow time spent to date to count as part of the in-kind match for funds. This record will also demonstrate a continuing commitment on the part of each community to the successful implementation of the master plan.

## **USDA Forest Service**

The USDA Forest Service has funded innovative bridge projects through the "Wood in Transportation " (WIT) program, and other environmental, recreation, and tourism-related projects through the "Rural Development" program. The grants are relatively small and best used for design, small construction, or interpretation. Grants must be matched 50-50 by nonfederal dollars or in-kind. Trail bridges in visible locations are potential applications, however the WIT program requires innovation or commercialization potential in the project.

The Forest Service Economic Action Program (EAP) strives to diversify local economies by providing the technical and financial assistance needed to expand the use of wood in transportation networks. By developing markets for underutilized wood; advancing modern timber bridge designs; improving wood composite products; and providing information to local community leaders, business officials, and engineers, the Program strives to improve local economies and infrastructure systems while utilizing America's primary renewable resource, wood.

## **The Conservation Fund**

The Conservation Fund forges partnerships to preserve our nation's outdoor heritage - America's legacy of wildlife habitat, working landscapes and community open-space. Pioneering a unique brand of conservation driven by effectiveness, efficiency, and environmental and economic balance, the Fund produces unsurpassed results.





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Since 1985, The Conservation Fund, through its partnership-driven approach, has protected more than three million acres of America's outdoor heritage, often leading other organizations in leverage and effectiveness.

### **Private Foundations**

There are various corporations and foundations, which support public works such as trail development. The competition for these funds has become brisk, but the opportunities should be researched. Funding must often be to non-profit organizations and projects which produce "on the ground" results are viewed favorably. A complete listing of potential foundation funding sources is presented in the attached Funding Source Matrix.

### **Schools**

Local schools may also be of assistance in several ways. The student body might get involved with clubs, fundraising events, and trail cleanup days. The faculty could incorporate the trail into various curricula with students helping to develop and possibly maintain the trail as part of a classroom assignment or after school club. While the amounts of funds raised may be relatively small, this process builds constituents and support for the trail, critical to its long-term success.







## **5.4 Operation, Maintenance and Security**

### **Maintenance Plan**

Annual trail maintenance costs as estimated by the National Park Service can typically run from \$500 per mile for low-use trails to \$5,000 per mile for high-use trails. It is anticipated that sections of the Chartiers Creek Trail will experience both levels of use across this range, equalizing in relative terms to an average “moderate” trail use. Trail use is anticipated to be highest from McKees Rocks to Carnegie. Carnegie to Bridgeville trail use would be next highest relative use. The Washington County sections of trail would receive the lightest use, both due to the proximity of population and lack of off-road facilities.

Typical trail maintenance tasks include clearing fallen trees across trails, removing dangerous trees or limbs, bridge inspections, maintaining adequate shoulder clearances along trail, cleaning drainage structures, repairing erosion and damaged trail surfaces, removing invasive plant species, trash pick-up and removal, undertaking periodic inspections and other associated tasks.

Experience on other trails has shown that with the aid of volunteers, these maintenance costs can be brought down significantly. The utilization of volunteer labor is an important component in managing a trail. The cleaning of drainage swales and structures and trash pick-up and removal along the trail are important volunteer tasks that can have immediate positive results.

### **Maintenance Task Schedule**

The following is a summary of trail maintenance tasks that should be performed annually to maintain the trails in safe condition. Some tasks such as trash collecting, drainage structure cleaning, plantings and other maintenance tasks can be completed by volunteers. This work should be coordinated by trail organization leaders in conjunction with the local municipality. Professional maintenance personnel from a local public works department best perform material-intensive tasks requiring larger equipment.

#### Winter

Trails maintenance work in the winter months can continue dependent on weather conditions. Typical winter trail work may include:

- Trash pick-up and removal.
- Removal of dangerous trees or tree limbs.
- Clearing free-hanging vines on trees in the right-of-way.
- Minor repairs to trails (erosion repair, etc.)
- Inspect and repair/replace signs, etc as needed.



- Minor repairs to structures, fences, and bridge railings.
- Keep drainage ways clear and clean out culverts as required.
- Keep bridge deck surfaces clear and fencing/railing free of vegetation.

#### Spring

- Trash pick-up and removal.
- Bridge / Structure inspections (every other year) - staggered schedule.
- Trail-wide inspection for winter damage. Schedule repair work over the next two to three months.
- Obtain and plant bare-root and other tree and shrub plant materials for revegetation projects.
- Distribute / post information about major trail repair and expansion projects for the spring.
- Remove downed trees as required.
- Keep drainage ways clear and clean out culverts as required.
- Keep bridge deck surfaces clear and fencing/railing free of vegetation.
- Prepare and seed areas as needed to prevent erosion.
- First spraying of invasive species.
- Mow and trim (first time).

#### Summer

- Trash pick-up and removal.
- Continue trail repair.
- Plan for fall planting.
- Removed downed trees as required.
- Keep drainage ways clear and clean out culverts as required.
- Keep bridge deck surfaces clear and fencing/railing free of vegetation.
- Undertake second spraying of invasive species.
- Mow and trim (second time).
- Distribute / post information about major trail repair and expansion projects for the fall

#### • Fall

- Trash pick-up and removal.
- Continue trail repair.
- Undertake third spraying of invasive species.
- Remove downed trees as required.
- Mow and trim (Third time).
- Keep drainage ways clear and clean out culverts as required.
- Keep bridge deck surfaces clear and fencing/railing free of vegetation.



- Complete fall planting of balled and burlapped trees in “structured” areas (i.e. trail heads, picnic areas.)
- Plan for spring planting.

It is preferable that trail related maintenance activities not be deferred. Deferring maintenance for short-term savings is a faulty strategy with a poor chance of long-term success. If the trail quality deteriorates and does not provide a high quality recreation experience, it will lose popular support and thus funding. Maintenance costs will only increase and must be planned for by the trail management entity and its partners. It should be noted that most funding agencies do not fund operational costs. This fact emphasizes the importance of developing a strong network of volunteers. Attendance at trail care events may be enhanced by centering them on seasonal or special events. These may include such concepts as spring or fall Cleanup Days, cookouts, musical events or rally days.

### **Security**

As uses of each trail section increases, both operations and security of the trail will become somewhat easier. Sections of trail that do not experience high volumes of use may be subject to a greater occurrence of undesirable activity. Littering, vandalism, all terrain vehicle (ATV) use and underage drinking are typical inappropriate activities that occur on some trails systems. As runners, hikers, cyclists and other trail users populate the trail, they will become the eyes and ears of authority, and these activities should subside. Increasing numbers of trail users will have cell phones which can be used to report undesirable or illegal activities.

A Trail Patrol program may be initiated for sections of trail that experience high use volumes or where there is a history of illicit or dangerous activity. A Trail Patrol program may be comprised of volunteers from the community, police, firemen, or governmental agency personnel. Trail patrol personnel should be heavily involved in positive trail activities such as giving directions, offering hiking or riding suggestions, and other service-oriented activities. Enforcement of regulations and resolution of user conflicts may also be performed as needed. Training should be provided to all patrol members and the actual patrol rounds should be conducted on bike, foot, or horseback.



*Volunteer Trail Patrol*

Trail users also help each municipality maintain and operate the trails. When there are problems, trail users may notify the trail owner or agency about the maintenance item. This can be accomplished by posting contact information at the trailheads by having a box for suggestions or contacts at various locations





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along the trail. Lighting should be provided in areas where trail use will extend into the evening and around comfort facilities.

To accommodate emergency vehicles, the multi-purpose sections of the trail will be designed to be accessible by police vehicles and ambulances. Vehicles such as pickup trucks will also access the trail for periodic inspections or maintenance. Bollards, gates and other vehicular controls will keep out private motor vehicles.

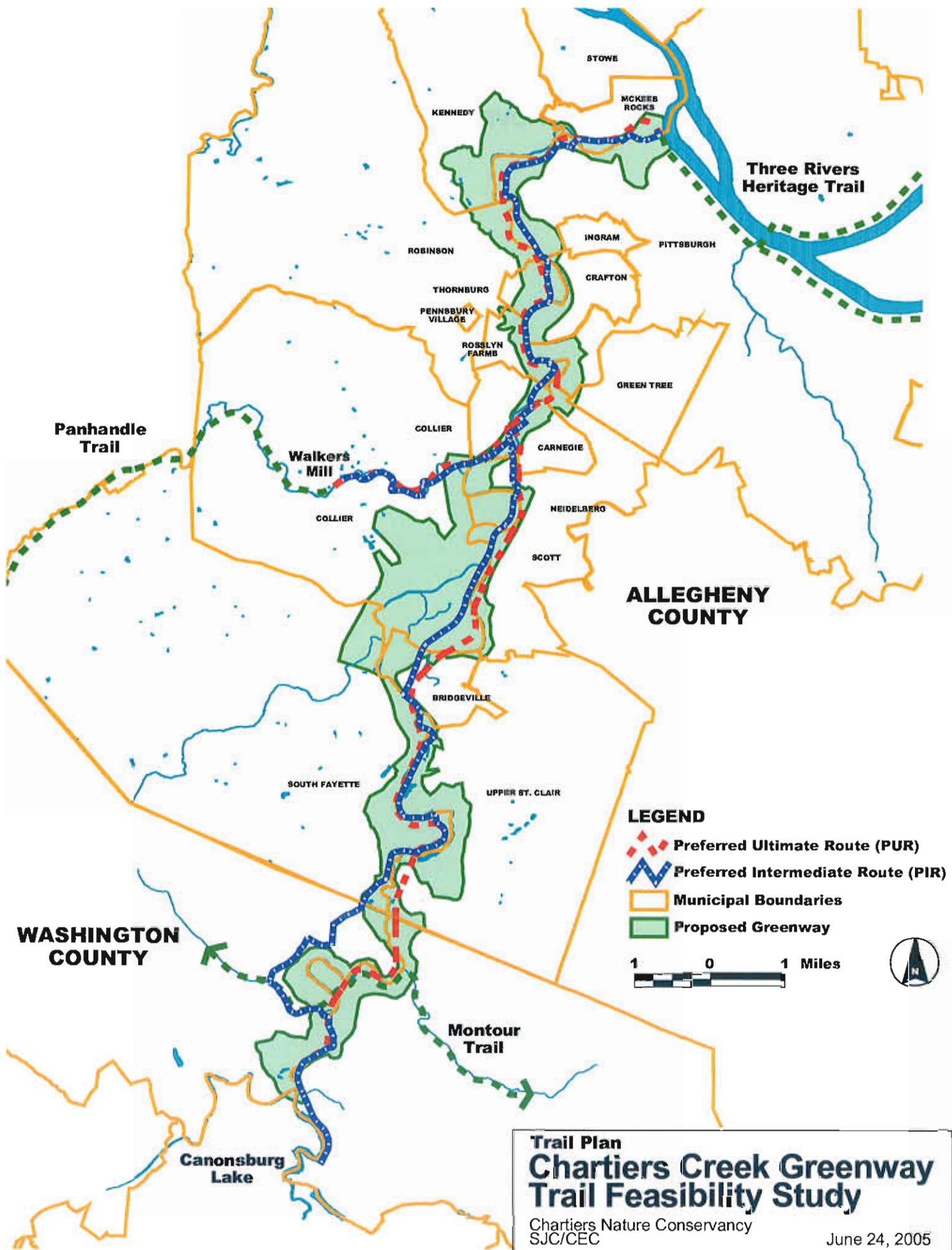




## FIGURES

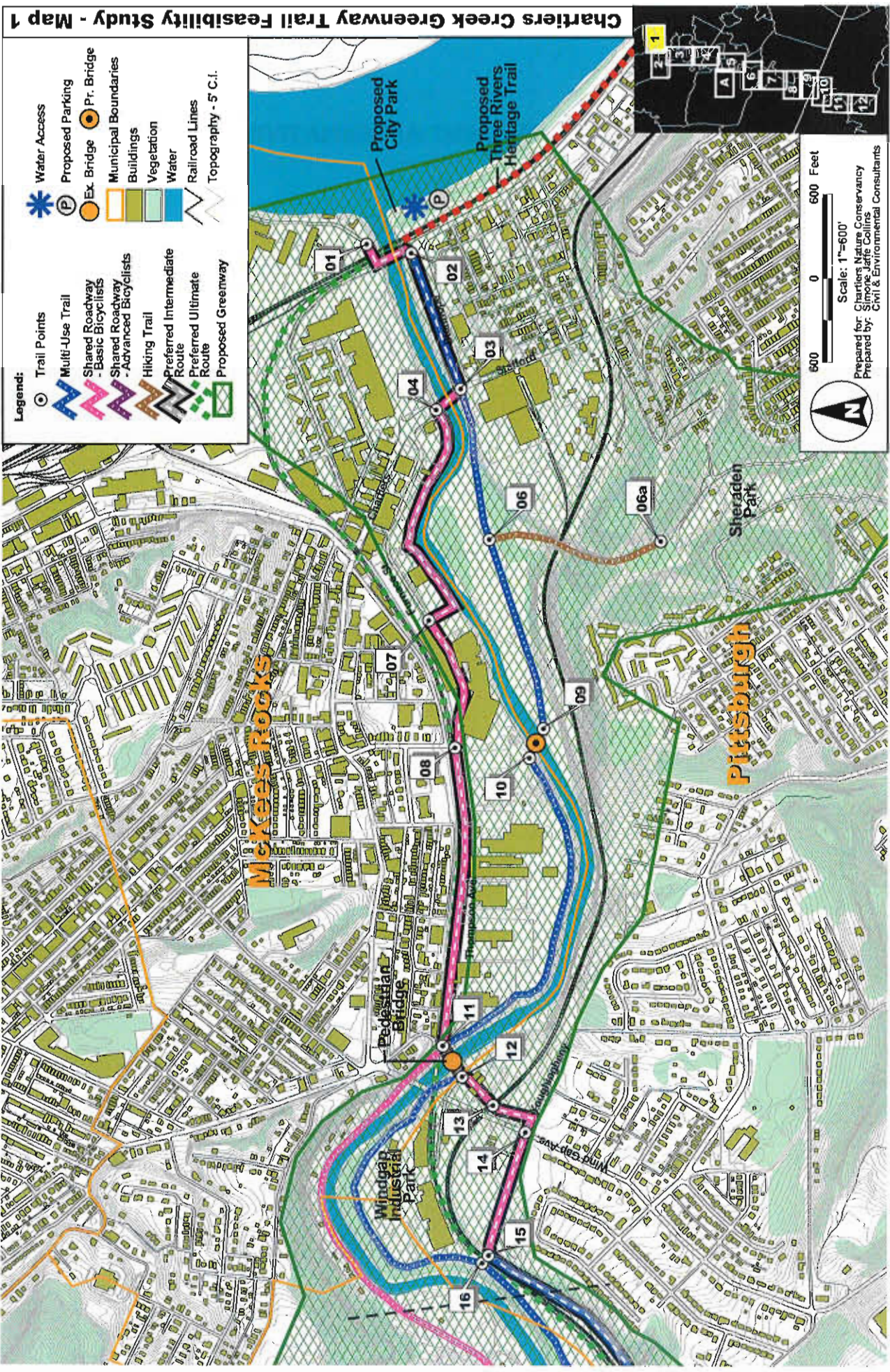
## STUDY OVERVIEW MAP





## TRAIL ALIGNMENT ALTERNATIVES





**Legend:**

- Trail Points
- Multi-Use Trail
- Shared Roadway - Basic Bicyclists
- Shared Roadway - Advanced Bicyclists
- Hiking Trail
- Preferred Intermediate Route
- Preferred Ultimate Route
- Proposed Greenway
- Water Access
- Proposed Parking
- Ex. Bridge
- Pr. Bridge
- Municipal Boundaries
- Buildings
- Vegetation
- Water
- Railroad Lines
- Topography - 5' C.I.

Scale: 1"=600'

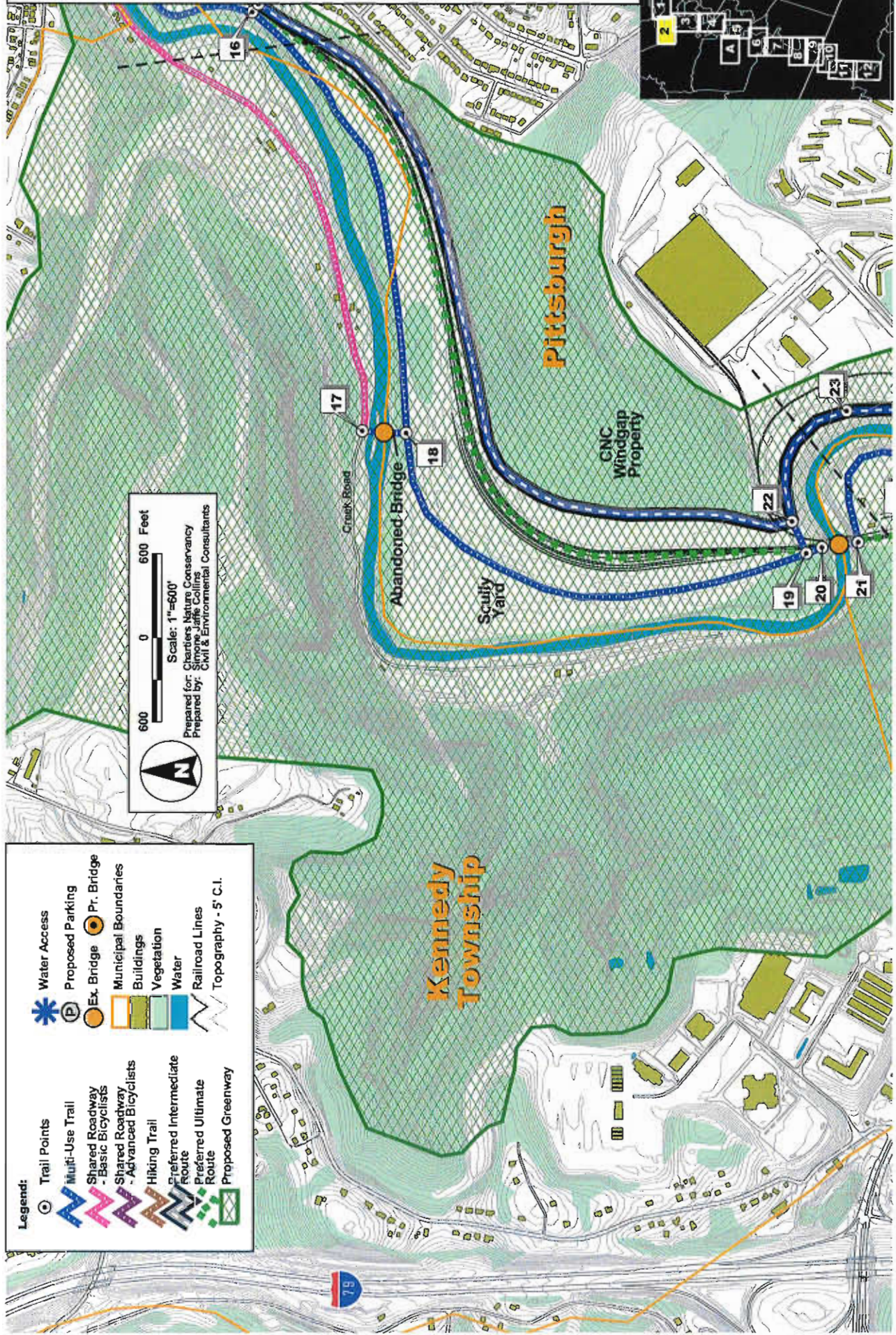
Prepared for: Chartiers Nature Conservancy

Prepared by: Simone Jarfe Collins

Civil & Environmental Consultants

Chartiers Creek Greenway Trail Feasibility Study - Map 1





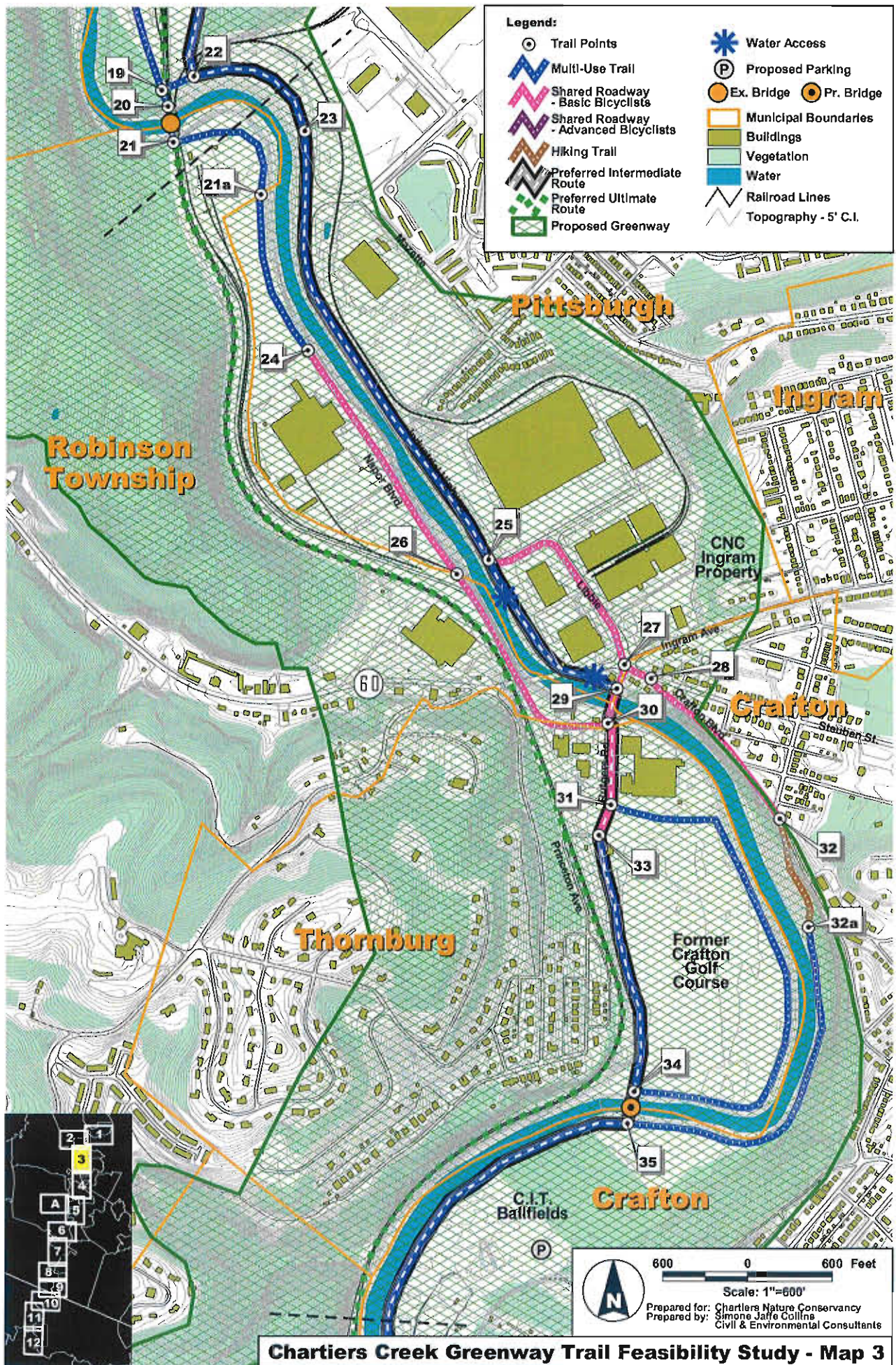
**Legend:**

- Trail Points
- Multi-Use Trail
- Shared Roadway - Basic Bicyclists
- Shared Roadway - Advanced Bicyclists
- Hiking Trail
- Preferred Intermediate Route
- Preferred Ultimate Route
- Proposed Greenway
- Water Access
- Proposed Parking
- Ex. Bridge
- Pr. Bridge
- Municipal Boundaries
- Buildings
- Vegetation
- Water
- Railroad Lines
- Topography - 5' C.I.

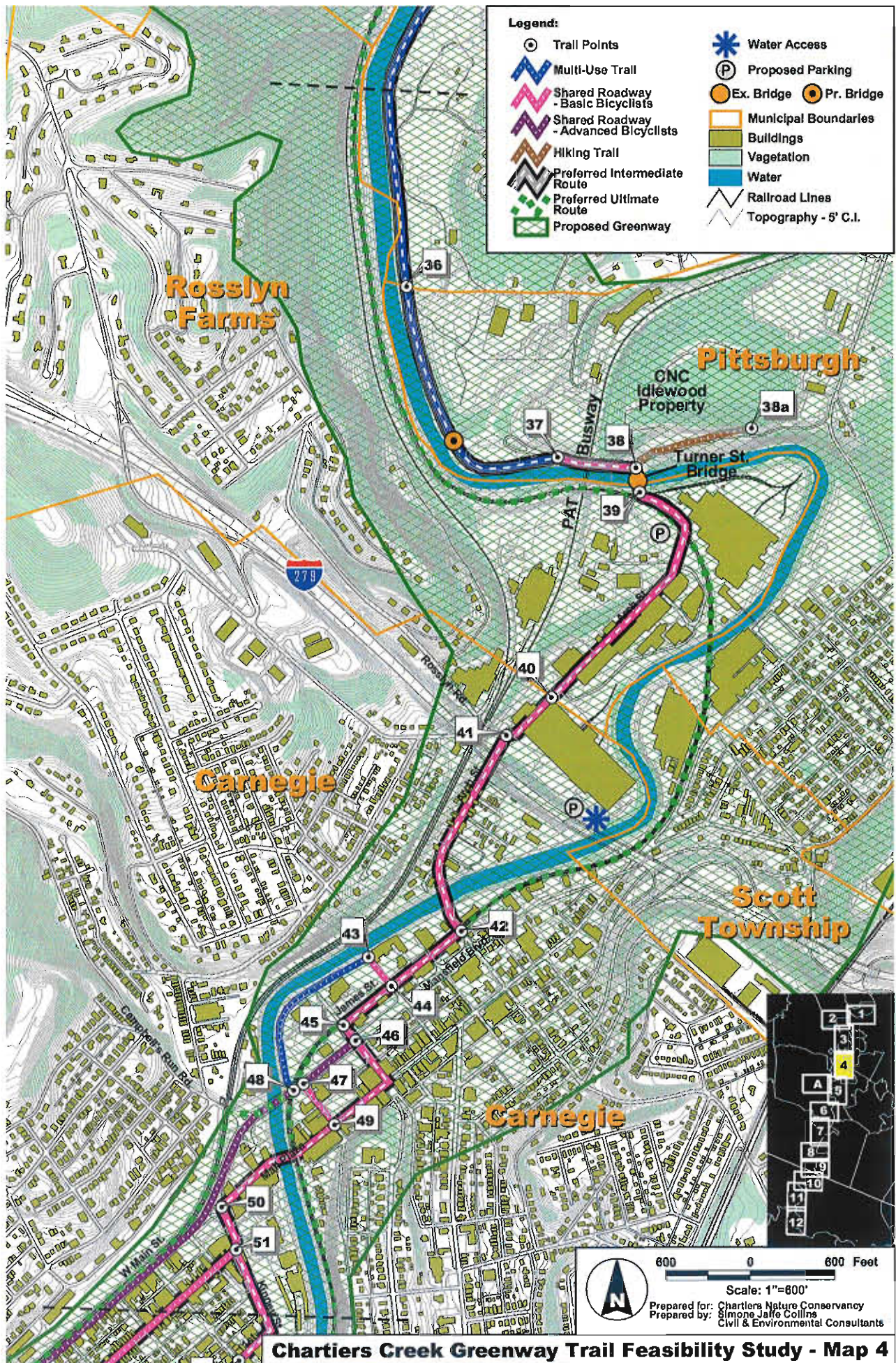
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Prepared for: Chartiers Nature Conservancy  
Prepared by: Sherry-Jane Collins, Consultants  
Chartiers & Environmental Consultants

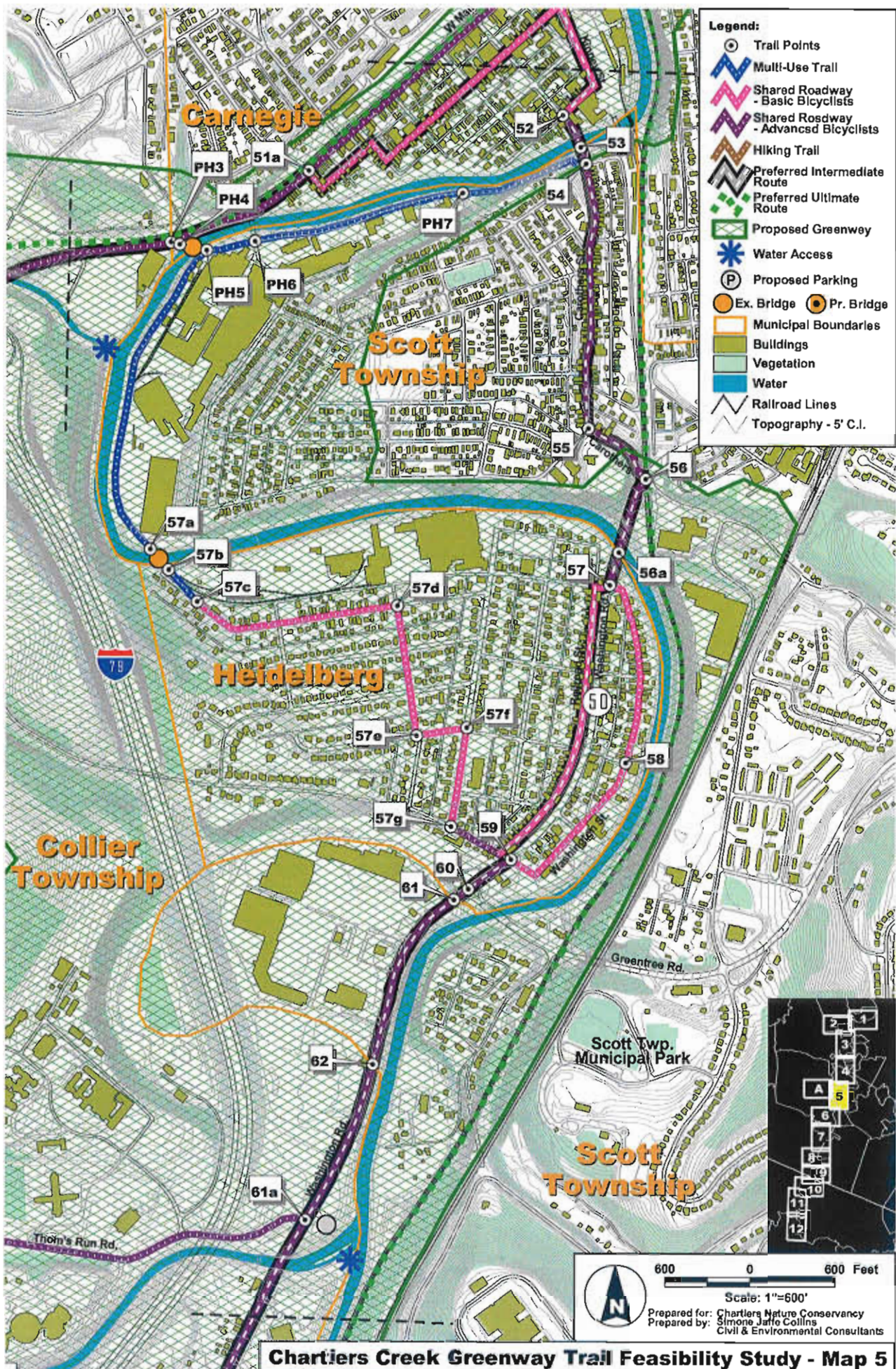






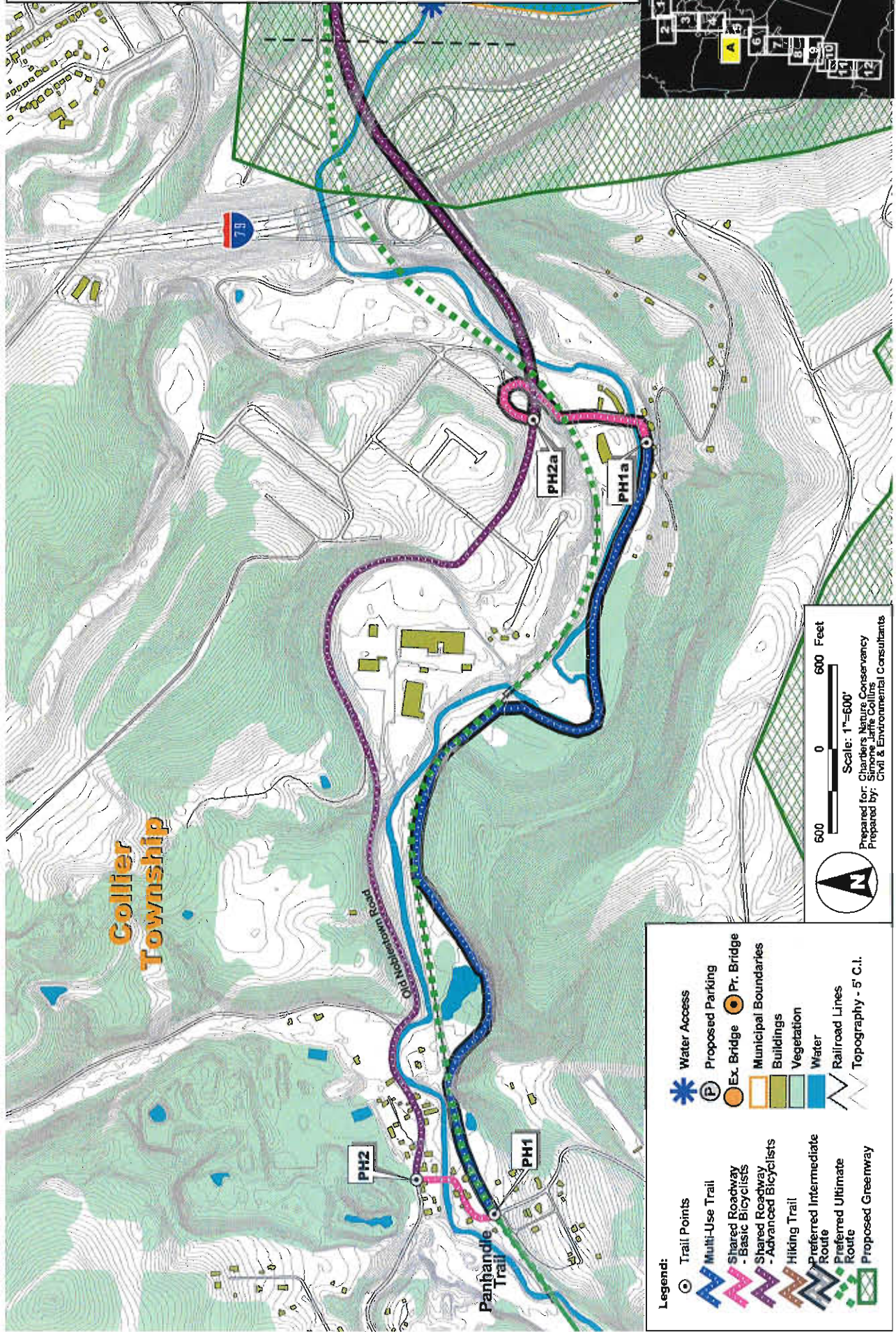




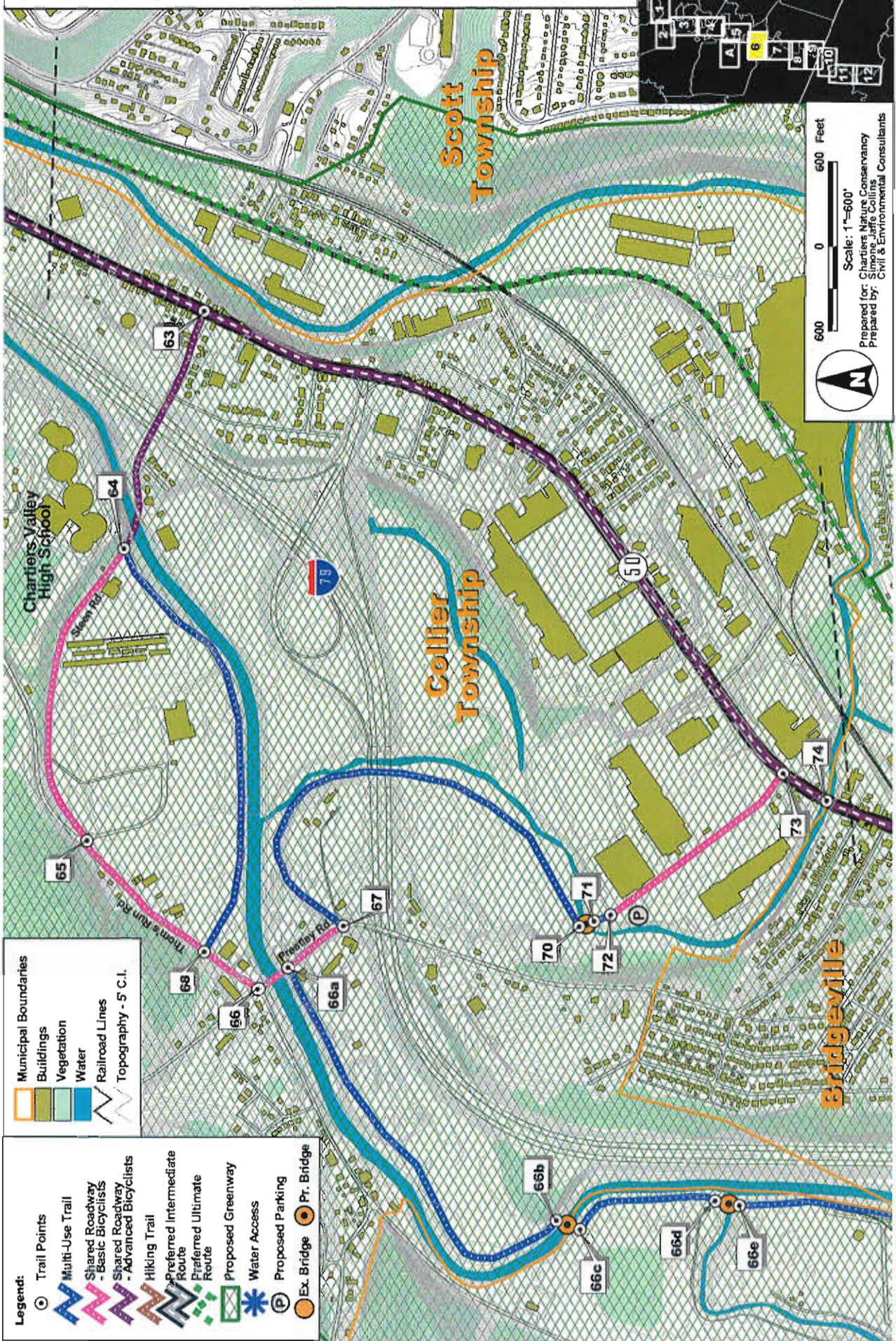


**Charters Creek Greenway Trail Feasibility Study - Map 5**













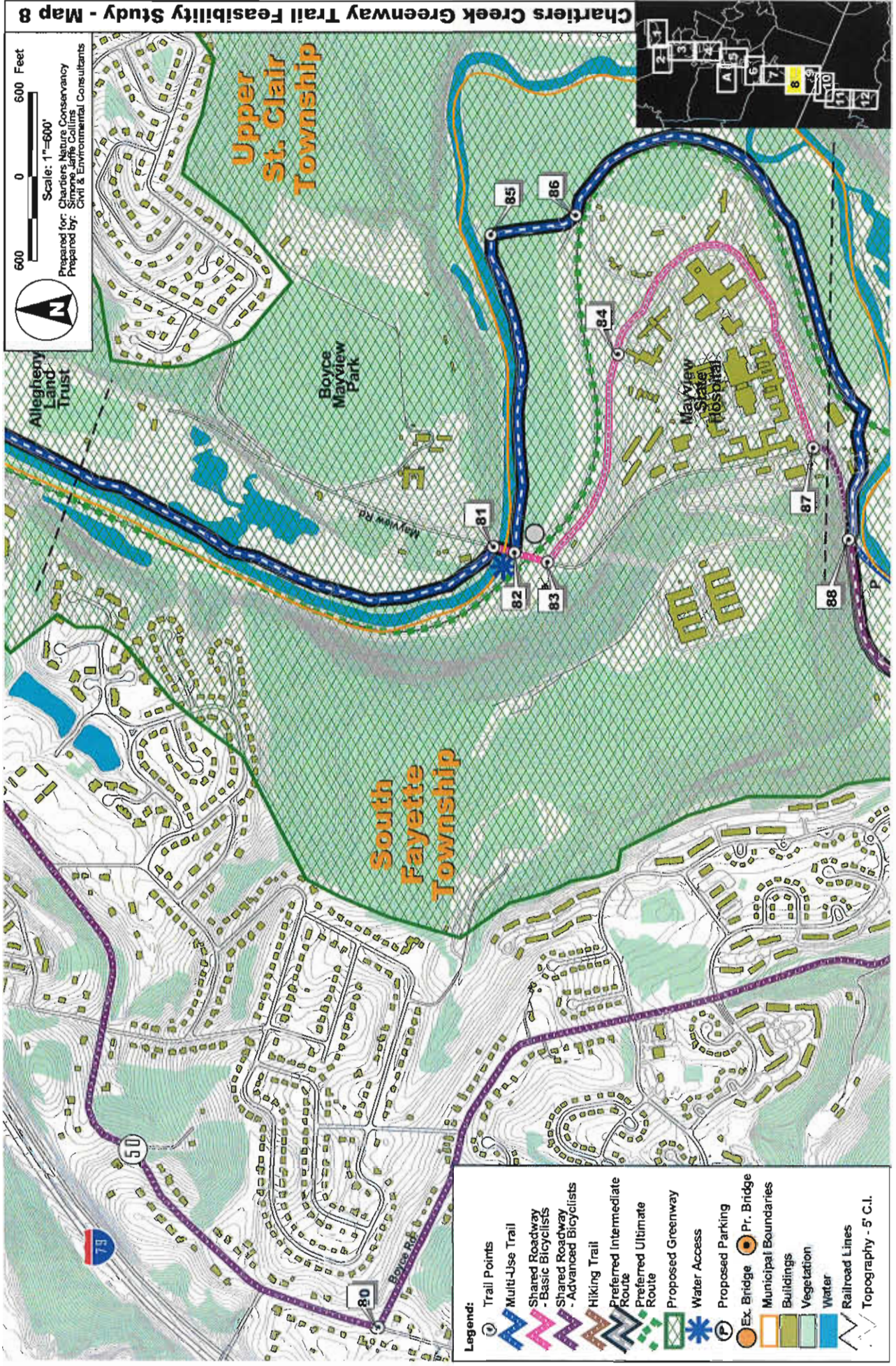
**Chartiers Creek Greenway Trail Feasibility Study - Map 7**



Scale: 1"=600'  
 Prepared for: Chartiers Nature Conservancy  
 Prepared by: S&B Environmental Consultants

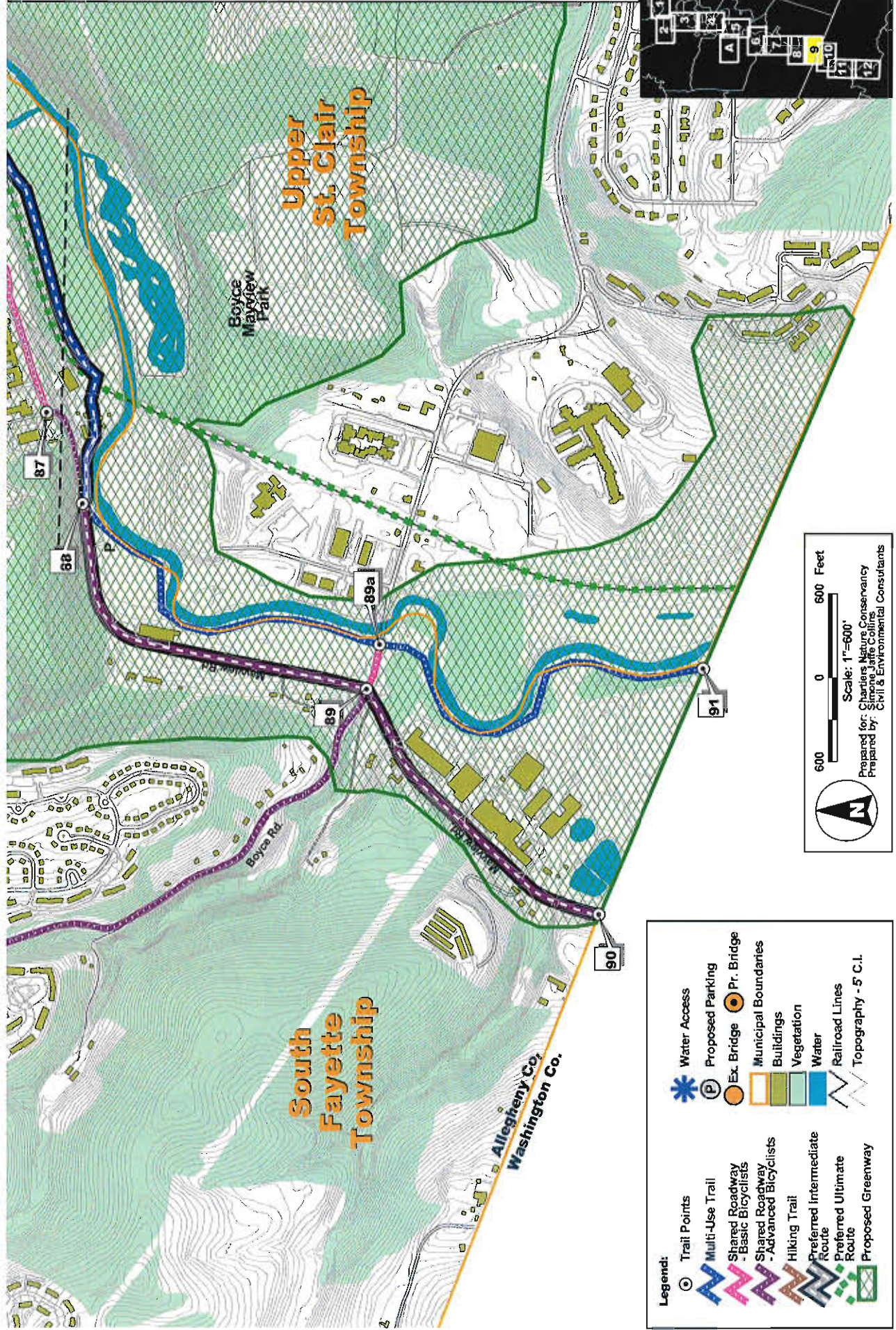


600 0 600 Feet



- Legend:**
- Trail Points
  - Multi-Use Trail
  - Shared Roadway - Basic Bicyclists
  - Shared Roadway - Advanced Bicyclists
  - Hiking Trail
  - Preferred Intermediate Route
  - Preferred Ultimate Route
  - Proposed Greenway
  - Water Access
  - Proposed Parking
  - Ex. Bridge
  - Pr. Bridge
  - Municipal Boundaries
  - Buildings
  - Vegetation
  - Water
  - Railroad Lines
  - Topography - 5' C.I.





South  
Fayette  
Township

Upper  
St. Clair  
Township

Allegheny Co.  
Washington Co.

Boyce  
Mayview  
Park

Boyce Rd.

87

88

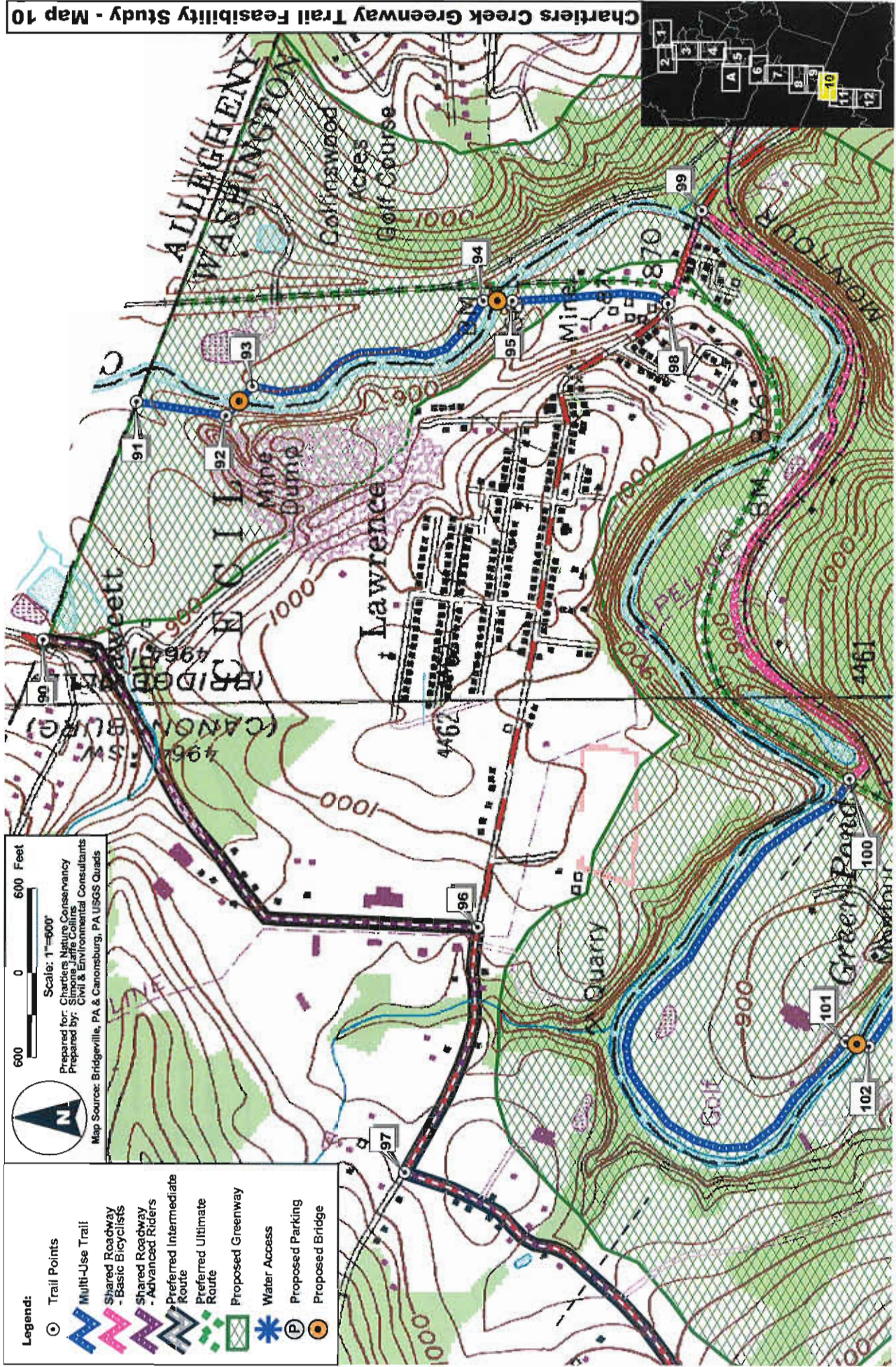
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89a

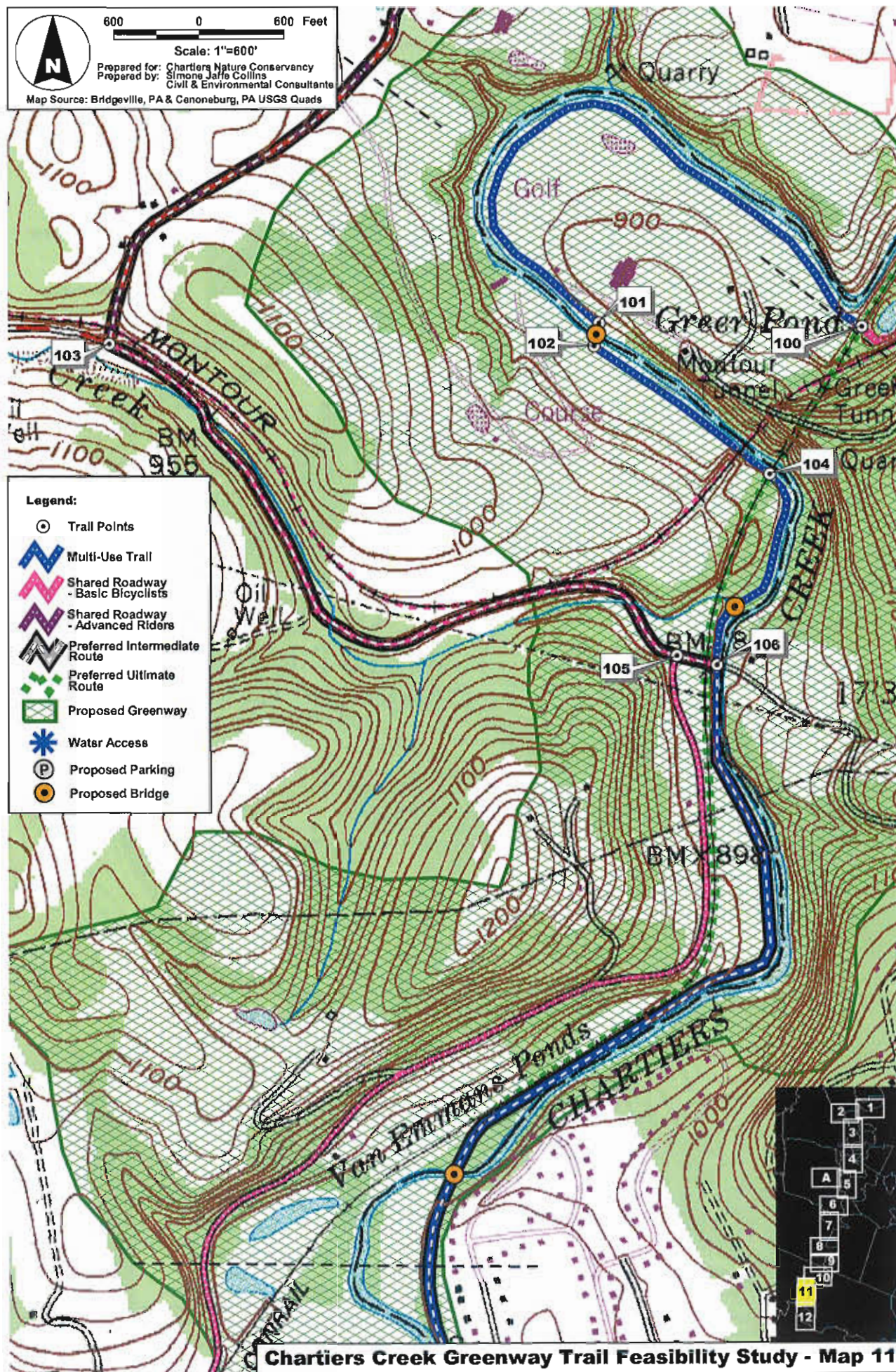
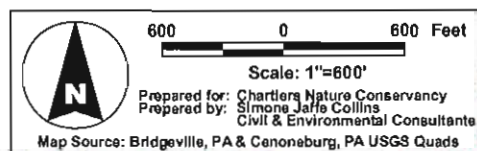
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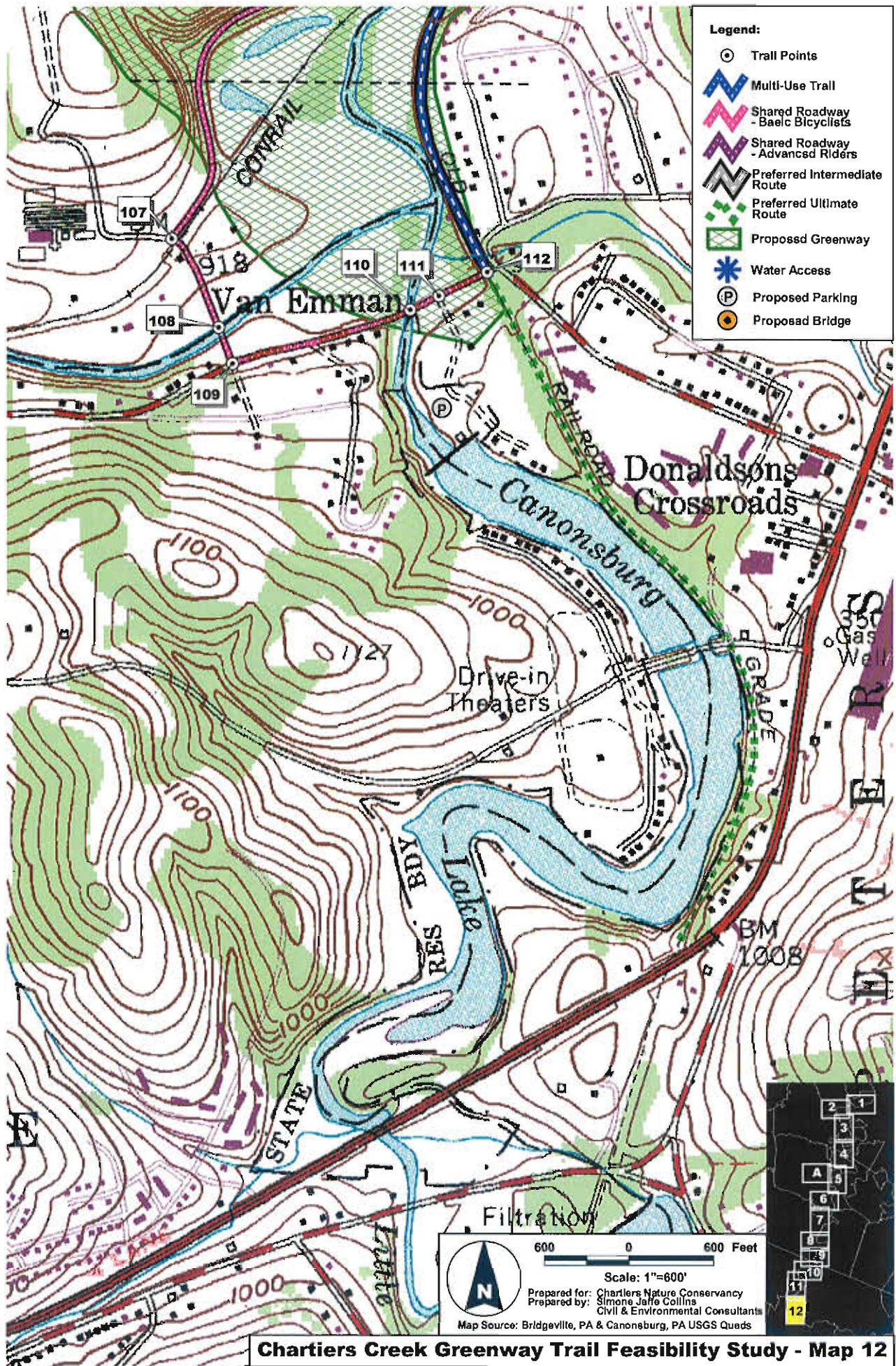






Charters Creek Greenway Trail Feasibility Study - Map 11







## TABLES



**ESTIMATED DEVELOPMENT COST MATRIX**

## Lower Chartiers Creek Greenway Trail Feasibility Study

## Estimated Development Cost Matrix

| Point ID | Segment ID | Length (FT) | Municipality | Type            | Existing Conditions                         | Ultimate Improvements*                   | QTY   | UNIT | UNIT COST | SUBTOTAL  | TOTAL     |
|----------|------------|-------------|--------------|-----------------|---------------------------------------------|------------------------------------------|-------|------|-----------|-----------|-----------|
| 01       | -          | -           | McKees Rocks | Terminus        | Proposed trail beginning.                   | Trail Information Board                  | 1     | LS   | \$5,000   | \$5,000   | \$15,000  |
| -        | 01-02      | 440         | M/RB/PGH     | On-Road - Basic | Roadway bridge over Chartiers Creek.        | Parking Improvements                     | 1     | LS   | \$10,000  | \$10,000  | \$15,000  |
| 02       | -          | -           | Pittsburgh   | Intersection    | Stanhope bridge over Chartiers Creek.       | Roadway Section                          | 440   | LF   | \$3       | \$1,320   | \$1,320   |
| -        | 02-03      | 1,030       | Pittsburgh   | Multi-Use       | Stanhope St./Carson St.                     | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$5,500   |
| 03       | -          | -           | Pittsburgh   | Intersection    | Undeveloped, grassy.                        | Security Measures                        | 1     | LS   | \$5,000   | \$5,000   | \$5,500   |
| -        | 03-04      | 230         | M/RB/PGH     | On-Road - Basic | Stanford/Stanhope                           | Multi-Use Section 1                      | 1,030 | LF   | \$7       | \$7,210   | \$7,210   |
| -        | 03-06      | 1,100       | Pittsburgh   | Multi-Use       | Roadway bridge over Chartiers Creek.        | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$5,500   |
| 04       | -          | -           | McKees Rocks | Intersection    | Duquesne Light right-of-way.                | Security Measures                        | 1     | LS   | \$5,000   | \$5,000   | \$5,500   |
| -        | 04-05      | 235         | McKees Rocks | On-Road - Basic | Stanford St.                                | Roadway Section                          | 230   | LF   | \$3       | \$690     | \$690     |
| -        | 04-07      | 1,800       | McKees Rocks | On-Road - Basic | Top of bank behind businesses.              | Multi-Use Section 1                      | 600   | LF   | \$7       | \$4,200   | \$15,200  |
| 05       | -          | -           | McKees Rocks | Intersection    | Chartiers St./Furnace St.                   | Multi-Use Section 3                      | 500   | LF   | \$17      | \$8,500   | \$15,200  |
| -        | 05-07      | 1,495       | McKees Rocks | On-Road - Basic | Furnace St.                                 | Split Rail Fencing                       | 500   | LF   | \$5       | \$2,500   | \$2,500   |
| 06       | -          | -           | Pittsburgh   | Junction        | Spur to Sheraden Park endpoint.             | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 06-06a     | 1,290       | Pittsburgh   | Hiking          | Duquesne Light right-of-way.                | Directional Signage                      | 235   | LF   | \$3       | \$705     | \$705     |
| -        | 06-09      | 1,405       | Pittsburgh   | Multi-Use       | Sheraden Park.                              | Roadway Section                          | 1,800 | LF   | \$3       | \$5,400   | \$5,400   |
| 06a      | -          | -           | Pittsburgh   | Terminus        | Furnace St.                                 | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| 07       | -          | -           | McKees Rocks | Intersection    | Furnace St.                                 | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 07-08      | 985         | McKees Rocks | On-Road - Basic | Furnace St.                                 | Directional Signage                      | 1,495 | LF   | \$3       | \$4,485   | \$4,485   |
| 08       | -          | -           | McKees Rocks | Intersection    | Furnace St./Thompson Ave.                   | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 08-11      | 2,140       | McKees Rocks | On-Road - Basic | Thompson Ave.                               | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| 09       | -          | -           | Pittsburgh   | Surface Change  | Proposed Abutment location.                 | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 09-10      | 235         | M/RB/PGH     | Multi-Use       | Proposed Bridge - New.                      | Roadway Section                          | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| 10       | -          | -           | McKees Rocks | Surface Change  | Proposed Abutment location.                 | New Abutment                             | 1     | LS   | \$250,000 | \$250,000 | \$250,000 |
| -        | 10-11      | 2,560       | McKees Rocks | Multi-Use       | Top of bank behind industrial buildings.    | Multi-Use Section 1                      | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| 11       | -          | -           | McKees Rocks | Intersection    | Thompson Ave./Creek Rd./Ped Bridge          | Directional Signage                      | 2,560 | LF   | \$7       | \$17,920  | \$17,920  |
| -        | 11-12      | 265         | McKees Rocks | Hiking          | Existing pedestrian bridge.                 | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 11-17      | 5,625       | M/RB/Kennedy | On-Road - Basic | Creek Rd.                                   | Approach, Deck, and Railing Improvements | 1     | LS   | \$25,000  | \$25,000  | \$25,000  |
| 12       | -          | -           | Pittsburgh   | Surface Change  | Ped. Bridge endpoint.                       | Roadway Section                          | 5,625 | LF   | \$3       | \$16,875  | \$16,875  |
| -        | 12-13      | 305         | Pittsburgh   | On-Road - Basic | Windgap Industrial Park Access Road         | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 12-16      | 2,430       | Pittsburgh   | Multi-Use       | Top of bank through Industrial Park.        | Roadway Section                          | 305   | LF   | \$3       | \$915     | \$915     |
| 13       | -          | -           | Pittsburgh   | Intersection    | Existing at-grade railroad crossing.        | Multi-Use Section 1                      | 2,430 | LF   | \$7       | \$17,010  | \$17,010  |
| -        | 13-14      | 410         | Pittsburgh   | On-Road - Basic | Windgap Industrial Park Access Road         | -                                        | -     | -    | -         | -         | -         |
| 14       | -          | -           | Pittsburgh   | Intersection    | Wind Gap Ave./Youghiogheny St.              | Roadway Section                          | 410   | LF   | \$3       | \$1,230   | \$1,230   |
| -        | 14-15      | 915         | Pittsburgh   | On-Road - Basic | Youghiogheny St.                            | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| 15       | -          | -           | Pittsburgh   | Surface Change  | Youghiogheny St.                            | Roadway Section                          | 915   | LF   | \$3       | \$2,745   | \$2,745   |
| -        | 15-22      | 6,100       | Pittsburgh   | Multi-Use       | Base of slope through CMC Windgap Property. | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| 16       | -          | -           | Pittsburgh   | Property Line   | Western end of Industrial Park.             | Security Measures                        | 1     | LS   | \$5,000   | \$5,000   | \$5,000   |
| -        | 16-18      | 3,335       | Pittsburgh   | Multi-Use       | Top of bank through Scully Yard             | Multi-Use Section 2                      | 3,335 | LF   | \$12      | \$40,020  | \$40,020  |
| 17       | -          | -           | Kennedy Twp. | Intersection    | Creek Road/Abandoned Bridge.                | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 17-18      | 315         | Kennedy/PGH  | Multi-Use       | Proposed Bridge - Rehab.                    | Security Measures                        | 1     | LS   | \$5,000   | \$5,000   | \$5,000   |
| 18       | -          | -           | Pittsburgh   | Surface Change  | Existing bridge endpoint.                   | Bridge Rehabilitation                    | 1     | EA   | \$50,000  | \$50,000  | \$50,000  |
| -        | 18-19      | 3,515       | Pittsburgh   | Multi-Use       | Scully Yard.                                | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| 19       | -          | -           | Pittsburgh   | Junction        | Scully Yard.                                | Multi-Use Section 2                      | 3,515 | LF   | \$12      | \$42,180  | \$42,180  |
| -        | 19-20      | 125         | Pittsburgh   | Multi-Use       | Scully Yard.                                | Directional Signage                      | 2     | EA   | \$250     | \$500     | \$500     |
| -        | -          | -           | -            | -               | -                                           | Multi-Use Section 1                      | 125   | LF   | \$7       | \$875     | \$875     |

## Estimated Development Cost Matrix

bold = Preferred Route

| Point ID | Segment ID | Length (FT) | Municipality      | Type               | Existing Conditions                           | Ultimate Improvements*       | QTY   | UNIT | UNIT COST | SUBTOTAL  | TOTAL     |
|----------|------------|-------------|-------------------|--------------------|-----------------------------------------------|------------------------------|-------|------|-----------|-----------|-----------|
| -        | 19-22      | 250         | Pittsburgh        | Multi-Use          | Will require railroad crossings.              | Multi-Use Section 1          | 250   | LF   | \$7       | \$1,750   | \$11,750  |
| 20       | -          | -           | Pittsburgh        | Surface Change     | Proposed Abutment.                            | RR Crossing Improvements     | 1     | LS   | \$10,000  | \$10,000  | \$15,000  |
| -        | 20-21      | 260         | Robinson/PGH      | Multi-Use          | Proposed Bridge - New. Parallel to railroad.  | New Abutment                 | 1     | LS   | \$15,000  | \$15,000  | \$270,000 |
| 21       | -          | -           | Robinson Twp.     | Surface Change     | Proposed Abutment.                            | New Abutment                 | 1     | LS   | \$15,000  | \$15,000  | \$15,500  |
| -        | 21-21a     | 1,000       | Robinson Twp.     | Multi-Use          | Existing rail yard.                           | Directional Signage          | 2     | EA   | \$250     | \$500     | \$7,000   |
| 21a      | -          | -           | -                 | Municipal Boundary | Robinson/Pittsburgh.                          | Multi-Use Section 1          | 1,000 | LF   | \$7       | \$7,000   | \$7,000   |
| -        | 21a-24     | 1,195       | Pittsburgh        | Multi-Use          | Existing rail yard.                           | -                            | -     | -    | -         | -         | -         |
| 22       | -          | -           | Pittsburgh        | Junction           | CNC Windgap Property                          | Multi-Use Section 1          | 1,195 | LF   | \$7       | \$8,365   | \$8,365   |
| -        | 22-23      | 1,030       | Pittsburgh        | Multi-Use          | Top of bank to Industrial Highway.            | Directional Signage          | 2     | EA   | \$250     | \$500     | \$500     |
| 23       | -          | -           | Pittsburgh        | Surface Change     | Industrial Highway endpoint.                  | Multi-Use Section 1          | 1,030 | LF   | \$7       | \$7,210   | \$7,210   |
| -        | 23-25      | 3,440       | Pittsburgh        | Multi-Use          | Industrial Highway                            | Trail Information Board      | 1     | LS   | \$5,000   | \$5,000   | \$5,000   |
| 24       | -          | -           | Pittsburgh        | Surface Change     | Napor Blvd. endpoint.                         | ***see GEC concept sketch*** | 3,440 | LF   | \$15      | \$51,600  | \$51,600  |
| -        | 24-26      | 1,920       | Pittsburgh        | On-Road - Basic    | Napor Blvd.                                   | Directional Signage          | 2     | EA   | \$250     | \$500     | \$500     |
| 25       | -          | -           | Pittsburgh        | Junction           | Industrial Highway endpoint.                  | Directional Signage          | 1,920 | LF   | \$3       | \$5,760   | \$5,760   |
| -        | 25-27      | 1,510       | Pittsburgh        | On-Road - Basic    | Libbe St.                                     | Security Measures            | 2     | EA   | \$250     | \$500     | \$5,500   |
| -        | 25-29      | 1,365       | Pittsburgh        | Multi-Use          | Top of bank Industrial Highway to Sharp Edge. | Multi-Use Section 1          | 1,510 | LF   | \$3       | \$4,530   | \$4,530   |
| 26       | -          | -           | -                 | Municipal Boundary | Robinson/Pittsburgh.                          | Multi-Use Section 1          | 1,365 | LF   | \$7       | \$9,555   | \$9,555   |
| -        | 26-30      | 1,685       | Robinson          | On-Road - Basic    | Napor Blvd.                                   | -                            | -     | -    | -         | -         | -         |
| 27       | -          | -           | -                 | Municipal Boundary | Crafton/Pittsburgh.                           | Roadway Section              | 1,685 | LF   | \$3       | \$5,055   | \$5,055   |
| -        | 27-28      | 220         | Crafton           | On-Road - Basic    | Sleuben St.                                   | -                            | -     | -    | -         | -         | -         |
| -        | 27-29      | 185         | Crafton           | On-Road - Basic    | W. Sleuben St.                                | Roadway Section              | 220   | LF   | \$3       | \$660     | \$660     |
| 28       | -          | -           | Crafton           | Intersection       | Crafton Blvd./Sleuben St.                     | Directional Signage          | 185   | LF   | \$3       | \$555     | \$555     |
| -        | 28-32      | 1,360       | Crafton           | On-Road - Basic    | Crafton Blvd.                                 | Directional Signage          | 2     | EA   | \$250     | \$500     | \$500     |
| 29       | -          | -           | -                 | Municipal Boundary | Crafton/Pittsburgh.                           | Roadway Section              | 1,360 | LF   | \$3       | \$4,080   | \$4,080   |
| -        | 29-30      | 255         | Robinson          | On-Road - Basic    | W. Sleuben St.                                | Security Measures            | 1     | LS   | \$5,000   | \$5,000   | \$5,000   |
| 30       | -          | -           | Robinson          | Intersection       | Napor Blvd./Rugers Rd.                        | Roadway Section              | 255   | LF   | \$3       | \$765     | \$765     |
| -        | 30-31      | 595         | Thornburg         | On-Road - Basic    | Rugers Rd.                                    | Directional Signage          | 2     | EA   | \$250     | \$500     | \$500     |
| 31       | -          | -           | Thornburg         | Junction           | Multi-Use to meet On-road.                    | Roadway Section              | 595   | LF   | \$3       | \$1,785   | \$1,785   |
| -        | 31-33      | 235         | Thornburg         | On-Road - Basic    | Rugers Rd.                                    | Directional Signage          | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 31-34      | 3,830       | Thornburg         | Multi-Use          | Loop around former Crafton Golf Course.       | Roadway Section              | 235   | LF   | \$3       | \$705     | \$705     |
| 32       | -          | -           | Crafton           | Intersection       | Crafton Blvd./Access Rd.                      | Multi-Use Section 1          | 3,830 | LF   | \$7       | \$26,810  | \$26,810  |
| -        | 32-32a     | 815         | Crafton           | Hiking             | Sleep, narrow access road.                    | None necessary.              | 2     | EA   | \$250     | \$500     | \$500     |
| 32a      | -          | -           | Crafton           | Surface Change     | Bottom of access road.                        | Directional Signage          | -     | -    | -         | -         | -         |
| -        | 32a-35     | 2,640       | Crafton           | Multi-Use          | Base of slope/top of bank.                    | Multi-Use Section 1          | 2,640 | LF   | \$6       | \$15,840  | \$15,840  |
| 33       | -          | -           | Thornburg         | Intersection       | Rugers Rd./Proposed Multi-Use Trail.          | Security Measures            | 2     | EA   | \$250     | \$500     | \$5,500   |
| -        | 33-34      | 1,865       | Thornburg         | Multi-Use          | Western edge of former Crafton Golf Course.   | Multi-Use Section 1          | 1,865 | LF   | \$7       | \$13,055  | \$13,055  |
| 34       | -          | -           | Thornburg         | Surface Change     | Proposed Abutment.                            | New Abutment                 | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| -        | 34-35      | 235         | Crafton/Thornburg | Multi-Use          | Proposed Bridge - New.                        | New Bridge                   | 1     | LS   | \$300,000 | \$300,000 | \$300,000 |
| 35       | -          | -           | Crafton           | Surface Change     | Proposed Abutment.                            | New Abutment                 | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| -        | 35-36      | 3,780       | Crafton           | Multi-Use          | Top of bank adjacent to CIT Ballfields.       | Multi-Use Section 1          | 3,780 | LF   | \$7       | \$26,460  | \$26,460  |
| 36       | -          | -           | -                 | Municipal Boundary | Crafton/Pittsburgh.                           | -                            | -     | -    | -         | -         | -         |
| -        | 36-37      | 1,995       | Pittsburgh        | Multi-Use          | Partially completed trail, requires bridge.   | Multi-Use Section 1          | 800   | LF   | \$7       | \$5,600   | \$5,600   |
| 37       | -          | -           | Pittsburgh        | Intersection       | Multi-Use trail/Turner St.                    | New Bridge over Bell's Run   | 1     | EA   | \$50,000  | \$50,000  | \$5,500   |
| -        | 37-38      | 570         | Pittsburgh        | On-Road - Basic    | Turner St.                                    | Directional Signage          | 2     | EA   | \$250     | \$500     | \$5,000   |
| 38       | -          | -           | Pittsburgh        | Junction           | Existing bridge endpoint.                     | Security Measures            | 1     | LS   | \$5,000   | \$5,000   | \$1,710   |
| -        | -          | -           | -                 | -                  | -                                             | Directional Signage          | 570   | LF   | \$3       | \$1,710   | \$1,710   |
| -        | -          | -           | -                 | -                  | -                                             | -                            | 2     | EA   | \$250     | \$500     | \$500     |



## Lower Chartiers Creek Greenway Trail Feasibility Study

## Estimated Development Cost Matrix

| Point ID | Segment ID | Length (FT) | Municipality | Type               | Existing Conditions                                | Ultimate Improvements*                  | QTY   | UNIT | UNIT COST | SUBTOTAL | TOTAL    |
|----------|------------|-------------|--------------|--------------------|----------------------------------------------------|-----------------------------------------|-------|------|-----------|----------|----------|
| -        | 38-38a     | 900         | Pittsburgh   | Hiking             | Spur to CNC Idlewood Property.                     | Hiking                                  | 900   | LF   | \$3       | \$2,700  | \$2,700  |
| -        | 38-39      | 175         | Pittsburgh   | Multi-Use          | Turner St. Bridge - proposed rehab.                | Bridge rehabilitation                   | 1     | LS   | \$60,000  | \$60,000 | \$60,000 |
| 38a      | -          | -           | Pittsburgh   | Terminus           | Idlewood Property.                                 | Trail Information Board                 | 1     | LS   | \$5,000   | \$5,000  | \$5,000  |
| 39       | -          | -           | Roslyn Farms | Surface Change     | Existing bridge endpoint.                          | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$15,500 |
| -        | 39-40      | 2,005       | Roslyn Farms | On-Road - Basic    | Arch St. w/ existing at-grade railroad crossing.   | Parking Improvements                    | 1     | LS   | \$10,000  | \$10,000 | \$15,500 |
| -        | 40         | -           | Roslyn Farms | Municipal Boundary | Arch St. w/ existing at-grade railroad crossing.   | Trail Information Board                 | 1     | LS   | \$5,000   | \$5,000  | \$15,500 |
| -        | 40-41      | 435         | Carnegie     | On-Road - Basic    | Roslyn Farms / Carnegie.                           | Roadway Section                         | 2,005 | LF   | \$3       | \$6,015  | \$6,015  |
| 41       | -          | -           | Carnegie     | Intersection       | Arch Street                                        | -                                       | -     | -    | -         | -        | -        |
| -        | 41-42      | 1,550       | Carnegie     | On-Road - Basic    | Arch Street crossing Chartiers via roadway bridge. | Roadway Section                         | 435   | LF   | \$3       | \$1,305  | \$1,305  |
| 42       | -          | -           | Carnegie     | Intersection       | Arch St./James St.                                 | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$10,500 |
| -        | 42-44      | 630         | Carnegie     | On-Road - Basic    | James St.                                          | Parking Improvements                    | 1     | LS   | \$10,000  | \$10,000 | \$10,500 |
| 43       | -          | -           | Carnegie     | Intersection       | Multi-Use Trail/Walnut St.                         | Directional Signage                     | 1,550 | LF   | \$3       | \$4,650  | \$4,650  |
| -        | 43-44      | 270         | Carnegie     | On-Road - Basic    | Walnut St.                                         | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| -        | 43-48      | 1,295       | Carnegie     | Multi-Use          | Top of bank behind businesses.                     | Roadway Section                         | 2     | EA   | \$250     | \$500    | \$500    |
| 44       | -          | -           | Carnegie     | Intersection       | James St./Walnut St.                               | Directional Signage                     | 1,295 | LF   | \$7       | \$9,065  | \$9,065  |
| -        | 44-45      | 445         | Carnegie     | On-Road - Basic    | James St.                                          | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| 45       | -          | -           | Carnegie     | Intersection       | James St./Broadway Ave.                            | Roadway Section                         | 445   | LF   | \$3       | \$1,335  | \$1,335  |
| -        | 45-46      | 140         | Carnegie     | On-Road - Basic    | Broadway Ave.                                      | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| 46       | -          | -           | Carnegie     | Intersection       | James St./Mansfield Blvd.                          | Roadway Section                         | 140   | LF   | \$3       | \$420    | \$420    |
| -        | 46-47      | 480         | Carnegie     | On-Road - Advanced | Mansfield Blvd.                                    | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| 47       | -          | -           | Carnegie     | Intersection       | Mansfield Blvd./Mansfield Blvd.                    | Roadway Section                         | 480   | LF   | \$3       | \$1,440  | \$1,440  |
| -        | 47-48      | 85          | Carnegie     | On-Road - Advanced | Mansfield Blvd.                                    | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| -        | 47-49      | 370         | Carnegie     | On-Road - Basic    | Mary St.                                           | Roadway Section                         | 85    | LF   | \$3       | \$255    | \$255    |
| 48       | -          | -           | Carnegie     | Intersection       | Mansfield Blvd. at-grade railroad crossing.        | Directional Signage                     | 370   | LF   | \$3       | \$1,110  | \$1,110  |
| -        | 48-PH3     | 4,360       | Carnegie     | On-Road - Advanced | West Main Street.                                  | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| 49       | -          | -           | Carnegie     | Intersection       | Mary St./Mall Plaza                                | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| -        | 49-50      | 1,005       | Carnegie     | On-Road - Basic    | Mall Plaza                                         | Roadway Section                         | 1,005 | LF   | \$3       | \$3,015  | \$3,015  |
| 50       | -          | -           | Carnegie     | Intersection       | Kinney St./Mall Plaza                              | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| -        | 50-51      | 330         | Carnegie     | On-Road - Basic    | Kinney St.                                         | Roadway Section                         | 330   | LF   | \$3       | \$990    | \$990    |
| 51       | -          | -           | Carnegie     | Intersection       | Kinney St./3rd Ave.                                | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| -        | 51-52      | 935         | Carnegie     | On-Road - Basic    | Kinney St./1st Ave.                                | Roadway Section                         | 935   | LF   | \$3       | \$2,805  | \$2,805  |
| 52       | -          | -           | Carnegie     | Intersection       | 1st & Carothers St.                                | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| -        | 52-53      | 265         | Carnegie     | On-Road - Advanced | Carothers St.                                      | Roadway Section                         | 265   | LF   | \$3       | \$795    | \$795    |
| 53       | -          | -           | Carnegie     | Municipal Boundary | Carnegie/Scott Township.                           | -                                       | -     | -    | -         | -        | -        |
| -        | 53-54      | 125         | Scott Twp.   | On-Road - Advanced | Carothers St.                                      | Roadway Section                         | 125   | LF   | \$3       | \$375    | \$375    |
| 54       | -          | -           | Scott Twp.   | Junction           | Multi-Use trail/Carothers St.                      | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| -        | 54-55      | 1,895       | Scott Twp.   | On-Road - Advanced | Carothers St.                                      | Security Measures (Panhandle Multi-Use) | 1     | LS   | \$5,000   | \$5,000  | \$5,000  |
| 55       | -          | -           | Scott Twp.   | Intersection       | Carothers St./Finley Ave.                          | Roadway Section                         | 1,895 | LF   | \$3       | \$5,685  | \$5,685  |
| -        | 55-56      | 565         | Scott Twp.   | On-Road - Advanced | Carothers St.                                      | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| 56       | -          | -           | Scott Twp.   | Intersection       | Carothers St./Washington Rd. (Rt.50)               | Roadway Section                         | 565   | LF   | \$3       | \$1,695  | \$1,695  |
| -        | 56-56a     | 560         | Scott Twp.   | On-Road - Advanced | Washington Rd. (Rt.50)                             | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| 56a      | -          | -           | Heidelberg   | Municipal Boundary | Washington Rd. (Rt.50)                             | Roadway Section                         | 560   | LF   | \$3       | \$1,680  | \$1,680  |
| -        | 56a-57     | 240         | Heidelberg   | On-Road - Advanced | Washington Rd. (Rt.50).                            | -                                       | -     | -    | -         | -        | -        |
| 57       | -          | -           | Heidelberg   | Intersection       | Washington Rd./Washington St.                      | Roadway Section                         | 240   | LF   | \$3       | \$720    | \$720    |
| -        | 57-58      | 1,350       | Heidelberg   | On-Road - Basic    | Washington St.                                     | Directional Signage                     | 2     | EA   | \$250     | \$500    | \$500    |
| -        | -          | -           | -            | -                  | -                                                  | Roadway Section                         | 1,350 | LF   | \$3       | \$4,050  | \$4,050  |

## Lower Chartiers Creek Greenway Trail Feasibility Study

## Estimated Development Cost Matrix

bold = Preferred Route

| Point ID | Segment ID | Length (FT) | Municipality          | Type               | Existing Conditions                         | Ultimate Improvements* | QTY   | UNIT | UNIT COST | SUBTOTAL  | TOTAL     |
|----------|------------|-------------|-----------------------|--------------------|---------------------------------------------|------------------------|-------|------|-----------|-----------|-----------|
| 57a      | 57-59      | 2,140       | Heidelberg            | On-Road - Advanced | Washington Rd. (Rt.50).                     | Roadway Section        | 2,140 | LF   | \$3       | \$6,420   | \$6,420   |
|          |            |             | Scott Twp.            | Surface Change     | Existing bridge endpoint.                   | Directional Signage    | 2     | EA   | \$250     | \$500     | \$6,920   |
|          |            |             |                       |                    |                                             | Security Measures      | 1     | LS   | \$5,000   | \$5,000   | \$11,920  |
| 57b      | 57a-PHS    | 2,370       | Scott Twp.            | Multi-Use          | Former rail line.                           | Multi-Use Section 1    | 2,370 | LF   | \$7       | \$16,590  | \$16,590  |
|          | 57a-57b    | 200         | Heidelberg            | Multi-Use          | Existing railroad bridge                    | Bridge rehabilitation  | 1     | LS   | \$100,000 | \$100,000 | \$116,590 |
|          |            |             |                       | Surface Change     | Existing bridge endpoint.                   | Directional Signage    | 2     | EA   | \$250     | \$500     | \$117,090 |
|          |            |             |                       |                    |                                             | Security Measures      | 1     | LS   | \$5,000   | \$5,000   | \$122,090 |
| 57c      | 57b-57c    | 305         | Scott Twp.            | Multi-Use          | Slope to roadway                            | Multi-Use Section 1    | 305   | LF   | \$7       | \$2,135   | \$2,135   |
|          |            |             | Heidelberg            | Intersection       | Proposed trail to meet roadway.             | Directional Signage    | 2     | EA   | \$250     | \$500     | \$2,635   |
|          |            |             |                       |                    |                                             | Security Measures      | 1     | LS   | \$5,000   | \$5,000   | \$7,635   |
| 57d      | 57c-57d    | 1,505       | Heidelberg            | On-Road - Basic    | Grant St.                                   | Roadway Section        | 1,505 | LF   | \$3       | \$4,515   | \$4,515   |
|          |            |             | Heidelberg            | Intersection       | Grant St./Hays St.                          | Directional Signage    | 2     | EA   | \$250     | \$500     | \$5,015   |
|          | 57d-57e    | 930         | Heidelberg            | On-Road - Basic    | Hays St.                                    | Roadway Section        | 930   | LF   | \$3       | \$2,790   | \$2,790   |
| 57e      |            |             | Heidelberg            | Intersection       | Hays St./Madison Ave.                       | Directional Signage    | 2     | EA   | \$250     | \$500     | \$3,290   |
|          | 57e-57f    | 360         | Heidelberg            | On-Road - Basic    | Madison Ave.                                | Roadway Section        | 360   | LF   | \$3       | \$1,080   | \$1,080   |
| 57f      |            |             | Heidelberg            | Intersection       | Madison Ave./Elisworth Ave.                 | Directional Signage    | 2     | EA   | \$250     | \$500     | \$1,580   |
|          | 57f-57g    | 715         | Heidelberg            | On-Road - Basic    | Elisworth Ave.                              | Roadway Section        | 715   | LF   | \$3       | \$2,145   | \$2,145   |
| 57g      |            |             | Heidelberg            | Intersection       | Elisworth Ave./Collier Ave.                 | Directional Signage    | 2     | EA   | \$250     | \$500     | \$2,645   |
|          | 57g-59     | 485         | Heidelberg            | On-Road - Advanced | Collier Ave.                                | Roadway Section        | 485   | LF   | \$3       | \$1,455   | \$1,455   |
| 58       |            |             | Heidelberg            | Intersection       | Washington St./3rd St.                      | Directional Signage    | 2     | EA   | \$250     | \$500     | \$1,955   |
|          | 58-59      | 1,240       | Heidelberg            | On-Road - Basic    | Washington St.                              | Roadway Section        | 1,240 | LF   | \$3       | \$3,720   | \$3,720   |
| 59       |            |             | Heidelberg            | Intersection       | Washington Rd./Washington St.               | Directional Signage    | 2     | EA   | \$250     | \$500     | \$4,220   |
|          | 59-60      | 370         | Heidelberg            | On-Road - Advanced | Washington Rd. (Rt.50)                      | Roadway Section        | 370   | LF   | \$3       | \$1,110   | \$1,110   |
| 60       |            |             | Heidelberg            | Intersection       | Washington Rd. (Rt.50)/Greentree Rd.        | Directional Signage    | 2     | EA   | \$250     | \$500     | \$1,610   |
|          | 60-61      | 130         | Heidelberg            | On-Road - Advanced | Washington Rd. (Rt.50)                      | Roadway Section        | 130   | LF   | \$3       | \$390     | \$390     |
| 61       |            |             | Scott Twp.            | Municipal Boundary | Heidelberg/Scott Twp.                       |                        |       |      |           |           |           |
|          | 61-62      | 1,350       | Scott Twp.            | On-Road - Advanced | Washington Rd. (Rt.50)                      | Roadway Section        | 1,350 | LF   | \$3       | \$4,050   | \$4,050   |
| 62       |            |             | Collier Twp.          | Municipal Boundary | Collier Twp./Scott Twp.                     |                        |       |      |           |           |           |
|          | 62-63      | 3,200       | Collier Twp.          | On-Road - Advanced | Washington Rd. (Rt.50)                      | Roadway Section        | 3,200 | LF   | \$3       | \$9,600   | \$9,600   |
| 63       |            |             | Collier Twp.          | Intersection       | Washington Rd./Steen Rd.                    | Directional Signage    | 2     | EA   | \$250     | \$500     | \$10,100  |
|          | 63-64      | 1,815       | Collier Twp.          | On-Road - Basic    | Steen Rd.                                   | Roadway Section        | 1,815 | LF   | \$3       | \$5,445   | \$15,545  |
|          | 63-73      | 5,470       | Collier Twp.          | On-Road - Advanced | Washington Rd. (Rt.50)                      | Roadway Section        | 5,470 | LF   | \$3       | \$16,410  | \$31,955  |
| 64       |            |             | Collier Twp.          | Junction           | Multi-Use Trail/Steen Rd.                   | Directional Signage    | 2     | EA   | \$250     | \$500     | \$32,455  |
|          | 64-65      | 2,310       | Collier Twp.          | On-Road - Basic    | Steen Rd.                                   | Security Measures      | 1     | LS   | \$5,000   | \$5,000   | \$37,455  |
|          | 64-68      | 2,490       | Collier Twp.          | Multi-Use          | Top of northern bank, behind Trader Jack's. | Roadway Section        | 2,310 | LF   | \$3       | \$6,930   | \$44,385  |
| 65       |            |             | Collier Twp.          | Intersection       | Thom's Run Rd./Steen Rd.                    | Multi-Use Section 2    | 2,490 | LF   | \$12      | \$29,880  | \$74,265  |
|          | 65-66      | 1,630       | Collier Twp.          | On-Road - Basic    | Thom's Run Rd.                              | Directional Signage    | 2     | EA   | \$250     | \$500     | \$74,765  |
|          | 66         |             | Collier Twp.          | Intersection       | Prestley Rd./Thom's Run Rd.                 | Roadway Section        | 1,630 | LF   | \$3       | \$4,890   | \$79,655  |
|          | 66-66a     | 265         | Collier Twp.          | On-Road - Basic    | Prestley Rd.                                | Directional Signage    | 2     | EA   | \$250     | \$500     | \$80,155  |
| 66a      |            |             | Collier Twp.          | Junction           | Multi-Use Trail/Prestley Rd.                | Roadway Section        | 265   | LF   | \$3       | \$795     | \$80,950  |
|          | 66a-67     | 495         | Collier Twp.          | On-Road - Basic    | Prestley Rd.                                | Directional Signage    | 2     | EA   | \$250     | \$500     | \$81,450  |
|          | 66a-66b    | 3,140       | Collier Twp.          | Multi-Use          | Top of eastern bank.                        | Security Measures      | 1     | LS   | \$5,000   | \$5,000   | \$86,450  |
| 66b      |            |             | Collier Twp.          | Surface Change     | Proposed Abutment.                          | Roadway Section        | 495   | LF   | \$3       | \$1,485   | \$87,935  |
|          | 66b-66c    | 180         | Collier/South Fayette | Multi-Use          | Proposed Abutment.                          | Multi-Use Section 2    | 3,140 | LF   | \$12      | \$37,680  | \$125,615 |
|          | 66c        |             | South Fayette         | Surface Change     | Proposed Bridge - New.                      | New Abutment           | 1     | LS   | \$15,000  | \$15,000  | \$140,615 |
|          | 66c-66d    | 1,050       | South Fayette         | Multi-Use          | Proposed Abutment.                          | New Bridge             | 1     | LS   | \$250,000 | \$250,000 | \$390,615 |
|          | 66d        |             | South Fayette         | Surface Change     | Top of western bank.                        | New Abutment           | 1     | LS   | \$15,000  | \$15,000  | \$405,615 |
|          | 66d-66e    | 180         | South Fayette         | Multi-Use          | Proposed Abutment.                          | Multi-Use Section 2    | 1,050 | LF   | \$12      | \$12,600  | \$418,215 |
| 66e      |            |             | South Fayette         | Surface Change     | Proposed Abutment.                          | New Abutment           | 1     | LS   | \$15,000  | \$15,000  | \$433,215 |
|          | 66e-66f    |             | South Fayette         | Multi-Use          | Proposed Bridge - New.                      | New Bridge             | 1     | LS   | \$250,000 | \$250,000 | \$683,215 |
|          |            |             | South Fayette         | Surface Change     | Proposed Abutment.                          | New Abutment           | 1     | LS   | \$15,000  | \$15,000  | \$698,215 |

## Lower Chartiers Creek Greenway Trail Feasibility Study

## Estimated Development Cost Matrix

bold = Preferred Route

| Point ID | Segment ID | Length (FT) | Municipality    | Type                     | Existing Conditions                           | Ultimate Improvements*  | QTY   | UNIT | UNIT COST | SUBTOTAL  | TOTAL     |
|----------|------------|-------------|-----------------|--------------------------|-----------------------------------------------|-------------------------|-------|------|-----------|-----------|-----------|
| -        | 66-66f     | 4,755       | South Fayette   | Multi-Use Junction       | Top of western bank.                          | Multi-Use Section 2     | 4,755 | LF   | \$12      | \$57,060  | \$57,060  |
| 66f      | -          | -           | South Fayette   | Junction                 | Multi-Use Trail/Washington Rd. (Rt.50)        | Directional Signage     | 2     | EA   | \$250     | \$500     | \$5,500   |
|          |            |             |                 |                          |                                               | Security Measures       | 1     | LS   | \$5,000   | \$5,000   |           |
| 67       | -          | -           | Collier Twp.    | Intersection             | Multi-Use Trail/Prestley Rd.                  | Directional Signage     | 2     | EA   | \$250     | \$500     | \$5,500   |
|          |            |             |                 |                          |                                               | Security Measures       | 1     | LS   | \$5,000   | \$5,000   |           |
| -        | 67-69      | 735         | Collier Twp.    | Multi-Use Surface Change | Connection to proposed bridge.                | Multi-Use Section 1     | 735   | LF   | \$7       | \$5,145   | \$5,145   |
| 68       | -          | -           | Collier Twp.    | Surface Change           | Proposed Abutment.                            | New Abutment            | 1     | LS   | \$20,000  | \$20,000  | \$20,000  |
| -        | 68-69      | 200         | Collier Twp.    | Multi-Use Surface Change | Proposed Bridge - New.                        | New Bridge              | 1     | LS   | \$275,000 | \$275,000 | \$275,000 |
| 69       | -          | -           | Collier Twp.    | Surface Change           | Proposed Abutment.                            | New Abutment            | 1     | LS   | \$20,000  | \$20,000  | \$20,000  |
| -        | 69-70      | 3,040       | Collier Twp.    | Multi-Use Surface Change | Western bank of former channel, beneath I-79. | Multi-Use Section 2     | 3,040 | LF   | \$12      | \$36,480  | \$36,480  |
| 70       | -          | -           | Collier Twp.    | Surface Change           | Proposed Abutment.                            | New Abutment            | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| -        | 70-71      | 115         | Collier Twp.    | Multi-Use Surface Change | Proposed Bridge - New.                        | New Bridge              | 1     | LS   | \$200,000 | \$200,000 | \$200,000 |
| 71       | -          | -           | Collier Twp.    | Surface Change           | Proposed Abutment.                            | New Abutment            | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| -        | 71-72      | 130         | Collier Twp.    | Multi-Use Surface Change | Connection to shopping center parking lot.    | Multi-Use Section 2     | 130   | LF   | \$12      | \$1,560   | \$1,560   |
| 72       | -          | -           | Collier Twp.    | Surface Change           | Multi-Use Trail/Parking lot.                  | Trail Information Board | 1     | LS   | \$5,000   | \$5,000   | \$10,000  |
|          |            |             |                 |                          |                                               | Security Measures       | 1     | LS   | \$5,000   | \$5,000   |           |
| -        | 72-73      | 1,585       | Collier Twp.    | On-Road - Basic Junction | Through existing parking lot.                 | Roadway Section         | 1,585 | LF   | \$3       | \$4,755   | \$4,755   |
| 73       | -          | -           | Collier Twp.    | Junction                 | Connection to Washington Rd. (Rt. 50).        | Directional Signage     | 2     | EA   | \$250     | \$500     | \$900     |
| -        | 73-74      | 365         | Collier Twp.    | On-Road - Advanced       | Washington Rd. (Rt.50).                       | Roadway Section         | 365   | LF   | \$3       | \$1,095   | \$1,095   |
| 74       | -          | -           | -               | Municipal Boundary       | Collier Twp./Bridgeville.                     | -                       | -     | -    | -         | -         | -         |
| -        | 74-75      | 3,405       | Bridgeville     | On-Road - Advanced       | Washington Rd. (Rt.50).                       | Roadway Section         | 3,405 | LF   | \$3       | \$10,215  | \$10,215  |
| 75       | -          | -           | -               | Municipal Boundary       | Bridgeville/South Fayette.                    | -                       | -     | -    | -         | -         | -         |
| -        | 75-76      | 2,870       | South Fayette   | On-Road - Advanced       | Washington Rd. (Rt.50).                       | Roadway Section         | 2,870 | LF   | \$3       | \$8,610   | \$8,610   |
| 76       | -          | -           | South Fayette   | Intersection             | Bursca Dr./Washington Rd. (Rt.50)             | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 76-77      | 900         | South Fayette   | On-Road - Basic          | Bursca Dr.                                    | Roadway Section         | 900   | LF   | \$3       | \$2,700   | \$2,700   |
| -        | 76-80      | 8,485       | South Fayette   | On-Road - Advanced       | Washington Rd. (Rt.50).                       | Roadway Section         | 8,485 | LF   | \$3       | \$25,455  | \$25,455  |
| 77       | -          | -           | South Fayette   | Surface Change           | Proposed Abutment.                            | New Abutment            | 1     | LS   | \$15,000  | \$15,000  | \$20,000  |
|          |            |             |                 |                          |                                               | Security Measures       | 1     | LS   | \$5,000   | \$5,000   |           |
| -        | 77-78      | 115         | SF/USC          | Multi-Use Surface Change | Proposed Bridge - New.                        | New Bridge              | 1     | LS   | \$200,000 | \$200,000 | \$200,000 |
| 78       | -          | -           | Upper St. Clair | Surface Change           | Proposed Abutment.                            | Multi-Use Section 1     | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| -        | 78-79      | 485         | Upper St. Clair | Multi-Use Intersection   | Chartiers Park.                               | Multi-Use Section 2     | 485   | LF   | \$12      | \$5,820   | \$5,820   |
| 79       | -          | -           | Upper St. Clair | Intersection             | Park road.                                    | Directional Signage     | 2     | EA   | \$250     | \$500     | \$15,500  |
|          |            |             |                 |                          |                                               | Parking Improvements    | 1     | LS   | \$10,000  | \$10,000  |           |
|          |            |             |                 |                          |                                               | Trail Information Board | 1     | LS   | \$5,000   | \$5,000   |           |
| -        | 79-81      | 6,985       | Upper St. Clair | Multi-Use                | Eastern bank through ALF property.            | Multi-Use Section 2     | 6,985 | LF   | \$12      | \$83,940  | \$83,940  |
| 80       | -          | -           | South Fayette   | Intersection             | Boyce Rd./Washington Rd.                      | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 80-89      | 7,515       | South Fayette   | On-Road - Advanced       | Boyce Rd.                                     | Roadway Section         | 7,515 | LF   | \$3       | \$22,545  | \$22,545  |
| 81       | -          | -           | Upper St. Clair | Junction                 | Mayview Rd./Multi-Use Trail.                  | Directional Signage     | 2     | EA   | \$250     | \$500     | \$5,500   |
|          |            |             |                 |                          |                                               | Security Measures       | 1     | LS   | \$5,000   | \$5,000   |           |
| -        | 81-82      | 180         | SF/USC          | On-Road - Basic Junction | Mayview Rd.                                   | Roadway Section         | 180   | LF   | \$3       | \$480     | \$480     |
| 82       | -          | -           | South Fayette   | Junction                 | Mayview Rd./Multi-Use Trail.                  | Directional Signage     | 2     | EA   | \$250     | \$500     | \$5,500   |
|          |            |             |                 |                          |                                               | Security Measures       | 1     | LS   | \$5,000   | \$5,000   |           |
| -        | 82-83      | 245         | South Fayette   | On-Road - Basic          | Mayview Rd.                                   | Roadway Section         | 245   | LF   | \$3       | \$735     | \$735     |
| -        | 82-85      | 2,305       | South Fayette   | Multi-Use Intersection   | Along southern bank of Chartiers.             | Multi-Use Section 2     | 2,305 | LF   | \$12      | \$27,660  | \$27,660  |
| 83       | -          | -           | South Fayette   | Intersection             | North hospital entrance.                      | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 83-84      | 1,625       | South Fayette   | On-Road - Basic          | Hospital road.                                | Roadway Section         | 1,625 | LF   | \$3       | \$4,875   | \$4,875   |
| 84       | -          | -           | South Fayette   | Intersection             | Hospital road intersection.                   | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 84-87      | 2,790       | South Fayette   | On-Road - Basic Junction | Hospital road.                                | Roadway Section         | 2,790 | LF   | \$3       | \$8,370   | \$8,370   |
| 85       | -          | -           | South Fayette   | Junction                 | End of utility/access road.                   | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 85-86      | 635         | South Fayette   | Multi-Use                | Former utility/access road.                   | Multi-Use Section 2     | 635   | LF   | \$12      | \$7,620   | \$7,620   |
| 86       | -          | -           | South Fayette   | Junction                 | At-grade railroad crossing.                   | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |



Estimated Development Cost Matrix

| Point ID | Segment ID | Length (FT) | Municipality  | Type               | Existing Conditions                          | Ultimate Improvements*  | QTY   | UNIT | UNIT COST | SUBTOTAL  | TOTAL     |
|----------|------------|-------------|---------------|--------------------|----------------------------------------------|-------------------------|-------|------|-----------|-----------|-----------|
| -        | -          | -           | -             | -                  | -                                            | -                       | -     | -    | -         | -         | -         |
| 87       | 86-88      | 4,450       | South Fayette | Multi-Use          | Between Creek and existing railroad.         | Multi-Use Section 2     | 4,450 | LF   | \$12      | \$53,400  | \$53,400  |
| -        | -          | -           | South Fayette | Intersection       | Mayview Rd./Hospital Rd.                     | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| 88       | 87-88      | 765         | South Fayette | On-Road - Advanced | Mayview Rd.                                  | Roadway Section         | 765   | LF   | \$3       | \$2,295   | \$2,295   |
| -        | -          | -           | South Fayette | Intersection       | Mayview Rd./Multi-Use Trail                  | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | -          | -           | -             | -                  | -                                            | Parking Improvements    | 1     | LS   | \$10,000  | \$10,000  | \$10,000  |
| -        | -          | -           | -             | -                  | -                                            | Trail Information Board | 1     | LS   | \$5,000   | \$5,000   | \$5,000   |
| -        | -          | -           | -             | -                  | -                                            | Security Measures       | 1     | LS   | \$5,000   | \$5,000   | \$5,000   |
| -        | 88-89      | 2,790       | South Fayette | On-Road - Advanced | Mayview Rd.                                  | Roadway Section         | 2,790 | LF   | \$3       | \$8,370   | \$8,370   |
| -        | 88-89a     | 2,665       | South Fayette | Multi-Use          | Along western bank of creek.                 | Multi-Use Section 2     | 2,665 | LF   | \$12      | \$31,980  | \$31,980  |
| 89       | -          | -           | South Fayette | Intersection       | Boyce Rd./Mayview Rd.                        | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 89-89a     | 335         | South Fayette | On-Road - Basic    | Boyce Rd.                                    | Roadway Section         | 335   | LF   | \$3       | \$1,005   | \$1,005   |
| -        | 89-90      | 2,390       | South Fayette | On-Road - Advanced | Mayview Rd.                                  | Roadway Section         | 2,390 | LF   | \$3       | \$7,170   | \$7,170   |
| 89a      | -          | -           | South Fayette | Junction           | Boyce Rd. crossing.                          | Directional Signage     | 2     | EA   | \$250     | \$500     | \$5,500   |
| -        | -          | -           | -             | -                  | -                                            | Security Measures       | 1     | LS   | \$5,000   | \$5,000   | \$5,500   |
| -        | 89a-91     | 2,945       | South Fayette | Multi-Use          | Along western bank of creek.                 | Multi-Use Section 2     | 2,945 | LF   | \$12      | \$35,340  | \$35,340  |
| 90       | -          | -           | -             | County Line        | Allegheny/Washington Counties                | -                       | -     | -    | -         | -         | -         |
| -        | 90-96      | 4,170       | Cecil Twp.    | On-Road - Advanced | Mayview Rd.                                  | Roadway Section         | 4,170 | LF   | \$3       | \$12,510  | \$12,510  |
| 91       | -          | -           | -             | County Line        | Allegheny/Washington Counties                | Special Sign            | 1     | LS   | \$1,000   | \$1,000   | \$1,000   |
| -        | 91-92      | 650         | Cecil Twp.    | Multi-Use          | Along western bank of creek.                 | Multi-Use Section 2     | 650   | LF   | \$12      | \$7,800   | \$7,800   |
| 92       | -          | -           | Cecil Twp.    | Surface Change     | Proposed Abutment.                           | New Abutment            | 1     | LS   | \$20,000  | \$20,000  | \$20,000  |
| -        | 92-93      | 285         | Cecil/Peters  | Multi-Use          | Proposed Bridge - New.                       | New Bridge              | 1     | LS   | \$300,000 | \$300,000 | \$300,000 |
| 93       | -          | -           | Peters Twp.   | Surface Change     | Proposed Abutment.                           | New Abutment            | 1     | LS   | \$20,000  | \$20,000  | \$20,000  |
| -        | 93-94      | 1,915       | Peters Twp.   | Multi-Use          | Along eastern bank of creek.                 | Multi-Use Section 2     | 1,915 | LF   | \$12      | \$22,980  | \$22,980  |
| 94       | -          | -           | Peters Twp.   | Surface Change     | Proposed Abutment.                           | New Abutment            | 1     | LS   | \$20,000  | \$20,000  | \$20,000  |
| -        | 94-95      | 210         | Cecil/Peters  | Multi-Use          | Proposed Bridge - New.                       | New Bridge              | 1     | LS   | \$275,000 | \$275,000 | \$275,000 |
| 95       | -          | -           | Cecil Twp.    | Surface Change     | Proposed Abutment.                           | New Abutment            | 1     | LS   | \$20,000  | \$20,000  | \$20,000  |
| -        | 95-98      | 1,125       | Cecil Twp.    | Multi-Use          | Private property.                            | Multi-Use Section 2     | 1,125 | LF   | \$12      | \$13,500  | \$13,500  |
| 96       | -          | -           | Cecil Twp.    | Intersection       | Georgetown Rd./Mayview Rd.                   | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 96-97      | 1,860       | Cecil Twp.    | On-Road - Advanced | Georgetown Rd.                               | Roadway Section         | 1,860 | LF   | \$3       | \$5,580   | \$5,580   |
| 97       | -          | -           | Cecil Twp.    | Intersection       | Baker Rd./Georgetown Rd.                     | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 97-103     | 4,320       | Cecil Twp.    | On-Road - Advanced | Georgetown Rd.                               | Roadway Section         | 4,320 | LF   | \$3       | \$12,960  | \$12,960  |
| 98       | -          | -           | Cecil Twp.    | Intersection       | Multi-Use Trail/Georgetown Rd.               | Directional Signage     | 2     | EA   | \$250     | \$500     | \$5,500   |
| -        | -          | -           | -             | -                  | -                                            | Security Measures       | 1     | LS   | \$5,000   | \$5,000   | \$5,500   |
| 98-99    | 705        | -           | Cecil Twp.    | On-Road - Basic    | Georgetown Rd.                               | Roadway Section         | 705   | LF   | \$3       | \$2,115   | \$2,115   |
| 99       | -          | -           | Peters Twp.   | Intersection       | Buckeye St./Georgetown Rd.                   | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 99-100     | 5,205       | Peters Twp.   | On-Road - Basic    | Buckeye St.                                  | Roadway Section         | 5,205 | LF   | \$3       | \$15,615  | \$15,615  |
| 100      | -          | -           | Peters Twp.   | Intersection       | Proposed railroad crossing.                  | Directional Signage     | 2     | EA   | \$250     | \$500     | \$5,500   |
| -        | -          | -           | -             | -                  | -                                            | Security Measures       | 1     | LS   | \$5,000   | \$5,000   | \$5,500   |
| -        | 100-101    | 4,780       | Peters Twp.   | Multi-Use          | Top of bank around Country Club golf course. | Multi-Use Section 1     | 4,780 | LF   | \$7       | \$33,460  | \$33,460  |
| 101      | -          | -           | Peters Twp.   | Surface Change     | Proposed Abutment.                           | New Abutment            | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| -        | 101-102    | 165         | Cecil/Peters  | Multi-Use          | Proposed Bridge - New.                       | New Bridge              | 1     | LS   | \$225,000 | \$225,000 | \$225,000 |
| 102      | -          | -           | Cecil Twp.    | Surface Change     | Proposed Abutment.                           | New Abutment            | 1     | LS   | \$15,000  | \$15,000  | \$15,000  |
| -        | 102-104    | 1,545       | Cecil Twp.    | Multi-Use          | Along southern/western bank of creek.        | Multi-Use Section 2     | 1,545 | LF   | \$12      | \$18,540  | \$18,540  |
| 103      | -          | -           | Cecil Twp.    | Intersection       | Georgetown Rd./Hahn Rd.                      | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 103-105    | 5,370       | Cecil Twp.    | On-Road - Basic    | Hahn Rd.                                     | Roadway Section         | 5,370 | LF   | \$3       | \$16,110  | \$16,110  |
| 104      | -          | -           | Cecil Twp.    | Intersection       | Multi-Use Trail/Railroad.                    | Directional Signage     | 2     | EA   | \$250     | \$500     | \$5,500   |
| -        | -          | -           | -             | -                  | -                                            | Security Measures       | 1     | LS   | \$5,000   | \$5,000   | \$5,500   |
| -        | 104-106    | 1,570       | Cecil Twp.    | Multi-Use          | Along western bank of creek.                 | Multi-Use Section 2     | 1,570 | LF   | \$12      | \$18,840  | \$18,840  |
| 105      | -          | -           | Cecil Twp.    | Intersection       | Hahn Rd./Maple Ln.                           | Directional Signage     | 2     | EA   | \$250     | \$500     | \$500     |
| -        | 105-106    | 295         | Cecil Twp.    | On-Road - Basic    | Maple Ln.                                    | Roadway Section         | 295   | LF   | \$3       | \$885     | \$885     |
| -        | 105-107    | 7,650       | Cecil Twp.    | On-Road - Basic    | Hahn Rd.                                     | Roadway Section         | 7,650 | LF   | \$3       | \$22,950  | \$22,950  |

## Lower Chartiers Creek Greenway Trail Feasibility Study

## Estimated Development Cost Matrix

bold = Preferred Route

| Point ID | Segment ID | Length (FT) | Municipality   | Type               | Existing Conditions                                 | Ultimate Improvements*         | QTY   | UNIT | UNIT COST | SUBTOTAL | TOTAL     |
|----------|------------|-------------|----------------|--------------------|-----------------------------------------------------|--------------------------------|-------|------|-----------|----------|-----------|
| 106      | -          | -           | Cecil Twp.     | Junction           | Multi-Use Trail/Maple Ln.                           | Directional Signage            | 2     | EA   | \$250     | \$500    | \$5,500   |
| -        | 106-112    | 7,035       | Cecil/Peters   | Multi-Use          | Along western bank of creek & former grade to Lake. | Security Measures              | 1     | LS   | \$5,000   | \$5,000  | \$334,420 |
| 107      | -          | -           | Cecil Twp.     | Intersection       | Atlas Ln./Hahn Rd.                                  | Multi-Use Section 2 New Bridge | 7,035 | LF   | \$12      | \$84,420 | \$250,000 |
| 108      | 107-108    | 735         | Cecil Twp.     | On-Road - Basic    | Hahn Rd.                                            | Directional Signage            | 2     | EA   | \$250     | \$500    | \$500     |
| -        | 108-109    | -           | -              | Municipal Boundary | Cecil/North Strabane                                | Roadway Section                | 735   | LF   | \$3       | \$2,205  | \$2,205   |
| 109      | -          | -           | North Strabane | On-Road - Basic    | Hahn Rd.                                            | Roadway Section                | 280   | LF   | \$3       | \$840    | \$840     |
| -        | 109-110    | 1,325       | North Strabane | Intersection       | Hahn Rd./W. McMurray Rd.                            | Directional Signage            | 2     | EA   | \$250     | \$500    | \$500     |
| 110      | -          | -           | -              | Municipal Boundary | North Strabane/Peters                               | Roadway Section                | 1,325 | LF   | \$3       | \$3,975  | \$3,975   |
| -        | 110-111    | -           | Peters Twp.    | On-Road - Basic    | W. McMurray Rd.                                     | -                              | -     | -    | -         | -        | -         |
| 111      | -          | -           | Peters Twp.    | Intersection       | W. McMurray Rd./Evergreen Rd.                       | Roadway Section                | 230   | LF   | \$3       | \$690    | \$690     |
| -        | 111-112    | 385         | Peters Twp.    | On-Road - Basic    | W. McMurray Rd.                                     | Directional Signage            | 2     | EA   | \$250     | \$500    | \$500     |
| 112      | -          | -           | Peters Twp.    | Junction           | Multi-Use Trail/W. McMurray Rd.                     | Roadway Section                | 385   | LF   | \$3       | \$1,155  | \$1,155   |
| -        | -          | -           | -              | -                  | -                                                   | Directional Signage            | 2     | EA   | \$250     | \$500    | \$15,500  |
| -        | -          | -           | -              | -                  | -                                                   | Parking Improvements           | 1     | LS   | \$10,000  | \$10,000 | \$5,000   |
| -        | -          | -           | -              | -                  | -                                                   | Security Measures              | 1     | LS   | \$5,000   | \$5,000  | \$5,000   |

## Panthandle Extension

| Point ID  | Segment ID | Length (FT) | Municipality        | Type               | Existing Conditions                              | Ultimate Improvements* | QTY    | UNIT | UNIT COST | SUBTOTAL | TOTAL    |
|-----------|------------|-------------|---------------------|--------------------|--------------------------------------------------|------------------------|--------|------|-----------|----------|----------|
| PH1       | -          | -           | Collier Twp.        | Junction           | Beginning of Panhandle Trail.                    | Directional Signage    | 2      | EA   | \$250     | \$500    | \$500    |
| -         | PH1-PH2    | 710         | Collier Twp.        | On-Road - Basic    | Walkers Mill Rd.                                 | Roadway Section        | 710    | LF   | \$3       | \$2,130  | \$2,130  |
| PH1a      | -          | 6,960       | Collier Twp.        | Multi-Use          | South bank of Robinson Run, paralleling RR line. | Multi-Use Section 2    | 6,960  | LF   | \$12      | \$83,520 | \$83,520 |
| PH1a-PH2a | -          | 1,580       | Collier Twp.        | Surface Change     | Cuxton Rd.                                       | Security Measures      | 1      | LS   | \$5,000   | \$5,000  | \$5,000  |
| PH2       | -          | -           | Collier Twp.        | On-Road - Basic    | Cuxton Rd.                                       | Roadway Section        | 1,580  | LF   | \$3       | \$4,740  | \$4,740  |
| PH2-PH3   | -          | 10,400      | Collier Twp.        | Intersection       | Walkers Mill Rd./Old Nobletown Rd.               | Directional Signage    | 2      | EA   | \$250     | \$500    | \$500    |
| PH3       | -          | -           | Collier Twp.        | On-Road - Advanced | Old Nobletown Road.                              | Roadway Section        | 10,400 | LF   | \$3       | \$31,200 | \$31,200 |
| -         | PH3-PH4    | -           | -                   | Municipal Boundary | Old Nobletown Road/Cuxton Rd.                    | Directional Signage    | 2      | EA   | \$250     | \$500    | \$500    |
| PH4       | -          | 70          | Carnegie            | Multi-Use          | Carnegie/Collier Twp.                            | -                      | -      | -    | -         | -        | -        |
| -         | PH4-PH5    | 195         | Carnegie            | Surface Change     | Bridge approach, former rail line.               | Multi-Use Section 1    | 70     | LF   | \$7       | \$490    | \$490    |
| PH5       | -          | -           | Carnegie/Scott Twp. | Multi-Use          | Existing Abutment.                               | Security Measures      | 1      | LS   | \$5,000   | \$5,000  | \$5,000  |
| -         | PH5-PH6    | 355         | Scott Twp.          | Surface Change     | Proposed Bridge - Rehab.                         | Bridge Rehabilitation  | 1      | LS   | \$80,000  | \$80,000 | \$80,000 |
| PH6       | -          | -           | Scott Twp.          | Multi-Use          | Existing Abutment.                               | Security Measures      | 1      | LS   | \$5,000   | \$5,000  | \$5,000  |
| -         | PH6-PH7    | 1,520       | Scott Twp.          | Intersection       | Former rail line.                                | Multi-Use Section 1    | 355    | LF   | \$7       | \$2,485  | \$2,485  |
| PH7       | -          | -           | Scott Twp.          | Multi-Use          | Multi-Use Trail/Gregg Rd.                        | Directional Signage    | 2      | EA   | \$250     | \$500    | \$500    |
| -         | -          | -           | -                   | Surface Change     | Former rail line.                                | Multi-Use Section 1    | 1,520  | LF   | \$7       | \$10,640 | \$10,640 |
| -         | -          | -           | -                   | Multi-Use          | End of rail line.                                | Directional Signage    | 2      | EA   | \$250     | \$500    | \$5,500  |
| -         | -          | -           | -                   | Surface Change     | Wooded slope.                                    | Security Measures      | 1      | LS   | \$5,000   | \$5,000  | \$5,500  |
| -         | PH7-54     | 935         | Scott Twp.          | Multi-Use          | -                                                | Multi-Use Section 4    | 935    | LF   | \$22      | \$20,570 | \$25,245 |
| -         | -          | -           | -                   | -                  | -                                                | Split Rail Fencing     | 935    | LF   | \$5       | \$4,675  | \$4,675  |

SUBTOTAL FOR ALL PROPOSED TRAIL POINTS AND SEGMENTS

\$5,773,170

SUBTOTAL FOR ALL PREFERRED INTERMEDIATE ROUTE TRAIL POINTS AND SEGMENTS

\$1,820,830

## FUNDING MATRIX



Potential Funding Sources Matrix  
Charlottesville Trail and Greenway  
Public Agencies

| FUNDING PROGRAM                                                                                                                                        | JURISDICTION                               | CONTACT INFORMATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | INTENDED FUNDING USES                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pennsylvania Department of Transportation (PennDOT)<br>Transportation Enhancement Program                                                              | Statewide<br>State and Federally<br>Funded | Dante C. Accardi<br>Bureau of Highway Safety & Traffic Engineering<br>400 North Street-8th Floor<br>P.O. Box 3365<br>Harrisburg PA 17105-3365<br>Ph: 717-783-2258; FAX: 717-787-5247<br>E-mail: daccardi@dot.state.pa.us<br>Kathy Franklin<br>Tracy Robinson<br>DCNR<br>1405 State Office Building<br>300 Liberty Ave.<br>Pittsburgh, PA 15222<br>(412) 880-0486<br>USDA Forest Service<br>180 Canfield St.<br>Morgantown, WV 26505<br>Tel: 304-285-1530<br>Fax: 304-285-1564<br>Email: ecora@fs.fed.us<br>http://www.fs.fed.us/natl/<br>Lary G. Williamson, Director<br>DCNR's Bureau of Recreation & Conservation<br>P.O. Box 9475<br>Harrisburg, PA 17105<br>(717) 783-2659<br>williamson@dcnr.state.pa.us | Eligible intermodal transportation projects                                                                                                                                                                                                                                     |
| DCNR Community Conservation Partnerships Program                                                                                                       | Statewide<br>State and Federally<br>Funded |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Land Acquisition, trails parks                                                                                                                                                                                                                                                  |
| Wood in Transportation Program                                                                                                                         | Nationwide<br>Funded                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Provide funding for projects which use wood materials.                                                                                                                                                                                                                          |
| Community Conservation Partnerships Program<br>(Growing Greener) / PADCNR                                                                              | Pennsylvania<br>State Funded               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | State Greenway and Trail Projects                                                                                                                                                                                                                                               |
| Community Development Block Grant Program (CDBG) / U.S.<br>Department of Housing and Urban Development (HUD)                                           | Nationwide<br>Funded                       | Barbara Neal, Director, Entitlement Communities Division<br>451 7th Street SW, Room 7282<br>Washington, DC 20410<br>(202) 706-1577                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CDBG provides eligible metropolitan cities and urban counties with annual grants to revitalize neighborhoods and improve community facilities. Several communities have used HUD funds to develop greenways, including the Bosobel Heights Safe Walk Greenway in Nashville, TN. |
| Economic Development Grants for Public Works and Development<br>Facilities / U.S. Department of Commerce, Economic Development<br>Administration (EDA) | Nationwide<br>Funded                       | David L. McKeon, Director<br>Works Division, EDA<br>Commerce, Room H7326<br>Washington, DC 20230                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | The EDA provides grants to states, counties and cities designated as redevelopment areas for public works projects that include developing trails and greenways.                                                                                                                |
| Healthy People 2010 Community Implementation Grants Program /<br>Federal Department of Health and Human Services                                       | Nationwide<br>Funded                       | Administrative Officer<br>Office of Disease Prevention and Health Promotion<br>Hubert H. Humphrey Building, Room 738-G<br>400 Independence Avenue, SW<br>Washington, DC 20201<br>(202) 260-7854                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | The Department of Health and Human Services plans to award hundreds of "micro-grants" (up to \$2,010) to community organizations for activities that support the goals of Healthy People 2010.                                                                                  |
| Land and Water Conservation Fund (LWCF) / National Park<br>Service                                                                                     | Nationwide<br>Funded                       | <a href="http://www.nps.gov/programs/lwcf/index.html">http://www.nps.gov/programs/lwcf/index.html</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Provide "close-to-home" park and recreation opportunities. LWCF grants can be used by communities to build a variety of facilities including trails and greenways.                                                                                                              |
| Recreational Trails Program / U.S. Department of Transportation's<br>Federal Highway Administration (FHWA)                                             | Nationwide<br>Funded                       | Christopher B. Douwes<br>Recreational Trails Program Manager<br>Highway Administration<br>400 Seventh St. SW, Room 3301<br>Washington, DC 20590<br>(202) 366-5013<br>Christopher.Douwes@fhwa.dot.gov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The Recreational Trails Program provides funds to the States to develop and maintain recreational trails and trail-related facilities for both nonmotorized and motorized recreational trail uses.                                                                              |

**Potential Funding Sources Matrix  
Chartiers Creek Trail and Greenway  
Private Agencies**

| FUNDING PROGRAM N                                                                  | JURISDICTION                   | CONTACT INFORMATION                                                                                                                                                                                                                                                              | INTENDED FUNDING USES                                                                                                                                              |
|------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AmeriCorps's National Civilian Community Corps (NCCC)                              | Nationwide<br>Privately Funded | arcc@cns.gov<br><a href="http://www.ameriacorps.org/nccc">http://www.ameriacorps.org/nccc</a><br>105 North Front Street<br>Harrisburg, PA 17101<br>Phone: 717-230-8166<br>Fax: 717-230-8167<br><a href="http://www.conservationsfund.org/">http://www.conservationsfund.org/</a> | Building or improving trails.                                                                                                                                      |
| The Conservation Fund - Pennsylvania Office                                        | Statewide                      | Bikes Belong Coalition 368 Beacon Street<br>Suite 102<br>Brookline, MA 02446<br>(617) 734-2800<br><a href="mailto:mail@bikesbelong.org">mail@bikesbelong.org</a>                                                                                                                 | Greenway and Land Conservation / Recreation                                                                                                                        |
| Bikes Belong Coalition                                                             | Nationwide<br>Privately Funded | Leigh Anne McDonald<br>Greenways Coord.<br>The Conservation Fund<br>North Kent Street, Suite 1120<br>22209 Arlington, VA<br>(703) 525-6300<br><a href="mailto:lmcdonald@conservationfund.org">lmcdonald@conservationfund.org</a>                                                 | Assist local organizations, agencies, and citizens in developing bicycle facility projects.                                                                        |
| Kodak American Greenways Awards / The Conservation Fund                            | Nationwide<br>Privately Funded | <a href="http://www.nccc.nps.gov/programs/awards/index.html">http://www.nccc.nps.gov/programs/awards/index.html</a>                                                                                                                                                              | Stimulate greenway planning and design linking the nation's natural areas, historic sites, parks and open space.                                                   |
| Land and Water Conservation Fund (LWCF) / National Park Service                    | Nationwide<br>Federally Funded | Liz Dooley, Alliance Programs Manager<br>1422 Fenwick Lane<br>Silver Spring, MD 20910<br>(301) 565-6704 x212<br><a href="mailto:NTE@AmericanHiking.org">NTE@AmericanHiking.org</a>                                                                                               | Provide "close-to-home" park and recreation opportunities. LWCF grants can be used by communities to build a variety of facilities including trails and greenways. |
| National Trails Fund / American Hiking Society                                     | Nationwide<br>Privately Funded | Oracle Giving<br>Oracle Parkway<br>50P11<br>CA 94065                                                                                                                                                                                                                             | National Trails Fund grants have been used for land acquisition, constituency building campaigns and traditional trail work projects.                              |
| Oracle Corporate Giving Program                                                    | Nationwide<br>Privately Funded | 500 Mail Stop<br>Redwood Shores, CA 94065                                                                                                                                                                                                                                        | Oracle provides grants to endangered animal protection, environmental protection, science and technology education. Past recipients have included trail groups.    |
| PowerBar's Direct Impact on Rivers and Trails Program (D.I.R.T.) / Powerfood, Inc. | Nationwide<br>Privately Funded | DIRT Program<br>Shattuck Avenue<br>CA 94710<br><a href="http://www.powerbar.com/whoWeAre/dirt/index.asp">http://www.powerbar.com/whoWeAre/dirt/index.asp</a>                                                                                                                     | D.I.R.T. supports efforts to protect, preserve and restore recreational lands and waterways.                                                                       |

**Potential Funding Sources Matrix**  
**Chartiers Creek Trail and Greenway**  
**State Government**

| State Senator                                                                                                                                                                  | Address                                                                                                | D.O. Address                                                                                                             | Municipalities                                                                                                     | Committees                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Senate District 37<br>Timothy F. Murphy @<br><a href="mailto:tmurphy@senate.gov">tmurphy@senate.gov</a><br><a href="http://murphy.senate.gov">http://murphy.senate.gov</a>     | Senate Box 203037<br>Harrisburg, PA 17120-3037<br>170 Capitol Building<br>(717) 787-5639               | 504 Washington Road<br>Pittsburgh, PA 15228                                                                              | Bridgeville, Collier Twp., South Fayette Twp.,<br>Upper St. Clair                                                  | Aging & Youth (Chair)<br>Public Health & Welfare (Vice Chair)<br>Appropriations<br>Banking & Insurance<br>Communications & High Technology<br>Education |
| Senate District 42<br>Jack Wagner (D)<br><a href="mailto:jwagner@senate.gov">jwagner@senate.gov</a><br><a href="http://www.senatorwagner.com">http://www.senatorwagner.com</a> | Senate Box 203042<br>Harrisburg, PA 17120-3042<br>458 Capitol Building<br>(717) 787-5300               | 1333 Banksville Road<br>Pittsburgh, PA 15216-3301<br>(412) 565-7974                                                      | Carnegie, Crafton, Heidelberg,<br>McKees Rocks, Pittsburgh, Robinson Twp.,<br>Rosslyn Farms, Scott Twp., Thornburg | Banking & Insurance - Minority Chair<br>Appropriations<br>Military & Veterans Affairs<br>Rules & Executive Nominations                                  |
| Senate District 46<br>J. Barry Stout (D)<br><a href="mailto:bstout@senate.gov">bstout@senate.gov</a>                                                                           | Senate Box 203046<br>Harrisburg, PA 17120-3046<br>10 East Wing<br>(717) 787-1463                       | 1024 Route 519, Suite 400<br>Eighty-Four, PA 15330                                                                       | Cecil Twp., North Strabane Twp. (Wash. Co)                                                                         | Transportation - Minority Chair<br>Appropriations<br>Environmental Resources And Energy<br>Rules & Executive Nominations                                |
| State Legislator                                                                                                                                                               | Address                                                                                                | D.O. Address                                                                                                             | Municipalities                                                                                                     | Committee Assignments                                                                                                                                   |
| Hon. Thomas C. Petrone (D)<br>27th Legislative District                                                                                                                        | 202 South Office Building<br>Harrisburg, PA 17120-2020<br>(717) 787-6410<br>FAX (717) 772-9966         | 179 Stauben Street<br>P.O. Box 8557<br>Pittsburgh, PA 15220<br>(412) 928-9514<br>FAX (412) 565-3170                      | Crafton, McKees Rocks<br>Pittsburgh 28th Ward                                                                      | Urban Affairs (Chairman)                                                                                                                                |
| Hon. John A. Maher - R<br>40th Legislative District                                                                                                                            | 160-B East Wing<br>Harrisburg, PA 17120-2020<br>(717) 783-1522<br>FAX (717) 783-8332                   | 711 Summerfield Commons<br>2547 Washington Road<br>Upper St. Clair, PA 15241<br>(412) 831-8080<br>FAX (412) 831-8083     | Upper St. Clair                                                                                                    | Appropriations<br>Finance<br>Local Government<br>Professional Licensure<br>Transportation                                                               |
| Hon. Thomas L. Stevenson - R<br>42nd Legislative District                                                                                                                      | Room 409<br>South Office Building<br>Harrisburg, PA 17120-2020<br>(717) 787-2047<br>FAX (717) 772-2468 | 246 Washington Road<br>Pittsburgh, PA 15216<br>(412) 571-2169<br>FAX (412) 429-5090                                      | Rosslyn Farms, Scott Twp.,<br>Thornburg                                                                            | Commerce & Econ Dev. (Sec)<br>Education<br>Ethics (Chairman)<br>Intergovernmental Affairs<br>Urban Affairs (Chairman)                                   |
| Hon. John A. Pippy - R<br>44th Legislative District                                                                                                                            | 402 South Office Building<br>Harrisburg, PA 17120-2020<br>(717) 787-6651<br>FAX (717) 783-3899         | 937 Beaver Grade Road<br>Moon Township, PA 15108<br>(412) 262-2260<br>FAX (412) 920-2052                                 | Collier Twp., Robinson Twp.<br>(Western Collier and<br>Nevillewood)                                                | Appropriations<br>Insurance (Secretary)<br>Transportation (Secretary)<br>Veterans Affairs & Emergency<br>Preparedness                                   |
| Hon. Nick Koik (D)<br>45th Legislative District                                                                                                                                | Not assigned yet                                                                                       | 1004 Fifth Avenue<br>Corapolis, PA 15108<br>(412) 264-4260                                                               | Bridgeville, Carnegie,<br>Collier Twp., McKees Rocks,<br>Scott Twp.                                                | Newly elected starts Jan 2003                                                                                                                           |
| Hon. Victor John Lescoz (D)<br>46th Legislative District                                                                                                                       | Room 414<br>Main Capitol<br>Harrisburg, PA 17120-2020<br>(717) 787-2769<br>FAX (717) 783-9755          | 1574 Smith Twp. State Rd.<br>Suite 4<br>P.O. Box 370<br>Atlasburg, PA 15004-0370<br>(724) 947-4422<br>FAX (724) 947-4471 | South Fayette Twp. (All Co.)<br>Cecil Twp. (Wash Co.)                                                              | Rules                                                                                                                                                   |
| Hon. Timothy J. Solobay (D)<br>48th Legislative District                                                                                                                       | 125B East Wing<br>Harrisburg, PA 17120-2020<br>(717) 787-1188<br>FAX (717) 705-1887                    | Canonsburg Borough Building<br>Room 205<br>Canonsburg, PA 15317<br>(724) 746-3762<br>FAX (724) 746-3509                  | North Strabane Twp. (Wash.)                                                                                        | Agriculture & Rural Affairs<br>Game & Fisheries<br>Local Government<br>Veterans Affairs & Emergency<br>Preparedness                                     |



**Potential Funding Sources Matrix  
Chartiers Creek Trail and Greenway  
County and City Government**

| <b>Council Member</b>                                                                                            | <b>Address</b>                                                                          | <b>Telephone Number / Email</b>                                                                                                                                                                                                      | <b>Municipalities</b>                                                                                 |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Pittsburgh City Council<br>District 2<br>Alan Hertzberg                                                          | City/County Building<br>Suite 510/5th Floor<br>414 Grant Street<br>Pittsburgh, PA 15219 | (412) 255-8963<br>FAX (412) 255-2821                                                                                                                                                                                                 | Pittsburgh 28th Ward                                                                                  |
| Allegheny County Council - District 1<br>Ron Francis, Jr.                                                        | 119 County Courthouse<br>436 Grant Street<br>Pittsburgh, PA 15219                       | (412) 350-6525<br><a href="mailto:francis@county.allegheny.pa.us">francis@county.allegheny.pa.us</a>                                                                                                                                 | Robinson Twp., Thornburg                                                                              |
| Allegheny County Council - District 4<br>Doug Price                                                              | 119 County Courthouse<br>436 Grant Street<br>Pittsburgh, PA 15219                       | (412) 350-6540<br><a href="mailto:dprice@county.allegheny.pa.us">dprice@county.allegheny.pa.us</a>                                                                                                                                   | Bridgeville, Carnegie, Collier,<br>Craffon, Heidelberg, Scott Twp.,<br>South Fayette, Upper St. Clair |
| Allegheny County Council - District 12<br>Wayne Fontana                                                          | 119 County Courthouse<br>436 Grant Street<br>Pittsburgh, PA 15219                       | (412) 350-6580<br><a href="mailto:wfontana@county.allegheny.pa.us">wfontana@county.allegheny.pa.us</a>                                                                                                                               | McKees Rocks, Pittsburgh 28th<br>Ward                                                                 |
| Washington County Commissioners<br>John P. Bevec, (Board Commissioner)<br>J. Bracken Burns, Sr.<br>Diana L. Irey | Courthouse Square<br>100 West Beau Street<br>Washington, PA 15301                       | <a href="mailto:burnsb@co.washington.pa.us">burnsb@co.washington.pa.us</a><br><a href="mailto:bevec@co.washington.pa.us">bevec@co.washington.pa.us</a><br><a href="mailto:jireyd@co.washington.pa.us">jireyd@co.washington.pa.us</a> |                                                                                                       |

## APPENDICES

**APPENDIX A**  
**LAND STEWARDSHIP FORM**



EXHIBIT M

**TRUST FOR LAND RESTORATION**  
**MONITORING FORM**

**TRUST FOR LAND RESTORATION**  
**Conservation Easement**  
**Annual Stewardship Monitoring Report**

*Departure Time* \_\_\_\_\_ *From where?* \_\_\_\_\_

*Return Time* \_\_\_\_\_ *Mileage* \_\_\_\_\_ *From where?* \_\_\_\_\_

*Property Name* \_\_\_\_\_ *Date* \_\_\_\_\_

Town \_\_\_\_\_ County \_\_\_\_\_

Landowner \_\_\_\_\_ Has owner changed? \_\_\_\_\_

Confirm Owner address \_\_\_\_\_

Confirm Owner telephone number \_\_\_\_\_

Owner e-mail address \_\_\_\_\_

**Easement Materials**

Maps to property \_\_\_\_\_ Maps of property \_\_\_\_\_

Landowner contacted \_\_\_\_\_?

Landowner (designee) accompany monitoring? \_\_\_\_\_

Easement document reviewed & summarized? \_\_\_\_\_

Baseline document reviewed & summarized? \_\_\_\_\_

File and past monitoring reports reviewed? \_\_\_\_\_

Any special monitoring activities required by the conservation easement? \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Easement Summary**

**Conservation Values:**

☐ Wildlife/Plant habitat

☐ River/stream

☐ Scenic

☐ Adjacent to conserved property/public land

☐ Other \_\_\_\_\_

Summary of Prohibited Uses (Describe specific prohibition):

- ☐ New roads \_\_\_\_\_
- ☐ Timber harvest \_\_\_\_\_
- ☐ Signs/billboards \_\_\_\_\_
- ☐ New structures/improvements \_\_\_\_\_
- ☐ Transfer of water rights \_\_\_\_\_
- ☐ Commercial/industrial uses \_\_\_\_\_
- ☐ Mineral development \_\_\_\_\_
- ☐ Recreational Uses \_\_\_\_\_
- ☐ Accumulation of trash \_\_\_\_\_
- ☐ New fences \_\_\_\_\_
- ☐ Domestic animals \_\_\_\_\_
- ☐ Overgrazing \_\_\_\_\_
- ☐ OHV use \_\_\_\_\_
- ☐ Subdivision \_\_\_\_\_
- ☐ \_\_\_\_\_
- ☐ \_\_\_\_\_
- ☐ \_\_\_\_\_
- ☐ \_\_\_\_\_
- ☐ \_\_\_\_\_

**Pre-Monitoring Checklist**

1. \_\_\_\_\_ Reviewed terms of easement with landowner.
2. \_\_\_\_\_ Reviewed any natural or man-made changes to the property with landowner.
3. \_\_\_\_\_ Discussed any problems with trespass or neighbors that affect conservation values or that may constitute a violation of the easement.

4. \_\_\_\_ Discussed landowner's plans for the property in the next 12 months, including plans to sell or lease the property.

Does landowner foresee any changes in the coming year? \_\_\_\_ If so, describe (road construction, structures to be built, logging, etc.) \_\_\_\_\_

Property Condition:

Boundaries identified? \_\_\_\_\_

Boundary survey reveal any conditions to note? \_\_\_\_\_

Visit to each photo point – any difference between Baseline should be noted here and new photo taken: \_\_\_\_\_

Photographs taken – Describe: \_\_\_\_\_

Describe land use practices \_\_\_\_\_

Any alterations to practices or land? \_\_\_\_ If so, describe: \_\_\_\_\_



Possible land management issues (over-grazing, degradation of riparian/wetland areas, timber harvesting, noxious weeds, etc.) \_\_\_\_\_

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General impression and description of inspected area (quality of vegetation, well-maintained land uses/structures, etc.)

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Any questions or observations that require follow-up? \_\_\_\_\_

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Comments: \_\_\_\_\_

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TLR's monitor name \_\_\_\_\_

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**APPENDIX B**

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